

SIEMENS



Industrial Controls

Load Feeders and Motor Starters

SIRIUS 3RM1 Motor Starter

Manual

Edition

09/2015

siemens.com

Industrial Controls

Load feeders and motor starters SIRIUS 3RM1 motor starter

Manual

<u>Introduction</u>	1
<u>Product-specific safety information</u>	2
<u>Description</u>	3
<u>Configuration</u>	4
<u>Mounting</u>	5
<u>Connection</u>	6
<u>Operator control and monitoring</u>	7
<u>Service and maintenance</u>	8
<u>Technical data</u>	9
<u>Dimension drawings</u>	10
<u>Circuit diagrams</u>	11
<u>Typical circuits</u>	A
<u>Directives</u>	B

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING

Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

Trademarks

All names identified by ® are registered trademarks of Siemens AG. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

Table of contents

1	Introduction.....	9
1.1	Responsibility of the user for system configuration and functionality	9
1.2	Required basic knowledge.....	9
1.3	Scope	10
1.4	Definition	10
1.5	Conformity.....	10
1.6	Further documentation.....	11
1.7	Siemens Industry Online Support.....	12
1.8	DataMatrix code.....	14
1.9	Quick navigation	15
2	Product-specific safety information.....	17
2.1	General safety notes.....	17
2.2	Safety information for hazardous areas.....	18
2.3	Safety instructions for safety-related applications	18
2.4	Intended use	20
2.5	Current information about operational safety	20
2.6	Declaration of conformity	21
3	Description.....	23
3.1	Overview	23
3.2	Applications.....	24
3.3	Hybrid technology	25
3.4	Device versions.....	26
3.5	Functions	29
3.5.1	Switching the 3RM1 motor starter under normal operating conditions.....	29
3.5.1.1	Direct-on-line starter	30
3.5.1.2	Reversing starter.....	30
3.5.2	Protective functions.....	32
3.5.2.1	Overload protection.....	32
3.5.2.2	Equipment protection	32
3.5.2.3	Phase failure protection	33
3.5.2.4	Asymmetry monitoring	34
3.5.2.5	Thermal calculation method (motor memory module)	34
3.5.2.6	ATEX-certified motor overload protection in the case of 3RM11/3RM13 Failsafe	36
3.5.3	Shutdown on malfunction	40
3.5.4	Safety-related shutdown with 3RM11/3RM13 Failsafe motor starters	41

3.5.4.1	Safety-related shutdown via the control supply voltage	42
3.5.5	Carrying out the function test	43
3.6	Accessories and article number scheme	45
3.6.1	Overview of all device components	45
3.6.2	Accessories	46
3.6.2.1	Infeed system	46
3.6.2.2	Sealable cover	48
3.6.2.3	Wall mounting	48
3.6.2.4	Device connectors	49
3.6.2.5	Terminals	52
3.6.2.6	Device identification label	52
3.6.3	Article number scheme for SIRIUS 3RM1 motor starter	53
4	Configuration	55
4.1	Rated operational current and derating	55
4.2	Ambient conditions	60
4.2.1	Application environment	60
4.2.2	Ambient temperature	60
4.2.3	Mounting position	60
4.2.4	Grounding measures	60
4.2.5	Permissible operating voltage	61
4.2.6	Minimum load current	61
4.3	Load feeders - protection against short circuit	62
4.3.1	Protection against short circuit	62
4.3.2	Configuration of load feeders	62
4.3.3	Configuration of a load feeder with 3RM1 motor starter in compliance with IEC	62
4.3.3.1	Fuseless design	63
4.3.3.2	Fused design	65
4.3.4	Configuration of a load feeder with 3RM1 motor starter in compliance with UL	67
4.3.4.1	Function according to UL 508	67
4.3.4.2	Configuration of individual load feeders	67
4.3.4.3	Configuring a group of load feeders (group installation)	70
4.4	Infeed for the main circuit	72
4.4.1	Infeed options	72
4.4.2	3RM19 3-phase infeed system	72
4.5	Configuration with device connectors	74
4.6	Examples/applications	74
4.6.1	Operation with the EMERGENCY-STOP function	74
5	Mounting	77
5.1	Warning notices	77
5.2	Mounting the devices on a level surface	78
5.3	Disassembling the devices from a level surface	79
5.4	Mounting the devices on a standard mounting rail	80
5.5	Disassembling devices from a standard mounting rail	81
5.6	Mounting the devices with device connectors on a standard mounting rail	82

5.7	Disassembling the devices with device connectors from a standard mounting rail	85
5.8	Mounting the devices with device connectors on a wall	88
5.9	Disassembling the devices with device connectors from a wall	91
5.10	Mounting the sealable cover	93
6	Connection	95
6.1	Connecting the screw-type terminals.....	95
6.2	Disconnecting the screw-type terminals	97
6.3	Connecting the push-in terminals	98
6.3.1	Wiring rules for spring-loaded terminals (with push-in technology)	98
6.3.2	Connecting the push-in terminals	99
6.4	Disconnecting the push-in terminals	101
6.5	Attaching the terminals	102
6.6	Removing the terminals	103
6.7	Connecting the infeed system (option)	104
7	Operator control and monitoring	107
7.1	Operator controls	107
7.1.1	Parameterizing the 3RM1 motor starter.....	108
7.1.2	Setting the rated operational current	109
7.1.3	Setting the RESET method.....	109
7.2	Displays and location of the connections.....	111
7.2.1	Alarms, faults and system events	114
7.2.1.1	LEDs	114
7.2.1.2	Status indicators of the 3RM1 motor starter	114
7.2.1.3	Signaling faults on external I/O	116
7.2.1.4	Fault rectification.....	116
8	Service and maintenance.....	119
8.1	Maintenance and service	119
8.2	Device replacement	120
9	Technical data	121
9.1	Data sheet.....	121
9.2	Connection cross sections	122
9.3	Number of starting operations	123
9.4	Overload protection/device protection characteristic.....	124
10	Dimension drawings.....	127
10.1	3RM1 dimension drawings	127
10.2	Dimension drawings for 3RM1 device connectors	130

11	Circuit diagrams.....	135
11.1	3RM10 circuit diagrams (direct-on-line starter, Standard).....	135
11.2	3RM11 circuit diagrams (direct-on-line starter, Failsafe).....	137
11.3	3RM12 circuit diagrams (reversing starter, Standard).....	138
11.4	3RM13 circuit diagrams (reversing starter, Failsafe).....	140
A	Typical circuits	141
A.1	Typical circuits for 3RM1.....	141
A.1.1	Direct-on-line starter 24 V DC with switch operation	141
A.1.2	Direct-on-line starter 24 V DC with switch operation and 230 V brake.....	142
A.1.3	Direct-on-line starter 24 V DC with switch operation and 400 V brake.....	143
A.1.4	Direct-on-line starter 24 V DC with switch operation and single-phase motor	144
A.1.5	Direct-on-line starter with group protection, 24 V DC and PLC operation	146
A.1.6	Reversing starter 24 V DC with PLC operation	148
A.2	Typical circuits for safety-related shutdown	149
A.2.1	General safety notes	149
A.2.2	3SK1 safety relay with 3RM13 motor starter via device connector	152
A.2.3	3SK1 safety relay wired with 3RM13 motor starter.....	154
A.2.4	3SK2 safety relay with 3RM11 motor starter via device connector	156
A.2.5	3SK2 safety relay wired to 3RM11 motor starter	157
A.2.6	3RM13 reversing starter with PLC control, F-DO pp-switching	158
A.2.7	3RM13 reversing starter with PLC control, F-DO pm-switching	160
B	Directives.....	161
B.1	ESD Guidelines.....	161
	Glossary	163
	Index	167

Introduction

1.1 Responsibility of the user for system configuration and functionality

The SIRIUS 3RM1 motor starters described here have been developed to carry out switching functions as part of a plant or machine.

The 3RM1 motor starters are available as direct-on-line starters in Standard design without safety-related shutdown (3RM10) and in Failsafe design with safety-related shutdown (3RM11 Failsafe), and also as reversing starters without safety-related shutdown (3RM12) and in Failsafe design with safety-related shutdown (3RM13 Failsafe).

The following must be noted when using 3RM11 Failsafe/3RM13 Failsafe safety-related motor starters:

A complete safety system consists of sensors, evaluation units, signaling devices and methods for safety-related shutdown.

The manufacturer is responsible for ensuring safe overall functioning of a plant or machine with safety-related components.

Siemens AG, its regional offices, and associated companies (hereinafter referred to as "Siemens") cannot guarantee all the properties of an overall installation or machine that has not been designed by Siemens.

Nor can Siemens assume liability for recommendations that appear or are implied in the following description. No new guarantee, warranty, or liability claims beyond the scope of the Siemens general terms of supply are to be derived or inferred from the following description.

1.2 Required basic knowledge

A general knowledge of the following areas is needed in order to understand this manual:

- Industrial controls
- Digital circuitry
- Automation technology
- Safety engineering

1.3 Scope

This manual is valid for SIRIUS 3RM1 motor starters. It contains a description of the motor starter and its functions. It provides information about configuration, commissioning and servicing. You will also find information on the infeed system, device connectors and further accessories in the manual.

To facilitate configuration, the manual contains dimension drawings, circuit diagrams and technical data of the system components.

SIEMENS reserves the right of including a Product Information for each new component, and for each component of a later version.

1.4 Definition

In this manual, "3RM1 motor starter" always refers to all variants of the SIRIUS 3RM1 motor starters.

1.5 Conformity

Standards

All 3RM1 motor starters comply with the following standards:

- IEC 60947-4-2:2011-05
- IEC 60947-4-3:2011-07

The 3RM11 Failsafe and 3RM13 Failsafe motor starters also comply with the following standards:

- EN 62061: 2005
- EN ISO 13849-1:2008
- IEC 61508-1:2010
- IEC 61508-2:2010
- IEC 61508-3:2010

Approvals, test reports, characteristics

Confirmation of approvals, test certificates and characteristic curves is available via the Internet (<http://www.siemens.com/sirius/approvals>).

Degree of protection

The 3RM1 motor starter's degree of protection is IP20.

The infeed system for the 3RM1 motor starter features IP20 degree of protection.

 DANGER
Hazardous voltage Will result in death, serious physical injury, or damage to property. To ensure protection against an electric shock hazard with the hinged cover open at the signaling contacts 95, 96, 98 at a voltage of ≥ 50 V, screw in all terminal screws that are not needed to clamp conductors.

1.6 Further documentation

Further documents that might be of interest for your configuration:

Title of the manual	Article number ¹⁾
SIRIUS 3SK1 safety relays (https://support.industry.siemens.com/cs/ww/de/view/67585885)	3ZX1012-0SK11-0AC0
SIRIUS 3SK2 Safety Relays (https://support.industry.siemens.com/cs/ww/de/view/109444336)	3ZX1012-0SK21-1AB1

¹⁾ The manuals are available in the Service&Support Portal for downloading free of charge.

1.7 Siemens Industry Online Support

Information and service

At Siemens Industry Online Support you can obtain up-to-date information from our global support database quickly and simply. To accompany our products and systems, we offer a wealth of information and services that provide support in every phase of the lifecycle of your machine or plant – from planning and implementation and commissioning, right through to maintenance and modernization:

- Product support
- Application examples
- Services
- Forum
- mySupport

Link: Siemens Industry Online Support (<https://support.industry.siemens.com/cs/ww/en/>)

Product support

Here you will find all the information and comprehensive know-how for your product:

- **FAQs**
Our replies to frequently asked questions.
- **Manuals/operating instructions**
Read online or download, available as PDF or individually configurable.
- **Certificates**
Clearly sorted according to approving authority, type and country.
- **Characteristics**
For support in planning and configuring your system.
- **Product announcements**
The latest information and news concerning our products.
- **Downloads**
Here you will find updates, service packs, HSPs and much more for your product.
- **Application examples**
Function blocks, background and system descriptions, performance statements, demonstration systems, and application examples, clearly explained and represented.
- **Technical data**
Technical product data for support in planning and implementing your project.

Link: Product support (<https://support.industry.siemens.com/cs/ww/en/ps>)

mySupport

With "mySupport", your personal work area, you get the very best out of your Industry Online Support experience. Everything enables you to find the right information - every time.

The following functions are now available:

- **Personal Messages**

Your personal mailbox for exchanging information and managing your contacts

- **Requests**

Use our online form for specific solution suggestions, or send your technical inquiry directly to a specialist in Technical Support

- **Notifications**

Make sure you always have the latest information - individually tailored to your needs

- **Filter**

Simple management and re-use of your filter settings from Product Support and the Technical Forum

- **Favorites / Tagging**

Create your own knowledge database by assigning "Favorites" and "Tags" to documents – simply and efficiently

- **Entries last viewed**

Clear presentation of your last viewed entries

- **Documentation**

Configure your individual documentation from different manuals – quickly and without complications

- **Personal data**

Change personal data and contact information here

- **CAX data**

Simple access to thousands of items of CAX data such as 3D models, 2D dimension drawings, EPLAN macros, and much more

Configurator

Various configurators are available on the Internet to assist you with configuration.

The configurator for 3RM1 motor starters and matching accessories is an easy-to-use selection and configuration tool. You can select the individual components and plan your system in accordance with your specific requirements. You can save your selection, export it as a text file or you can order it directly.

The configurator automatically compiles a document list of the information available in Service&Support for every component. You can use it as the basis for putting together your system documentation.

Link: Configurator (<http://www.siemens.com/sirius/configurators>)

1.8 DataMatrix code

A DataMatrix code is lasered onto the lower terminal cover of all devices of this series.

DataMatrix codes are standardized in ISO/IEC 16022. The DataMatrix codes on Siemens devices use ECC200 coding.

The following device information is encoded in the DataMatrix codes as a bit stream:

- Article number
- Serial number

This information is stored in the following format in the DataMatrix code:

1P	Article number	+	S	Serial number
Data identifier	Net content	Separator	Data identifier	Net content

Note

The information content is displayed without spaces.

This machine-readable information simplifies and accelerates handling of the respective devices.

As well as fast access to the serial numbers of the respective devices for unique identification, the DataMatrix codes simplify communication with Siemens Technical Support.

SIEMENS Industry Support App

DataMatrix codes primarily enable extremely fast and convenient access to all device-specific information relating to an article number in the SIEMENS Service&Support Portal (<https://support.industry.siemens.com/cs/ww/en/>), such as operating instructions, manuals, data sheets, FAQs, etc.

We provide the SIEMENS Industry Support app free for this purpose and it can be used on most commercially available smart phones and tablets.

The SIEMENS Industry Support app is available for iOS and Android-based devices and can be accessed via the following links:



Link for Android



Link for iOS



Link for Windows phone

1.9 Quick navigation

Note

Reading the manual

The quick navigation is provided to help you find important topics quickly. It does not substitute for reading the manual.

Always read the entire manual.

Configuring

- Functions (Page 29)
- Technical data (Page 121)
- Derating (temperature, installation altitude, installation) Rated operational current and derating (Page 55)
- Applications Operation with the EMERGENCY-STOP function (Page 74)
- Circuit diagrams (Page 135)
- Product-specific safety information (Page 17)
- Overload trip:
 - Thermal calculation method (motor memory module) (Page 34)
 - Setting the RESET method (Page 109)
- Terminal designations Terminals (Page 52)
- Also note the following: Number of starts Number of starting operations (Page 123), switching times Switching the 3RM1 motor starter under normal operating conditions (Page 29), pauses Technical data (Page 121), minimum current Rated operational current and derating (Page 55) and Minimum load current (Page 61)

Commissioning:

- Diagnosis Alarms, faults and system events (Page 114)
- Fault rectification (Page 116)
- Terminal designation Terminals (Page 52)
- Connection method Device versions (Page 26) and Connection cross sections (Page 122)
- Removable terminals Terminals (Page 52)
- Setting the rated operational current (Page 109)

Service:

- Diagnosis Alarms, faults and system events (Page 114)
- Setting the rated operational current (Page 109)
- Removable terminals Terminals (Page 52)
- Terminal designation Terminals (Page 52)

FAQ (<https://support.industry.siemens.com/cs/ww/en/view/81525494>)

Product-specific safety information

2.1 General safety notes

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

To ensure protection against an electric shock hazard with the hinged cover open at the signaling contacts 95, 96, 98 at a voltage of ≥ 50 V, screw in all terminal screws that are not needed to clamp conductors.

NOTICE

Protection against electrostatic charge

Damage to property can occur

When handling and installing the 3RM1 motor starters, ensure that the components are protected from electrostatic charge. Changes to the system configuration and wiring are only permissible after disconnection from the power supply.

2.2 Safety information for hazardous areas

WARNING

Hazardous voltage

Can result in death, serious physical injury, or damage to property.

Installation of the motor starters in hazardous areas

The components of the 3RM1 motor starters are not suitable for installation in Class I and Class II Hazardous Locations.

Please contact your ATEX specialist.

You will find additional information in Chapter "ATEX-certified motor overload protection in the case of 3RM11/3RM13 Failsafe (Page 36)."

2.3 Safety instructions for safety-related applications

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

WARNING

Function test interval for 3RM11/3RM13 Failsafe motor starters

Can result in death, serious physical injury, or damage to property.

In continuous operation, the key safety values apply in the case of a function test interval (state change of the outputs) ≤ 1 year.

Annual function test

- Actuate the connected sensors.
- Check their effect on the safety relay and the downstream actuators.
- Activate the safety relay via the connected sensors.
- Check their effect on the safety relay and the downstream actuators.
- Defective devices must be replaced.

⚠ WARNING**Bypassing the safety function in the event of a fault in the case of 3RM11/3RM13 Failsafe motor starters with 110 to 230 V AC/110 V DC control supply voltage**

Can result in death, serious physical injury, or damage to property.

The control signal for the control inputs in the case of 3RM11/3RM13 Failsafe motor starters with 110 to 230 V AC control supply voltage must come from A1. Otherwise, the safety function is bypassed in the event of a fault. Thus, only relay outputs are admissible when using a PLC.

Do not use a separate control voltage. Only use the relay outputs when using a PLC.

⚠ WARNING**Bypassing the safety function when using device connectors**

Can result in death, serious physical injury, or damage to property.

When the product is operated with a 3SK safety relay and a device connector, the supply voltage for 3RM1 motor starters is fed in via the device connectors.

In this case, do not connect anything to terminals A1 and A2 of the 3RM1 motor starters, in order to prevent bypassing of the safety function.

⚠ WARNING**Failure of the safety function during hybrid operation of Failsafe and Standard motor starters**

Can result in death, serious physical injury, or damage to property.

Hybrid operation of 3RM10/3RM12 Standard motor starters with 3RM11/3RM13 Failsafe motor starters in safety-related applications is not admissible.

Only ever use safety-related motor starters (3RM11 Failsafe and 3RM13 Failsafe) in safety-related applications.

Note**SILCL 3 to EN 62061:2005, PL e/Cat. 4 to EN ISO 13849-1:2008**

The 3RM11 Failsafe and 3RM13 Failsafe safety-related motor starters are designed in such a way as to allow implementation of safety-related applications up to SILCL 3 in accordance with EN 62061 and PL e/Cat. 4 in accordance with EN ISO 13849-1.

Note**Use of PELV/SELV power supply units**

Power to the 3RM11 Failsafe and 3RM13 Failsafe safety-related motor starters must be supplied from PELV/SELV power supply units.

2.4 Intended use

 WARNING
Hazardous voltage Can result in death, serious physical injury, or damage to property. Proper use of hardware products This equipment is only allowed to be used for the applications described in the catalog and in the technical description, and only in conjunction with non-Siemens equipment and components recommended by Siemens. Correct transport, storage, installation and assembly, as well as careful operation and maintenance, are required to ensure that the product operates safely and without faults. EU note: Commissioning is absolutely prohibited until it has been ensured that the machine in which the component described here is to be installed complies with the stipulations of the Directive 2006/42/EC.

2.5 Current information about operational safety

Important note for maintaining operational safety of your system

 WARNING
Hazardous voltage Can result in death, serious physical injury, or damage to property. Please take note of our latest information Systems with safety-related characteristics are subject to special operational safety requirements on the part of the operator. The supplier is also obliged to comply with special product monitoring measures. For this reason, we publish a special newsletter containing information on product developments and features that are (or could be) relevant to operation of safety-related systems. By subscribing to the appropriate newsletter, you will ensure that you are always up-to-date and able to make changes to your system, when necessary: Siemens newsletter (http://www.industry.siemens.com/newsletter) Sign on to the following newsletter under "Products & Solutions": • Control Components and System Engineering News • Safety Integrated Newsletter

2.6 Declaration of conformity

The manufacturer declares that the safety components of the 3RM1 motor starter series in the designs marketed by us comply with the applicable basic health and safety requirements of the EC Directives* stated (including amendments), and that the stated standards* were applied in their design and construction.

* You can download the complete EC Declaration of Conformity from the Internet (<http://www.siemens.com/sirius/approvals>) as a PDF.

2.6 Declaration of conformity

Description

3.1 Overview

The 3RM1 motor starter is a compact device with a width of 22.5 mm. The 3RM1 motor starter comprises combinations of relay contacts, power semiconductors, and a solid-state overload relay for switching three-phase motors up to 3 kW (at 400 V) and resistive loads up to 10 A (at AC voltages to 500 V) under normal operating conditions.



3.2 Applications

The 3RM1 motor starters can be used wherever combinations of contactors and overload relays were previously used.

Thanks to the additional functionality of safety-related shutdown, the 3RM11 Failsafe and 3RM13 Failsafe motor starter variants are ideally suited to safety-related applications up to SILCL 3 in accordance with EN 62061 and PL e/Cat. 4 in accordance with EN ISO 13849-1.

Short-circuit protection in a load feeder designed in this way must be implemented by appropriate upstream short-circuit protection devices. These can be, for example, circuit breakers or appropriate fuses.

You achieve maximum space-savings benefits by means of a group configuration of the 3RM1 motor starters. This enables several 3RM1 motor starters to be protected by one short-circuit protection device.

The 3RM1 motor starters are used in the following areas, for example:

- Conveyor technology
- Logistics systems
- Production machines
- Machine tools
- Small elevators

3.3 Hybrid technology

The 3RM1 motor starter combines the benefits of semiconductor technology and relay technology.

This combination is known as hybrid technology. The hybrid technology in the 3RM1 motor starter is characterized by the following features:

Switching on

The inrush current in the case of motorized loads is conducted briefly via the semiconductors.

Advantage: The relay contacts are protected. Longer service life is achieved thanks to reduced wear and tear.

Current conducting

The continuous current is conducted via relay contacts.

Advantage: Relay contacts cause less thermal losses than semiconductors.

Switching off

Switch-off is implemented again via the semiconductor.

Advantage: The contacts are only slightly burdened by arcs. This results in increased service life.

Circuit diagram

The semiconductors switch the motor on and off in two phases. In the switched-off state, all three phases are opened by a relay contact.

To achieve the highest safety level SILCL 3 in accordance with EN 62061, PL e/Cat. 4 in accordance with EN ISO 13849-1 in the case of 3RM11/3RM13 fail-safe motor starters, further relay contacts are installed in series with the semiconductors.

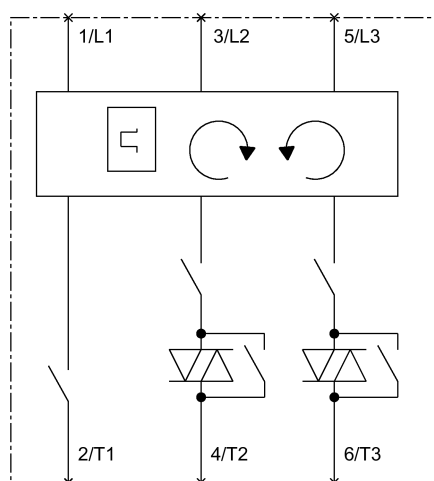


Figure 3-1 Circuit diagram

3.4 Device versions

The 3RM1 motor starters are characterized by their compact design and narrow width.

They can be used for easy assembly of fuseless load feeders with SIRIUS motor starter protectors. Alternatively, combinations with fuses or other short-circuit protection devices are possible.

This allows implementation of load feeders with coordination type 1 for short-circuit currents of up to 55 kA at 400 V.

The 3RM1 motor starters are available in four different versions:

Motor starter version	3RM10	3RM11	3RM12	3RM13
Product symbol				
Direct-on-line starter	✓	✓	-	-
Reversing starter	-	-	✓	✓
Overload protection with wide setting range	✓	✓	✓	✓
ATEX certification overload protection	-	✓	-	✓
Safety-related shutdown up to SILCL 3 to EN 62061, PL e/Cat. 4 to EN ISO 13849-1	-	✓	-	✓

Connection systems

The 3RM1 motor starters are optionally available with screw connections or push-in connections.

Push-in connections are a form of spring-loaded terminals allowing fast wiring without tools for rigid conductors or conductors equipped with end sleeves. For wiring finely-stranded or stranded conductors without end sleeves on push-in connections, a screwdriver (with 3.0 x 0.5 mm blade) is required. A screwdriver (with 3.0 x 0.5 mm blade) is also required to disconnect a conductor.

The advantages of the push-in terminals are found, as with all spring-loaded terminals, in speed of assembly and disassembly and vibration-proof connection. There is no need for the checking and tightening required with screw connections.

Current ranges

The 3RM1 motor starters are designed for the following rated operational currents of the loads:

Versions	Adjustable response value current [A]	Maximum permissible motor power at 400 V AC [kW]
3RM1.01-.....	0.1 ... 0.5	0.12
3RM1.02-.....	0.4 ... 2	0.75
3RM1.07-.....*	1.6 ... 7	3

* The versions of the 3RM1.07-..... motor starter are also suitable for operating resistive loads (e.g. heaters) up to a rated operational current of $I_{AC51} = 10$ A.

Control supply voltages

The 3RM1 motor starters are designed for the following control supply voltages:

Versions	Control supply voltage
3RM1...-AA0.	24 V DC
3RM1...-AA1.	110 ... 230 V AC 50/60 Hz; 110 V DC

Note the following information for the different device versions:

Note

Motor starters with 24 V DC control supply voltage

As is usual on solid-state motor starters, a buffering capacitance of 250 μ F is installed in the 24 V DC control circuit of the 3RM1 Standard and Failsafe motor starters. Its purpose is to specifically shut down the motor with the semiconductors in the event of failure or disconnection of the control supply voltage. As a result, charging currents can arise briefly upon activation of the control supply voltage. The buffering capacitance is decoupled by a diode. This means that there is no influence on the dark test of safety-related outputs.

You can find more information on the Internet

(<https://support.industry.siemens.com/cs/ww/en/view/91372998>).

Note

3RM10/3RM12 Standard motor starters with 24 V DC control supply voltage

The same voltage source (potential) must be used for the control supply voltage and the control inputs.

The reference point for the control inputs is terminal A2.

Note

3RM10/3RM12 Standard motor starters with 110 to 230 V AC/110 V DC control supply voltage

The same voltage source (potential) must be used for the control supply voltage and the control inputs.

The reference point for the control inputs is terminal A2.

When controlling with a PLC, control must be via relay outputs.

There must be no parallel load (e.g. a lamp) connected at the control inputs in the case of motor starters with the product version E01.

Note

3RM11/3RM13 Failsafe motor starters with 24 V DC control supply voltage

The control inputs are electrically isolated from the control supply voltage (A1, A2).

The terminals M1 and M2 are the reference point for the control inputs.

For controlling the control inputs, e.g. via the digital outputs of a PLC, a voltage from an SELV/PELV Class 2 supply is required.

Note

3RM11/3RM13 Failsafe motor starters with 110 to 230 V AC/110 V DC control supply voltage

The same voltage source (potential) must be used for the control supply voltage and the control inputs in safety-related applications.

The reference point for the control inputs is terminal A2.

When controlling with a PLC, control must be via relay outputs.

3.5 Functions

3.5.1 Switching the 3RM1 motor starter under normal operating conditions

A typical application area of the 3RM1 motor starter is the switching and protection of motors. You can also operate the 3RM1 motor starter on resistive loads such as heaters.

During operational switching of the 3RM1 motor starter, a pause time is stored after shutdown of the motor. The motor starter does not accept a new start until the pause time has elapsed.

Table 3- 1 Pause time of 3RM1 motor starters

Version	Article number	Pause time
Standard	3RM10..-....; 3RM12..-....	• 500 ms (E01) • 100 ms (E02)
Fail-safe	3RM11..-....; 3RM13..-....	• 500 ms (E01)

No distinction is made between direct-on-line and reversing starters in terms of pause time.

Note

Minimum loads must be observed for the 3RM1 motor starters.

You will find additional information in the chapter "Minimum load current (Page 61)".

NOTICE

Damage from operating capacitive loads

When using capacitive loads, the switching components in the 3RM1 motor starter can be damaged by excessively high making currents.

Do not operate capacitive loads such as frequency converters on the 3RM1 motor starter. Operation of single-phase capacitor motors is admissible.

NOTICE

Damage during operation on non-sinusoidal voltages

When operating frequency converters, the switching components in the 3RM1 motor starter can be damaged by non-sinusoidal voltages.

Do not operate the 3RM1 motor starter on outputs of frequency converters.

3.5.1.1 Direct-on-line starter

Using the 3RM1 motor starter as a direct-on-line starter

All 3RM1 motor starters can be used as direct-on-line starters.

To switch the 3RM1 motor starter on, apply the control voltage at input IN1.

After the motor is switched off, the motor starter accepts a new start after a pause time.

You can also use the versions of the 3RM1 motor starter with a rated operational current up to 7 A for switching resistive loads up to $I_{AC51} = 10$ A under normal operating conditions.

3.5.1.2 Reversing starter

Using the 3RM1 motor starter as a reversing starter

The 3RM12 and 3RM13 motor starters can be used as reversing starters.

The reversing starter starts the motor when the control voltage is applied at input IN1 in direction of rotation 1 (no phase swapping), when controlling input IN2 in direction of rotation 2 (phase change from L1 and L3).

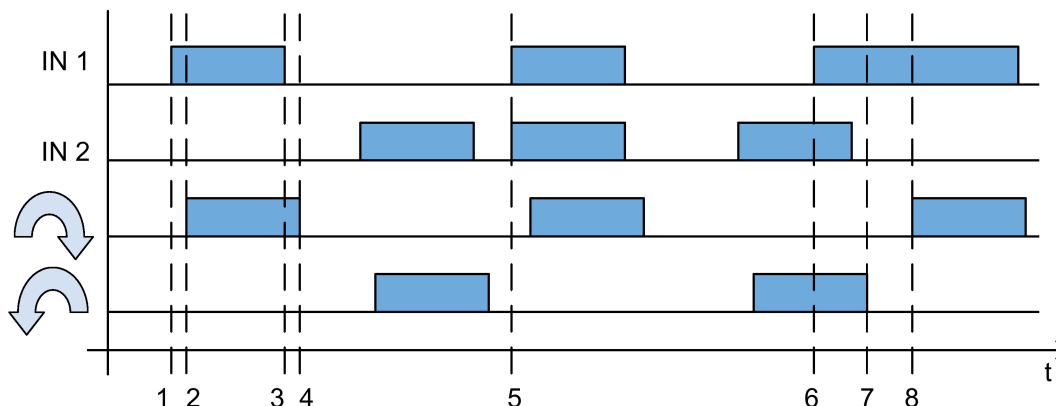
Interlocking of the control inputs

After the motor is switched off, the motor starter accepts a new start after a pause time.

If you apply control inputs IN1 and IN2 simultaneously, control input IN1 has priority. The motor starts up in this direction of rotation.

When the motor is running, activation of the second input has no effect.

The figure below shows an example of the effects of the signals at the control inputs on the direction of rotation of the motor:



- 1 Controller 1 active
- 2 Motor ON
- 1-2 Switch-on delay
- 3 Control input 1 inactive
- 4 Motor OFF
- 3-4 Switch-off delay
- 5 With simultaneous activation of the control inputs, the motor is started in the clockwise direction.
- 6 Additional activation of the second control input has no influence.
- 7-8 After motor OFF, the motor starter accepts a new start (motor ON) after a pause time.

Note

Please observe the information about the control supply voltage and the control inputs in the chapter "Device versions (Page 26)".

3.5.2 Protective functions

3.5.2.1 Overload protection

The 3RM1 motor starter protects three-phase motors against overload. If the current exceeds the set value, the 3RM1 motor starter switches off within the specified tripping time and signals the fault via the LEDs and the fault signaling output. You will find the tripping characteristic in the chapter "Overload protection/device protection characteristic (Page 124)".

Note

ATEX-certified motor overload protection in the case of 3RM11/3RM13 Failsafe

Overload protection of safety-related 3RM11 Failsafe and 3RM13 Failsafe motor starters is ATEX-certified up to SILCL 2; see Chapter "ATEX-certified motor overload protection in the case of 3RM11/3RM13 Failsafe (Page 36)."

3.5.2.2 Equipment protection

In addition to the motor protection function, the 3RM1 motor starters also protect themselves against overload. This means that in the case of 3RM1 motor starters with a rated operational current of 7 A, an overload trip may occur in the upper current range before the motor protection trips.

If, for example, 8-fold current flows in the case of an unloaded motor memory module and a set current of 7 A, tripping takes place after approximately one second.

You will find the tripping characteristic in the chapter "Overload protection/device protection characteristic (Page 124)".

3.5.2.3 Phase failure protection

The 3RM1 motor starter is provided with phase failure protection to prevent excessive temperature rise in the load in double phasing if phase failure occurs.

Depending on whether the phase failure already occurs before the ON command or during the ON state, the devices respond as described in the table and signal a phase failure.

	Standard motor starters 3RM10/3RM12	Failsafe motor starters 3RM11/3RM13
Phase failure before the ON command	The motor starter switches off after 5 s.	The motor starter does not switch on.
Phase failure during the ON state	The motor starter switches off after 5 s.	The motor starter switches off after 5 s.

The fault state is indicated by LEDs and the fault signaling output.

NOTICE

Damage due to asymmetrical current consumption by built-in brakes

When connecting and operating motors with built-in brakes, the energy for releasing the brakes is taken from the motor connection cables. This can result in substantial asymmetry in power consumption.

Observe the following measures for all 3RM1 motor starters:

- If the current for the braking facility is taken from two phases, connect the motor cables that are additionally loaded with braking current to terminals T1 and T3. Observe the additional braking current when setting the motor current.
- Alternatively, the braking facility can be fed externally.

A further braking facility is possible for 3RM10 and 3RM12 Standard motor starters:

- If the current for the braking facility is taken from just one phase and the neutral conductor, connect the motor cable that is loaded with the braking current to terminal T2 of the motor starter.

Do not use 3RM11 Failsafe and 3RM13 Failsafe motor starters for such braking facilities.

NOTICE

Damage from incorrect connection

Incorrect connection of the motor can result in damage.

Connect the motor as specified in the typical circuit diagrams.

You can find circuit diagrams for operating motors with brakes in the chapter "Typical circuits (Page 141)".

3.5.2.4 Asymmetry monitoring

Asymmetry detection will only work if all three phases are directly dependent upon one another, i.e. when no external current flow is possible.

Asymmetry is detected as soon as one of the three phases deviates by more than 40 % from the mean value of all phases. As soon as this happens, the 2-pole trip curve CLASS 10 A is applied (see Chapter "Overload protection/device protection characteristic (Page 124)"). This means: The function trips as soon as a minimum value of 85.5 % of the set rated current is reached. If the current is below 85.5 % of the set rated current in all three phases, the function will not trip. For values above 85.5 % of the set current, the higher the current, the faster the function will trip.

For exact values, see the curve in Chapter "Overload protection/device protection characteristic (Page 124)." Some values are provided as examples from this curve in the table.

Table 3- 2 Tripping times for tripping due to detected asymmetry

Maximum current in one of the 3 phases with reference to the set rated current [%]	Tripping time [s]
86	132
90	77
100	50
120	32

The asymmetry detection is not active for setting value 10 A for resistive loads; in this case the overload trip occurs according to the table Table 9-1 Tripping for resistive loads at a setting value of 10 A. (Page 126)

3.5.2.5 Thermal calculation method (motor memory module)

Principle of operation

The electronics continuously calculate a model of the thermal load on the motor dependent on the operating time and the current load. The motor memory module charges when the motor is switched on. The motor memory module discharges after the motor is switched off.

Following an overload tripping operation, the motor memory module is fully discharged after approximately three minutes. You must wait for this cooling time to elapse before you can acknowledge the fault. If the control supply voltage fails, the 3RM1 motor starter stores the remaining cooling time. When the control supply voltage is restored, the remaining cooling time elapses before the motor can be switched on again.

If you initiate a restart within a very short time after switching off the motor, it may be that the motor memory module has not yet fully discharged. This can result in an extremely fast overload trip after the restart. During continuous operation ("Warm" motor memory module), tripping times are reduced depending on the prior load.

If voltage is present at terminals A1/A2 and at control inputs IN1/IN2 after resetting the overload trip, the 3RM1 motor starter restarts.

The rated current of 10 A must not be set for motorized loads but only for resistive loads. In this setting, the tripping characteristics deviate from Class 10A; see Chapter "Overload protection/device protection characteristic (Page 124)".

You can find information about typical switching frequencies in Chapter "Number of starting operations (Page 123)".

 WARNING
Hazardous voltage Can result in death, serious physical injury, or damage to property. Health hazard from automatic hot restart If the DEVICE LED shows a yellow light and the FAILURE LED a red light, and a manual RESET or auto RESET takes place following an overload trip, the machine starts up immediately if there is a control command at IN1/IN2. Personnel in the danger area can be injured. Make sure that the danger area of the machine is kept clear of people.

Reset in manual operation

If the RESET method is set to manual RESET, acknowledge an overload trip as follows:

1. Wait for the cooling time to elapse.
2. Press the "TEST/RESET/RESET MODE" button to acknowledge.

As an alternative, in the case of the 3RM10/3RM12 Standard motor starters, you can use the RESET control input IN3 for acknowledgment.

Note

If you briefly switch the control supply voltage off and on again, there is no automatic acknowledgment.

3.5.2.6 ATEX-certified motor overload protection in the case of 3RM11/3RM13 Failsafe

Standards

Increased danger in hazardous areas means it is necessary to carefully observe the following standards:

- **EN 60079-14 / VDE 0165-1** Electrical apparatus for explosive gas atmospheres.
- **EN 60079-17** Inspection and maintenance of electrical installations in hazardous areas.
- **EN 50495** Safety devices required for the safe functioning of equipment with respect to explosion risks.

Certification of 3RM11/3RM13 Failsafe motor starters

The 3RM11 Failsafe and 3RM13 Failsafe motor starters are approved under Device Group II, Category (2) in the area "GD" (areas in which explosive gas, steam, mist, air mixtures and combustible dust are present):

BVS 12 ATEX F 002 X 
 II (2) G [Ex e] [Ex d] [Ex px]^{*)}
 II (2) D [Ex t] [Ex p]
 I (M2) [Ex d]

^{*)} For the px applications, pressure and flow rate monitoring are additionally required for the over-pressure encapsulation systems.

WARNING

Hazardous voltage

Failure to follow proper procedures may result in death, serious physical injury, or damage to property.

All work involved in connecting, commissioning and maintenance must be carried out by qualified personnel.

Control cabinet degree of protection in hazardous areas

WARNING

Hazardous voltage

Failure to follow proper procedures may result in death, serious physical injury, or damage to property.

The 3RM1 motor starter is not suitable for installation in hazardous areas.

The device may only be installed in a control cabinet with the minimum degree of protection of IP 4x.

In addition to the motor protection function, the 3RM1 motor starters also protect themselves against overload. This means that in the case of 3RM1 motor starters with a rated operational current of 7 A, an overload trip may occur in the upper current range before the motor protection trips.

If, for example, 8-fold current flows in the case of an unloaded motor memory module and a set current of 7 A, tripping takes place after approximately one second.

You will find the tripping characteristic in the chapter "Overload protection/device protection characteristic (Page 124)".

Setting the rated motor operational current

Set the 3RM1 motor starter to the rated motor operational current (according to the type plate or design test certificate of the motor).

On this topic, refer to the Chapter "Setting the rated operational current (Page 109)".

Note

Observe the trip class or the tripping characteristic of the 3RM1 motor starter in Chapter "Overload protection/device protection characteristic (Page 124)". The motor and cables must be designed for the selected trip class.

RESET settings

Note

Restriction in the hazardous area

In applications for the protection of motors in hazardous areas, a 3RM11/3RM13 Failsafe motor starter may only be operated with the "Manual RESET" setting. The "RESET Mode" LED must be off.

Short-circuit protection

Short-circuit protection must be handled by separately arranged overcurrent protection devices.

On this topic, refer to the Chapter "Load feeders - protection against short circuit (Page 62)".

Cable protection

Avoid impermissibly high cable surface temperatures by correctly dimensioning the cross sections. Select an adequate cable cross-section.

Cyclic test of the protection functions

The tests are run by means of the "TEST/RESET/RESET MODE" button and can take place when the motor is either on or off. The test consists of three function tests.

Carry out the tests during commissioning and cyclically every 36 months at the latest.

You will find out how to run the test in Chapter "Carrying out the function test (Page 43)".

Ambient conditions

See the chapter "Ambient conditions (Page 60)".

ATEX-specific safety data

The motor protection function according to ATEX in the case of 3RM11/3RM13 motor starters has the following safety data:

Safety integrity level (SILCL)	2
Hardware fault tolerance (HFT)	0
Interval for testing the protective functions	3 years
Probability of failure on demand (PFD)	5×10^{-4}
Probability of a dangerous failure per hour (PFH)	5×10^{-8} 1/h

Note

See the data sheet for other specifications. You will find the data sheet in Service&Support (<http://www.siemens.com/sirius/support>).

Maintenance and repairs

These devices are maintenance-free.

Note

Repairs to the device may only be carried out by the manufacturer.

Warranty

NOTICE
To meet the conditions of the warranty, you must observe the safety and commissioning instructions.

Further information

You will find further information on the Internet at

- Internet (<http://www.siemens.com/sirius>)
- Information and Download Center (<http://www.siemens.com/sirius/infomaterial>)
- Product Support (<http://www.siemens.com/sirius/support>)
- Service and Support (<http://www.siemens.com/sirius/technical-assistance>)
- Certificates (<http://www.siemens.com/sirius/approvals>)

3.5.3 Shutdown on malfunction

Shutdown via integrated monitoring

The 3RM1 motor starter has integrated monitoring of the main circuit components. During operation, the 3RM1 motor starter detects when a switching element fails to switch, or when a phase has failed.

Monitoring before switching on in the case of 3RM11/3RM13 Failsafe motor starters

In the case of the 3RM11 Failsafe and 3RM13 Failsafe motor starters, monitoring prevents the load from switching on in the case of the following malfunctions:

- Damage to the power semiconductors
- Damage to the relay contacts
- Faults in the control electronics

Acknowledging

A malfunction is acknowledged by switching the control supply voltage back on and then pressing the "TEST/RESET/RESET MODE" pushbutton.



DANGER

Checking the safety function

Will result in death, serious physical injury, or damage to property.

Check the safety function after acknowledging a disruption in the case of 3RM11/3RM13 Failsafe motor starters.

Note

If a malfunction is repeatedly indicated, the 3RM1 motor starter is defective and must be replaced.

Safety-related shutdown in the case of 3RM11/3RM13 Failsafe motor starters

Safety-related shutdown is achieved for the 3RM11 Failsafe and 3RM13 Failsafe motor starters thanks to the special arrangement and checking of the switching elements. Safety-related shutdown takes place when the control supply voltage is deactivated or by switching off the control inputs. Malfunctions result in automatic shutdown. You will find further information in Chapter "Safety-related shutdown with 3RM11/3RM13 Failsafe motor starters (Page 41)."

Display concept

The following events indicate the "Malfunction" status:

- DEVICE LED flashes red
- FAILURE LED shows a red light
- Fault signaling output active

3.5.4 Safety-related shutdown with 3RM11/3RM13 Failsafe motor starters

General description

In the case of the 3RM11/3RM13 Failsafe motor starters, safety-related shutdown takes place by removing the control voltage (A1/A2). As an alternative in the case of units with a 24 V DC supply voltage, safety-related shutdown can be performed by canceling the ON command (IN1/IN2).

SILCL 3 to EN 62061, PL e/Cat. 4 to EN ISO 13849-1 can be achieved with both shutdown methods. To achieve SILCL 3/PL e/Cat. 4 for the safety function, all components of the safety function (detecting/evaluating/reacting) must be designed accordingly.

In the case of 3RM11/3RM13 Failsafe motor starters, the OFF state is defined as the safe state. 3RM11/3RM13 Failsafe motor starters are self-monitoring in compliance with SILCL 3/PL e and therefore do not need to be monitored in the feedback circuit of the upstream evaluation unit/control.

 WARNING
Failure of the safety function
Can result in death, serious physical injury, or damage to property.
Hybrid operation of 3RM10/3RM12 Standard motor starters with 3RM11/3RM13 Failsafe motor starters in safety-related applications is not admissible.
Only ever use safety-related motor starters (3RM11 Failsafe and 3RM13 Failsafe) in safety-related applications.

Note

Fault signaling output (95-96-98) behavior in the event of safety-related shutdown

The fault signaling output (95-96-98) resumes its original state in the event of safety-related shutdown via the control supply voltage. If the fault is still present after reconnection of the control supply voltage, the fault signaling unit becomes active again. If an overload was detected before shutdown, the remaining cooling time is saved and elapses after reconnection of the control supply voltage.

3.5.4.1 Safety-related shutdown via the control supply voltage

Direct link to the 3SK safety relay

To ensure safety-related shutdown via the control supply voltage, you can directly connect up to 5 motor starters to the 3SK safety relay via the device connector. In this case, the 24 V DC control supply voltage is supplied via the device connector. In a safety case, the 3RM11/3RM13 Failsafe motor starters will be shut down by removal of the control supply voltage by the 3SK safety relay. No additional wiring is required between the 3SK safety relay and the 3RM1 motor starter. Nothing may be connected to terminals A1 and A2 of the 3RM1 motor starters.

**WARNING****Bypassing the safety function when using device connectors**

Can result in death, serious physical injury, or damage to property.

When operating with a 3SK safety relay and a device connector, the supply voltage for 3RM1 motor starters is established via the device connectors.

In this case, do not connect anything to terminals A1 and A2 of the 3RM1 motor starters, in order to prevent bypassing of the safety function.

Note

The device connector can only be used in combination with motor starters that have a 24 V DC supply voltage.

Wire connection to a safe output

Shutdown of the 3RM11 / 3RM13 Failsafe motor starter by removal of the control supply voltage can also be performed by wiring to a safe output of a fail-safe control / evaluation unit. This method can be used to shut down a single motor starter or a group of up to 5 motor starters safely. For safe group shutdown, the 3RM11 / 3RM13 Failsafe motor starters are connected to each other via the device connector. In this case, the control supply voltage is wired from the safe output to the first motor starter in the group.

If it can be ensured that the cables are installed in such way that they are protected against a cross-circuit and short-circuit to a power cable, safety-related shutdown of A1 (I+) via a safety-related output of the control will suffice. If this cannot be ensured, or if a PM-switching output is used, both A1 (L+) and A2 (M) must be shut down via a safety-related output. In both cases, safety-related shutdown in compliance with SILCL 3 per EN 62061, PL e / Cat. 4 per EN ISO 13849-1 is achieved.

Safety-related shutdown via inputs IN1 and IN2

As an alternative in the case of 3RM11 / 3RM13 Failsafe motor starters with a 24 V DC supply voltage, safety-related shutdown via the IN1 and IN2 inputs is possible. The safety-related output of the fail-safe control / evaluation unit performs operational switching.

Depending on peripherals used, shutdown is in one or two channels:

- PM (PPM)-switching: shutdown is in two channels.
- PP-switching: shutdown is in one channel.

Single-channel switching achieves SILCL 3 per EN 62061, PL e/Cat. 4 per EN ISO 13849-1 if steps are taken to ensure that the cables are installed in such a way that they are protected against a cross-circuit/line-to-line fault.

3.5.5 Carrying out the function test

You can start the different individual tests for the function test with the "TEST/RESET/RESET MODE" button.

Procedure

Depending on how long the "TEST/RESET/RESET MODE" button is pushed, a specific test is started:

Actuation time for starting the test	Explanation
< 2 s	LED-Test All four LEDs are activated. ¹⁾
2 s ... 5 s	Current measuring test Current measuring is tested in all 3 phases.
> 5 s	Motor protection test disconnection The entire control chain is tested, including the switching elements and the fault signaling output. If the device responds as standard, an overload trip takes place. This can be reset immediately without a cooling time by pressing the "TEST/RESET/RESET MODE" pushbutton again.

¹⁾ Exception in the case of 3RM11/3RM13 Failsafe motor starters: in the case of 3RM11/3RM13 Failsafe motor starters, the LED test is only carried out in the OFF state (without load current). In the ON state (with load current), the LEDs stay unchanged.

3.5 Functions

Actuation time for starting the test	Status	OK	Fault
LED test			
< 2 s	"DEVICE" LED	Shows a yellow light ¹⁾	Off
	"FAILURE" LED	Shows a red light ¹⁾	Off
	"STATE" LED	Shows a green light ¹⁾	Off
	"RESET MODE" button	Shows a green light ¹⁾	Off
Current measuring test			
2 s ... 5 s	"DEVICE" LED	- Without load current: Flickering red - With load current: Flashing red	Off
Motor protection trip test			
> 5 s	"DEVICE" LED	Shows a yellow light	Flashing red
	"FAILURE" LED	Shows a red light	Shows a red light
	Fault signaling output	Active	Not active
	Switching elements/motor	Deactivated	Unchanged

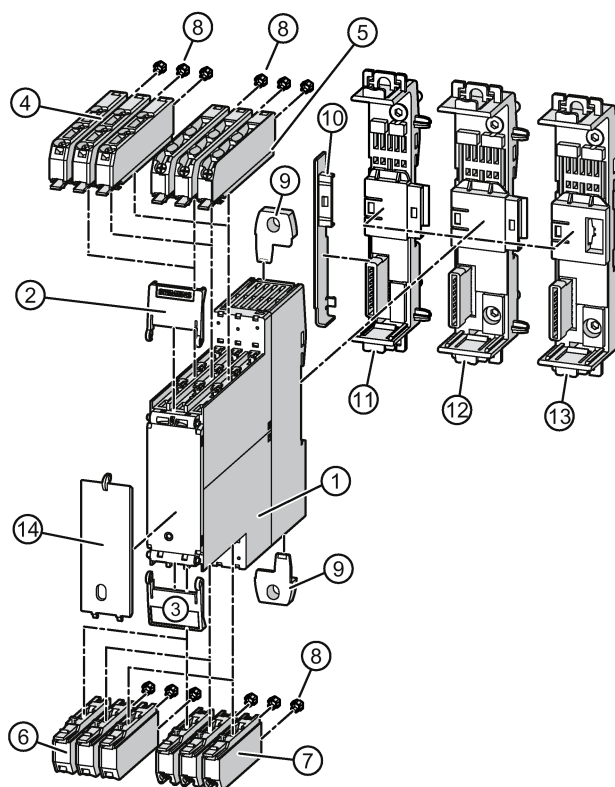
- ¹⁾ Exception in the case of 3RM11/3RM13 Failsafe motor starters: in the case of 3RM11/3RM13 Failsafe motor starters, the LED test is only carried out in the OFF state (without load current). In the ON state (with load current), the LEDs stay unchanged.

Note

If the 3RM1 motor starter does not respond as in the table above during the function test, check the connections and the supply to the 3RM1 motor starter. If the fault persists after correct disconnection, the device must be replaced.

3.6 Accessories and article number scheme

3.6.1 Overview of all device components



- ① Basic unit/expansion unit, e.g. 3RM1 motor starter
- ② Top cover flap
- ③ Bottom cover flap
- ④ Terminals, 3-pole, push-in, 1 x 2.5 mm²
- ⑤ Terminals, 3-pole, screw-type, 1 x 2.5 mm²
- ⑥ Terminals, 3-pole, push-in, 1 x 4 mm²
- ⑦ Terminals, 3-pole, screw-type, 1 x 4 mm²

For connection, there is an infeed system (not shown in the overview diagram) for these terminals.

- ⑧ Coding pins
- ⑨ Push-in lugs for wall mounting
- ⑩ Cover
- ⑪ Device connector
- ⑫ Device connector
- ⑬ Device termination connector
- ⑭ Sealable cover

3.6.2 Accessories

The following accessories are offered for the 3RM1 motor starter:

- Infeed system (Page 46)
- Sealable cover (Page 48)
- Wall mounting (Page 48)
- Device connectors (Page 49)
- Terminals (Page 52)
- Device identification label (Page 52)

3.6.2.1 Infeed system

The infeed system comprises 3-phase busbars, one infeed terminal, and one protective cap for covering unused connections. The infeed system is suitable for 3RM1 motor starters with screw connections.

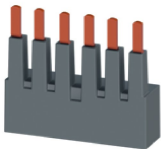
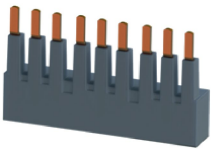
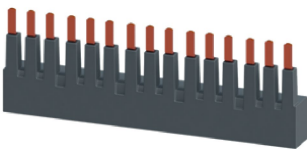
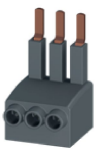
Time-saving wiring between 3RM1 motor starters with screw terminals to the supply connection is possible thanks to the infeed system with 3-phase busbars.

These busbars are available in three lengths, thus allowing 2, 3 or 5 motor starters (arranged side-by-side) to be connected at the same time. More than 5 devices can be connected by clamping the connection tags of an additional busbar rotated by 180° (e.g. 6 devices using one 5-pole busbar and one 2-pole busbar).

A single motor starter can be removed from the assembly without loosening the terminal screws of neighboring motor starters.

The maximum summation current must not exceed 25 A (UL: 16 A). Primary infeed is connected via a three-phase infeed terminal. The three-phase busbars are finger-safe but empty connection tags must be fitted with covers.

Table 3- 3 Infeed system article numbers

	Description	Article number
	3-phase busbar for two 3RM1 motor starters	3RM1910-1AA
	3-phase busbar for three 3RM1 motor starters	3RM1910-1BA
	3-phase busbar for five 3RM1 motor starters	3RM1910-1DA
	3-phase infeed terminal	3RM1920-1AA
	Cover caps for terminal lugs of the 3-phase busbars	3RM1910-6AA

3.6.2.2 Sealable cover

The sealable cover prevents unauthorized access to the rotary coding switch.

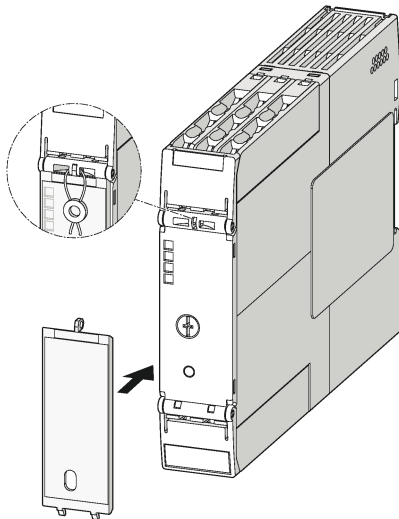


Table 3- 4 Article number of the sealable cover

Description	Article number
SIRIUS sealable cover 22.5 mm, cutout for pushbuttons at bottom	3ZY1321-2AA00

3.6.2.3 Wall mounting

The push-in lugs for wall mounting of the 3RM1 motor starter are inserted at the top and bottom of the motor starter enclosure. This method is used to screw the motor starter onto level mounting surfaces.

The wall fixture is inserted into the enclosure at the top and bottom for screw-mounting the 3RM1 motor starter to level mounting surfaces.

Table 3- 5 Article numbers of the push-in lugs for wall mounting

Description	Article number
SIRIUS push-in lugs for wall mounting	3ZY1311-0AA00

3.6.2.4 Device connectors

Function

The device connectors can be used to loop through a common control supply voltage of 24 V DC to reduce wiring complexity between the individual motor starters (3RM1...-AA04). Moreover, Failsafe motor starters with a 24 V DC control supply voltage (3RM11...-AA04 / 3RM13...-AA04) can be interfaced to one 3SK safety relay as a group via device connectors. The control signals are connected via the device terminals on the motor starter.

The device connectors can either be snapped onto a standard mounting rail or can be screwed onto a flat mounting wall.

If the motor starters are not to be interconnected side by side, device connectors for looping through signals must be used for the gaps, e.g. for thermal reasons (see "Derating (Page 55)").

Table 3- 6 Article numbers for device connectors for 3RM1 motor starters

	Description	Article number
	Device connector for 3RM1 motor starters, 22.5 mm wide	3ZY1212-2EA00
	Device connector for looping through signals, 22.5 mm wide	3ZY1212-2AB00
	Device termination connector for 3RM1 motor starters, 22.5 mm wide including left cover for the first device connector in the system configuration	3ZY1212-2FA00

Use of device connectors exclusively for the 24 V DC control supply voltage

By using device connectors, several 3RM1...-AA04 motor starters can be jointly supplied with a control supply voltage of 24 V DC. This requires the control supply voltage to be applied to the A1 and A2 terminals of only one motor starter.

Up to ten motor starters can be connected to one 24 V DC control supply voltage with device connectors. The 24 V DC control supply voltage must be within the operating range of 0.9 to 1.1 for this purpose. If the full operating range of 0.8 to 1.25 is to be used, no more than five 3RM1...-AA04 motor starters can be used. Otherwise, the infeed must be repeated for every 5th 3RM1 motor starter.

The last motor starter in a row can be placed on a device termination connector.

Using device connectors with 3SK safety relays

Multiple 3RM11/3RM13 Failsafe motor starters can be connected in a group for joint safety-related shutdown by a 3SK safety relay. Safety-related applications up to SILCL 3 to EN 62061, PL e/Cat. 4 to EN ISO 13849-1 can be realized.

**WARNING****Failure of the safety function during hybrid operation of Failsafe and Standard motor starters**

Can result in death, serious physical injury, or damage to property.

Hybrid operation of 3RM10/3RM12 Standard motor starters with 3RM11/3RM13 Failsafe motor starters in safety-related applications is not admissible.

Only ever use safety-related motor starters (3RM11 Failsafe and 3RM13 Failsafe) in safety-related applications.

3RM1 motor starters are shut down by interrupting the control supply voltage by means of the 3SK safety relay. For this reason, the control supply voltage must not also be applied to the 3RM1 motor starters.

**WARNING****Bypassing the safety function when using device connectors**

Can result in death, serious physical injury, or damage to property.

When the product is operated with a 3SK safety relay and a device connector, the supply voltage for 3RM1 motor starters is fed in via the device connectors.

In this case, do not connect anything to terminals A1 and A2 of the 3RM1 motor starters, in order to prevent bypassing of the safety function.

With device connectors, up to five Failsafe motor starters can be operated on one 3SK safety relay. For configuration using 3SK safety relays, also follow the configuration rules in the manuals for the "3SK Safety Relays"; see Chapter "Further documentation (Page 11)."

The last motor starter in a row must be placed on a device termination connector. This closes the circuits that were built up with the connectors.

Overview of all device connectors for 3RM1 motor starters and 3SK safety relays

Article number	Type of device connector	Units
3ZY1212-2EA00	Device connector for 3RM1 motor starters, 22.5 mm wide	3RM1 motor starter
3ZY1212-2FA00	Device termination connector for 3RM1 motor starters, 22.5 mm wide	3RM1 motor starter
3ZY1212-2AB00	Device connector for looping through signals, 22.5 mm wide	3RM1 motor starter
3ZY1212-1BA00	Device connector for 3SK1 safety relay, 17.5 mm wide	3SK1120 safety relay 3SK1220 input expansion
3ZY1212-2BA00	Device connector for 3SK1 safety relay, 22.5 mm wide	3SK1121 safety relay 3SK1122 safety relay 3SK1211 output expansion 3SK1230 power supply
3ZY1212-2DA00	Device termination connector for 3SK1 safety relay, 22.5 mm wide	3SK1121 safety relay 3SK1122 safety relay 3SK1211 output expansion
3ZY1212-0FA01	Device termination connector set for 3SK1 safety relay, > 45 mm wide	3SK1213 output expansion
3ZY1212-2GA00	Device connectors for 3SK2 safety relay, width 22.5 cm	3SK2112 safety relay
3ZY1212-2AB00	Device connectors for 3SK2 safety relay for looping through signals, width 22.5 cm	3SK21 safety relay
3ZY1212-4GA01	Device connector set for 3SK2 safety relay, width 45 cm	3SK2122 safety relay

Reference

You will find further information on using device connectors in:

- "FAQs on the 3RM1 motor starter
(<https://support.industry.siemens.com/cs/ww/en/view/81525494>)"
- Manual "SIRIUS 3SK1 safety relays
(<https://support.industry.siemens.com/cs/ww/de/view/67585885>)"
- Manual "SIRIUS 3SK2 Safety Relays
(<https://support.industry.siemens.com/cs/ww/de/view/109444336>)"

3.6.2.5 Terminals

Each 3RM1 motor starter has three main current terminals and three control current terminals. The terminals can be replaced with ease.

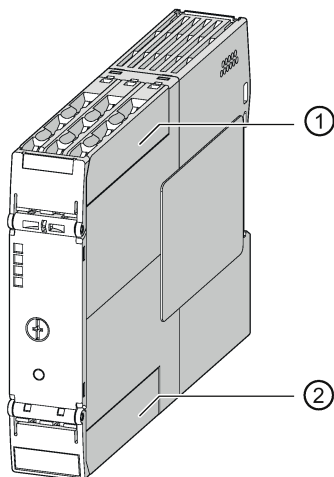


Table 3- 7 Article numbers for the terminals

Description			Article number
Control current terminals	①	Detachable terminal, 3-pole, screw terminals up to 1x2.5 mm ²	3ZY1131-1BA00
		Detachable 3-pole terminal, push-in terminals up to 2x1.5 mm ²	3ZY1131-2BA00
Main current terminals	②	Detachable 2-pole terminal, screw terminals up to 1x4 mm ²	3ZY1122-1BA00
		Detachable 2-pole terminal, push-in terminals up to 1x4 mm ² or 2x1.5 mm ²	3ZY1122-2BA00

3.6.2.6 Device identification label

Table 3- 8 Article numbers for the device identification labels

Description	Article number
SIRIUS device identification label, titanium gray, 17 mats each with 20 labels for snapping into place, size 20 x 7 mm, in box	3RT2900-1SB20
SIRIUS device identification label, titanium gray, 10 sheets each with 306 adhesive labels, size 19 x 6 mm, in transparent bag	3RT2900-1SB60

3.6.3 Article number scheme for SIRIUS 3RM1 motor starter

Digit of the Article No.	1. - 3. □□□	4. □	5. □	6. □	7. □	-	8. □	9. □	10. □	11. □	12. □
SIRIUS 3RM1 motor starter	3RM										
Generation (1)		☐									
Function (Direct-on-line starter = 0, direct-on-line starter Failsafe & ATEX certification = 1; reversing starter = 2; reversing starter Failsafe & ATEX certification = 3; accessories = 9)			☐								
Setting range rated motor current (0.1 ... 0.5 A = 01; 0.4 ... 2.0 A = 02; 1.6 ... 7.0 A = 07)				☐	☐						
Connection type (screw terminal = 1, push-in spring-loaded terminal = 2, hybrid connection method*) = 3)							☐				
Reserved (A)								☐			
Width (22.5 mm = A)									☐		
Rated control supply voltage (24 V DC = 0; 110 ... 230 V AC / 110 V DC = 1)										☐	
Rated operating voltage (48 ... 500 V = 4)											☐
Example	3RM	1	0	0	1	-	1	A	A	0	4

*) Hybrid connection method: control circuit realized as push-in spring-loaded terminal and main circuit as screw terminal

Configuration

4.1 Rated operational current and derating

Values for rated operational current I_e

Depending on the current setting range, the following rated operational currents apply for the 3RM1 motor starters:

Motor starter version	3RM1.01-.....	3RM1.02-.....	3RM1.07-.....
Rated operational current I_{eAC53a} [A]	0.5	2	7
Setting range for I_e [A]	0.1 ... 0.5	0.4 ... 2	1.6 ... 7 (10)
Minimum current per phase of the motorized load [A]	0.05	0.14	0.6

Note

The devices with $I_{eAC53a} = 7$ A have in addition to this setting value an I_{eAC51} setting value for 10 A in the highest setting level. This setting is intended for resistive loads to 10 A and is not suitable for operating motors. With this setting, there is no overload trip in accordance with Class 10A. Instead, this takes place in accordance with the description in the chapter "Overload protection/device protection characteristic (Page 124)".

As soon as the minimum current limit is violated, fault detection (phase failure) picks up.

When using the 3RM11 Failsafe and 3RM13 Failsafe motor starters, note that no neutral conductor may be connected if the load circuit contains 3-phase loads, e.g. a 230 V brake or resistive loads.

Motors with a high efficiency and high motor starting currents

High starting currents may have to be taken into consideration when using 3RM1 motor starters on motors with a high efficiency (IE3 or IE4). According to the product standard IEC 60947-4-2, the 3RM1 motor starters are designed for motors with a maximum 8-fold starting current.

If motors are operated that have a higher starting current, refer to the following table for the maximum adjustable motor current:

Motor starter version I_{eAC53a} [A] at 40°C max. motor starting current	3RM1.01-.....	3RM1.02-.....	3RM1.07-.....
≤ 8 -fold I_e	0.5	2	7
8.5-fold I_e	0.47	1.9	6.6
9-fold I_e	0.45	1.8	6.2
9.5-fold I_e	0.42	1.7	5.9
10-fold I_e	0.4	1.6	5.6

The device protection may respond sooner in the case of motors with a higher starting current. You will find details in "Figure 9-2 Device protection characteristic (Page 125)".

Dependence of the rated operational current on the assembly type, the design version and the ambient temperature

The permissible rated operational current of the 3RM1 motor starter reduces depending on the assembly type, the design version, and the ambient temperature T_A as follows:

Motor starter version 3RM1.01-..... to 0.5 A and 3RM1.02-..... to 2 A:

For the versions 0.5 A and 2 A, the following ambient temperature limits apply depending on the assembly type:

Assembly type	Max. ambient temperature for 0.5 A and 2 A device for AC 53 a	
	Side-by-side	Individually/with 22.5 mm clearance
Vertically on horizontal rail	50 °C	60 °C
Horizontally on vertical rail	45 °C	60 °C
Upright on horizontal rail/mounting surface	45 °C	60 °C

Motor starter version 3RM1.07-..... to 7 A:

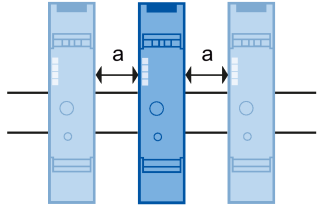
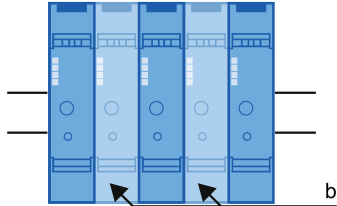
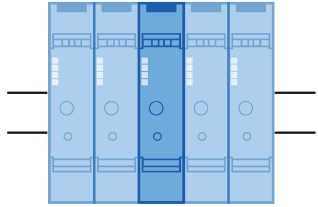
For the 7 A version, the following derating values apply depending on the assembly type:

Assembly type	Ambient temperature	Max. setting values for 7 A device in A for AC 53 a		Side-by-side
		Individually/with 22.5 mm clearance	Alternately with 2 A device	
Vertically on horizontal rail	40 °C	7	4.6	4
	45 °C	6.1	4	3.6
	50 °C	6.1	3.2 / 2.8 ¹⁾	3.2 / 2.4 ¹⁾
	55 °C	5.2	0	0
	60 °C	4.6	0	0
Horizontally on vertical rail	40 °C	5.2	4	3.2
	45 °C	4.6	2	2
	50 °C	4	0	0
	55 °C	3.2	0	0
	60 °C	2.0	0	0
Upright on horizontal rail/mounting surface	40 °C	6.1	4	3.6
	45 °C	5.2	3.6	3.2
	50 °C	5.2	0	0
	55 °C	4	0	0
	60 °C	3.6	0	0

¹⁾ When using device connectors

Design versions for 7 A devices

For the 7 A device without clearance, a distinction is made between the following design versions:

Design version	
Individually with clearance $a = 22.5 \text{ mm}$ Can be implemented with signal loop-through connectors when using device connectors. Note: When using the 3-phase busbar systems, the necessary clearances can also be implemented by leaving the connections free. Cover the free connections with cover caps (accessories) to protect the free terminal lugs against touch.	
7 A device alternately with 2 A device or 0.5 A device (b) without clearance	
Devices with the same rated operational current without clearance: Side-by-side	

Note

Please note the assembly examples in Chapter "Typical circuits (Page 141)".

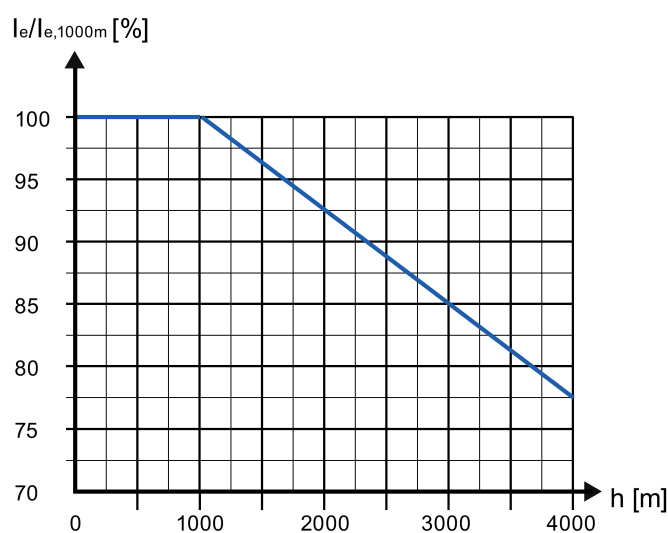
Dependence of the rated operational current on the installation altitude

The permissible rated operational current of the 3RM1 motor starter is reduced as shown in the diagram depending on the installation altitude:

Note

Maximum installation altitude of 3RM11/3RM13 Failsafe motor starters: 2000 m

Installation altitudes of more than 2000 m are not admissible for 3RM11/3RM13 Failsafe motor starters.



From an installation altitude in excess of 2000 m, in the case of Standard 3RM10/3RM12 motor starters the maximum permissible operational voltage must be reduced due to the limited insulation strength. You can obtain details on request from SIEMENS Technical Assistance (<http://www.siemens.com/sirius/technical-assistance>).

4.2 Ambient conditions

4.2.1 Application environment

The 3RM1 motor starters are designed for use in enclosed spaces in which no severe operating conditions prevail (e.g. dust, caustic vapors, hazardous gases).

If they are to be installed in dusty or damp rooms, suitable control cabinets must be provided.

Note

If you operate the 3RM1 motor starter in a control cabinet, ensure that the cabinet is adequately ventilated.

4.2.2 Ambient temperature

The 3RM1 motor starter has been designed for use under ambient temperatures ranging from -25 °C to +40 °C, singly in a vertical mounting position, and without a reduction in the permissible rated operational current. With the deratings given in the section "Rated operational current and derating (Page 55)", you can operate the 3RM1 motor starters at ambient temperatures of up to 60 °C.

4.2.3 Mounting position

Installation on a vertical mounting surface is recommended for the 3RM1 motor starter. Please observe the power reduction levels/deratings for other mounting positions given in the section "Rated operational current and derating (Page 55)".

For information on clearances to neighboring components, please refer to the chapter "Technical data (Page 121)".

4.2.4 Grounding measures

No grounding measures are required for the 3RM1 motor starter.

4.2.5 Permissible operating voltage

To be able to ensure safe isolation of the circuits from each other in accordance with IEC 60947-1, the following operating voltages are permissible:

Table 4- 1 Safe isolation of the 3RM1 motor starters

... to ...	Control circuit (A1/A2, OUT, IN1-3, M1/M2)	Signaling contacts (98/95/96)	Main circuit (1L1/3L2/5L3 2T1/4T2/6T3)
Control circuit (A1/A2, OUT, IN1-3, M1/M2)	-	250 V	500 V
Signaling contacts (98/95/96)	250 V	-	500 V
Main circuit (1L1/3L2/5L3 2T1/4T2/6T3)	500 V	500 V	-

4.2.6 Minimum load current

The minimum load current is 20% of the set motor current, but at least the absolute minimum current specified in the tables below.

The minimum load current differs in the case of Standard 3RM10/3RM12 and Failsafe 3RM11/3RM13 motor starters:

Table 4- 2 Minimum load current in the case of Standard 3RM10/3RM12 motor starters

Motor current	0.5 A	2 A	7 A
Motorized load	0.05 A	0.14 A	0.6 A
Resistive load	0.05 A	0.14 A	0.6 A

Table 4- 3 Minimum load current in the case of Failsafe 3RM11/3RM13 motor starters

Motor current	0.5 A	2 A	7 A
Motorized load	0.05 A	0.14 A	0.6 A
Resistive load	0.25 A	1 A	3.5 A

As soon as the minimum current limit is violated, fault detection (phase failure) picks up.

4.3 Load feeders - protection against short circuit

4.3.1 Protection against short circuit

The short-circuit strength in combination with fuses is up to 100 kA at 400 V and with SIRIUS 3RV2 motor starter protectors up to 55 kA at 400 V. Type of coordination 1 is maintained here.



DANGER

Hazardous voltage on the motor

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

After every short circuit in the case of short-circuit protection according to **type of coordination 1**, a 3RM1 motor starter is defective.

Replace a 3RM1 motor starter including the terminals after a short circuit in the case of short-circuit protection according to type of coordination 1.

With 3RM11/3RM13 Failsafe motor starters, you must check the safety function in addition to replacing the unit.

With 3RM11/3RM13 Failsafe motor starters, also check the safety function after a short circuit in the case of short-circuit protection according to **type of coordination 2**.

4.3.2 Configuration of load feeders

The 3RM1 motor starters can be combined with any short-circuit protection devices. Using SIRIUS 3RV20 motor starter protectors offers a convenient solution for the configuration of fuseless load feeders. However, combinations with fuses are also possible, mastering particularly high short-circuit currents. Type of coordination 2 (see Glossary (Page 163)) can also be achieved.

4.3.3 Configuration of a load feeder with 3RM1 motor starter in compliance with IEC

Short-circuit protection for load feeders in compliance with IEC can also be realized by motor starter protectors or fuses. Depending on the design of the protection device, the two types of coordination 1 or 2 can be achieved. The protection devices listed are the maximum permissible versions and, as an alternative, protection devices of the same type with a lower rated current can be used.

4.3.3.1 Fuseless design

Combining 3RM1 motor starters with motor starter protectors

! DANGER**Hazardous voltage on the motor**

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

After every short circuit in the case of short-circuit protection according to **type of coordination 1**, a 3RM1 motor starter is defective.

Replace a 3RM1 motor starter including the terminals after a short circuit in the case of short-circuit protection according to type of coordination 1.

With 3RM11/3RM13 Failsafe motor starters, you must check the safety function in addition to replacing the unit.

With 3RM11/3RM13 Failsafe motor starters, also check the safety function after a short circuit in the case of short-circuit protection according to **type of coordination 2**.

The 3RM1 motor starters can be combined with the following motor starter protectors for short circuits according to type of coordination 1:

Table 4- 4 Motor starter protectors according to type of coordination 1 for 3RM1 motor starters with 400 V

Motor starter $T_A = 40\text{ °C}$		Short-circuit current	Maximum permitted motor starter protectors for 400 V + 10%	
Article number	Rated current [A]	Current [kA]	Article number	Rated current [A]
3RM1...	0.5 / 2.0 / 7.0	55	3RV2011-1HA..	8
3RM1...	0.5 / 2.0 / 7.0	12.5	3RV2011-1KA..	12.5
3RM1...	0.5 / 2.0 / 7.0	10	3RV2021-4AA..	16
3RM1...	0.5 / 2.0 / 7.0	7.5	3RV2021-4BA..	20
3RM1...	0.5 / 2.0 / 7.0	5	3RV2021-4DA..	25

Table 4- 5 Motor starter protectors according to type of coordination 1 for 3RM1 motor starters with 500 V

Motor starter $T_A = 40\text{ °C}$		Short-circuit current	Maximum permitted motor starter protectors for 500 V + 10%	
Article number	Rated current [A]	Current [kA]	Article number	Rated current [A]
3RM1.01...	0.5	10	3RV2011-1HA..	8
3RM1.02...	2.0	10	3RV2011-1HA..	8
3RM1.07...	7.0	10	3RV2011-1HA..	8

Combining 3RM1 motor starters with miniature circuit breakers

**DANGER****Hazardous voltage on the motor**

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

After every short circuit in the case of short-circuit protection according to **type of coordination 1**, a 3RM1 motor starter is defective.

Replace a 3RM1 motor starter including the terminals after a short circuit in the case of short-circuit protection according to type of coordination 1.

With 3RM11/3RM13 Failsafe motor starters, you must check the safety function in addition to replacing the unit.

With 3RM11/3RM13 Failsafe motor starters, also check the safety function after a short circuit in the case of short-circuit protection according to **type of coordination 2**.

The 3RM1 motor starters can be combined with the following miniature circuit breakers for short circuits according to type of coordination 1:

Table 4- 6 Miniature circuit breakers according to type of coordination 1 for 3RM1 motor starters with 400 V

Motor starter T _A = 40 °C		Short-circuit current	Maximum permitted miniature circuit breakers with 400 V + 10%	
Article number	Rated current [A]	Current [kA]	Article number	Rated current [A]
3RM1.01...	0.5	6	5SY4325-6	25
3RM1.02...	2.0	6	5SY4325-6	25
3RM1.07...	7.0	6	5SY4325-6	25

4.3.3.2 Fused design

On tested assemblies, the type of coordination defines the permissible condition of devices following a short-circuit.

Type of coordination 1 means that the load feeder may be non-operational after a short circuit has been cleared.

Type of coordination 2 means that the 3RM1 motor starter is still operational following a short-circuit.

With the 3RM1 motor starter, short-circuit protection is only provided by fuses up to 400 V (+10%) for type of coordination 2.

Combining 3RM1 motor starters with fuses for type of coordination 1

! DANGER

Hazardous voltage on the motor

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

After every short circuit in the case of short-circuit protection according to **type of coordination 1**, a 3RM1 motor starter is defective.

Replace a 3RM1 motor starter including the terminals after a short circuit in the case of short-circuit protection according to type of coordination 1.

With 3RM11/3RM13 Failsafe motor starters, you must check the safety function in addition to replacing the unit.

With 3RM11/3RM13 Failsafe motor starters, also check the safety function after a short circuit in the case of short-circuit protection according to **type of coordination 2**.

The 3RM1 motor starters can be combined with the following fuses for short circuits of 55 kA according to type of coordination 1:

Table 4- 7 Fuses according to type of coordination 1 for 3RM1 motor starters

Motor starter T _A = 40 °C		Maximum permitted fuse links for 500 V + 10% DIAZED operating class gG	
	Rated current [A]	Article number	Rated current [A]
3RM1.01...	0.5	5SB281	25
3RM1.02...	2.0	5SB281	25
3RM1.07...	7.0	5SB281	25

Combining 3RM1 motor starters with fuses for type of coordination 2

**DANGER****Hazardous voltage on the motor**

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

With 3RM11/3RM13 Failsafe motor starters, check the safety function after a short circuit in the case of short-circuit protection according to **type of coordination 2**.

The 3RM1 motor starters can be combined with the following fuses for short-circuits according to type of coordination 2: In a fused configuration with 3RM1 motor starters, please observe the dimensioning information of the fuse manufacturer.

Table 4- 8 Fuses according to type of coordination 2 for 3RM1 motor starters at 400 V

Motor starter $T_A = 40\text{ °C}$		Short-circuit current	Maximum permissible fuse links at 400 V + 10% operational class gG	
Article number	Rated current [A]	Current [kA]	Article number	Rated current [A]
3RM1.01...	0.5	2	FR10GR69V20 (make: Mersen)	20
3RM1.02...	2.0	2	FR10GR69V20 (make: Mersen)	20
3RM1.07...	7.0	2	FR10GR69V20 (make: Mersen)	20

4.3.4 Configuration of a load feeder with 3RM1 motor starter in compliance with UL

4.3.4.1 Function according to UL 508

The 3RM1 motor starters can be combined individually with a circuit breaker (motor starter protector) or fuses. A group installation is also possible according to UL.

4.3.4.2 Configuration of individual load feeders

A circuit breaker or appropriate fuses is/are assigned to each motor starter.

SCCR short circuit current ratings

The SCCR (short circuit current rating) indicates the maximum permissible short-circuit current for a switching device. A higher value can only be achieved in combination with protection devices such as fuses or circuit breakers.

This rating is required for mounting accessories in control cabinets (control panels) conforming to UL 508a. The switching device or switchgear assembly with the lowest value specifies the value for the entire control panel.

A low value makes engineering the supply voltage for the control panel difficult.

According to UL 508a, the 3RM1 motor starters have a standard short circuit current rating of 1 kA for the versions up to 2 A rated current and 5 kA for the version with 7 A.

Tests were also carried out with a short circuit rating of 100 kA (high capacity short circuit current ratings). The UL reports (<https://support.industry.siemens.com/cs/ww/en/view/66427510>) include details of possible combinations with Class J fuses, as well as with a number of circuit breakers.

Motor starter protectors for protecting against short-circuit currents



DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The 3RM1 motor starter and the safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

Replace a defective 3RM1 motor starter after a short circuit and check the safety function of 3RM11/3RM13 Failsafe motor starters.

Standard short circuit according to UL 508 for 460 V + 5 %

Table 4- 9 Motor starter protectors for protecting against short-circuit currents

Motor starter T _A = 50 °C		Short-circuit current	Circuit breaker	
Article number	Rated current [A]	[kA]	Type	[A]
3RM1.01...	0.5	1	3RV2711	6.3
3RM1.02...	2.0	1	3RV2711	6.3
3RM1.07...	6.1	5	3RV2711 / 2721	22

High fault according to UL 508 for 460 V + 5 %

Table 4- 10 Circuit breakers for protecting against high short-circuit currents

Motor starter T _A = 50 °C		Short-circuit current	Circuit breaker	
Article number	Rated current [A]	[kA]	Type	Max. Size [A]
3RM1.01...	0.5	65	3RV2711	2.5
3RM1.02...	2.0	65	3RV2711	2.5
3RM1.07...	6.1	18	3RV2711	10

Fuses for protection from short-circuit currents

! DANGER**Hazardous voltage**

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The 3RM1 motor starter and the safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

Replace a defective 3RM1 motor starter after a short circuit and check the safety function of 3RM11/3RM13 Failsafe motor starters.

Standard short circuit according to UL 508 for 460 V + 5 %

Table 4- 11 Fuses for protecting against short-circuit currents

Motor starter T _A = 50 °C		Short-circuit current	Fuse	
Article number	Rated current [A]	[kA]	Type	Max. size [A]
3RM1.01...	0.5	5	RK 5	8
3RM1.02...	2.0	5	RK 5	8
3RM1.07...	6.1	5	RK 5	30

High fault according to UL 508 for 460 V + 5 %

Table 4- 12 Fuses for protecting against high short-circuit currents

Motor starter T _A = 50 °C		Short-circuit current	Fuse	
Article number	Rated current [A]	[kA]	Type	Max. size [A]
3RM1.01...	0.5	100	CLASS J	35
3RM1.02...	2.0	100	CLASS J	35
3RM1.07...	6.1	100	CLASS J	35

4.3.4.3 Configuring a group of load feeders (group installation)

A "group installation" is a combination of several 3RM1 motor starters with one common short-circuit protection device (branch circuit protection device). This common branch circuit protection device can be a circuit breaker according to UL489, e.g. a 3RV27 or fuses according to UL248.

The 3RM19 infeed system is suitable for a group installation according to UL because it can be loaded up to a total current of 16 A. As an alternative, distribution of the main circuit can be realized with individual cables that have an AWG 12 cross-section or with infeed system 16 A, with a different distribution, for example, "Power Distribution blocks" 5ST2-5 distribution blocks. Whether loading with a total current of >16 A is possible depends on the tested assembly.

For further information, see Certificate of Compliance CoC
(<https://support.industry.siemens.com/cs/ww/en/view/66427510>).

SCCR short-circuit current rating

The SCCR (short circuit current rating) indicates the maximum permissible short-circuit current for a switching device. A higher value can only be achieved in conjunction with protection devices such as fuses or circuit breakers.

This rating is required for mounting accessories in control cabinets (control panels) conforming to UL 508a. The switchgear assembly with the lowest value specifies the value for the entire control panel.

Motor starter protectors for protecting against short-circuit currents

DANGER

Hazardous Voltage.

Can Cause Death, Serious Injury, or Property Damage.

Health hazard from automatic hot restart.

The 3RM1 motor starter and the safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

Replace a defective 3RM1 motor starter after a short circuit and check the safety function of 3RM11/3RM13 Failsafe motor starters.

Standard short circuit according to UL 508 for 460 V + 5 %

Table 4- 13 Motor starter protectors for protecting against short-circuit currents

Motor starter T _A = 50 °C		Short-circuit current	Circuit breaker	
Article number	Rated current [A]	[kA]	Type	Max. size [A]
3RM1.01...	0.5	5	3RV2711 / 2721	22
3RM1.02...	2.0	5	3RV2711 / 2721	22
3RM1.07...	6.1	5	3RV2711 / 2721	22

High fault according to UL 508 for 460 V + 5 %

Table 4- 14 Circuit breakers for protecting against high short-circuit currents

Motor starter $T_A = 50\text{ °C}$		Short-circuit current	Motor starter protector	
Article number	Rated current [A]	[kA]	Type	Max. size [A]
3RM1.01...	0.5	10	3RV2711	12.5
3RM1.02...	2.0	10	3RV2711	12.5
3RM1.07...	6.1	10	3RV2711	12.5

Fuses for protection from short-circuit currents

 **DANGER**

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The 3RM1 motor starter and the safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

Replace a defective 3RM1 motor starter after a short circuit and check the safety function of 3RM11/3RM13 Failsafe motor starters.

Standard short circuit according to UL 508 for 460 V + 5 %

Table 4- 15 Fuses for protecting against short-circuit currents

Motor starter $T_A = 50\text{ °C}$		Short-circuit current	Fuse	
Article number	Rated current [A]	[kA]	Type	Max. size [A]
3RM1.01...	0.5	5	RK 5	30
3RM1.02...	2.0	5	RK 5	30
3RM1.07...	6.1	5	RK 5	30

High fault according to UL 508 for 460 V + 5 %

Table 4- 16 Fuses for protecting against high short-circuit currents

Motor starter $T_A = 50\text{ °C}$		Short-circuit current	Fuse	
Article number	Rated current [A]	[kA]	Type	Max. size [A]
3RM1.01...	0.5	100	CLASS J	35
3RM1.02...	2.0	100	CLASS J	35
3RM1.07...	6.1	100	CLASS J	35

4.4 Infeed for the main circuit

4.4.1 Infeed options

You have the following options for implementing the infeed for the main circuit of the 3RM1 motor starter:

- Infeed via parallel wiring of individual 3RM1 motor starters
- Infeed via the 3RM19 infeed system (for 3RM1...-1AA.4 motor starters with screw connections and 3RM1...-3AA.4 motor starters with hybrid connections)

4.4.2 3RM19 3-phase infeed system

The 3RM19 3-phase infeed system is designed for 3RM1...-1AA.4 motor starters with screw connections and 3RM1...-3AA.4 motor starters with hybrid connections. Push-in terminals can therefore not be connected with this infeed system.

Note

Use of several busbars

More than five devices can be connected by clamping the terminal lugs of a second busbar rotated by 180°.

A motor starter must be plugged into the connection points of the busbars. Observe this in the case of gaps stipulated for derating purposes, and select appropriate busbars.

The special 3-phase busbar enables fault-free and fast main circuit wiring when using several 3RM1 motor starters arranged side-by-side. The energy is routed from an upstream short-circuit protection device to the system via an infeed terminal. The 3-phase busbar is designed in such a way that the maximum number of 3RM1 motor starters is only limited by the rated operational current of the infeed system.

Rated operational current

For the components of the 3RM19 infeed system, the maximum permissible rated operational current is 25 A (UL: 16 A).

If you use several 3RM1 motor starters that together require more than 25 A, form groups that each have a maximum requirement of 25 A (UL: 16 A).

Note

Multiple infeed

If you plan multiple infeed in the case of a respective requirement of more than 25 A, you must observe a potential current asymmetry in the infeeds.

Mounting

Attach the busbar from below to the rear terminals L1, L2 and L3 of the 3RM1 motor starter. The outgoing feeders to the motors thus remain accessible.

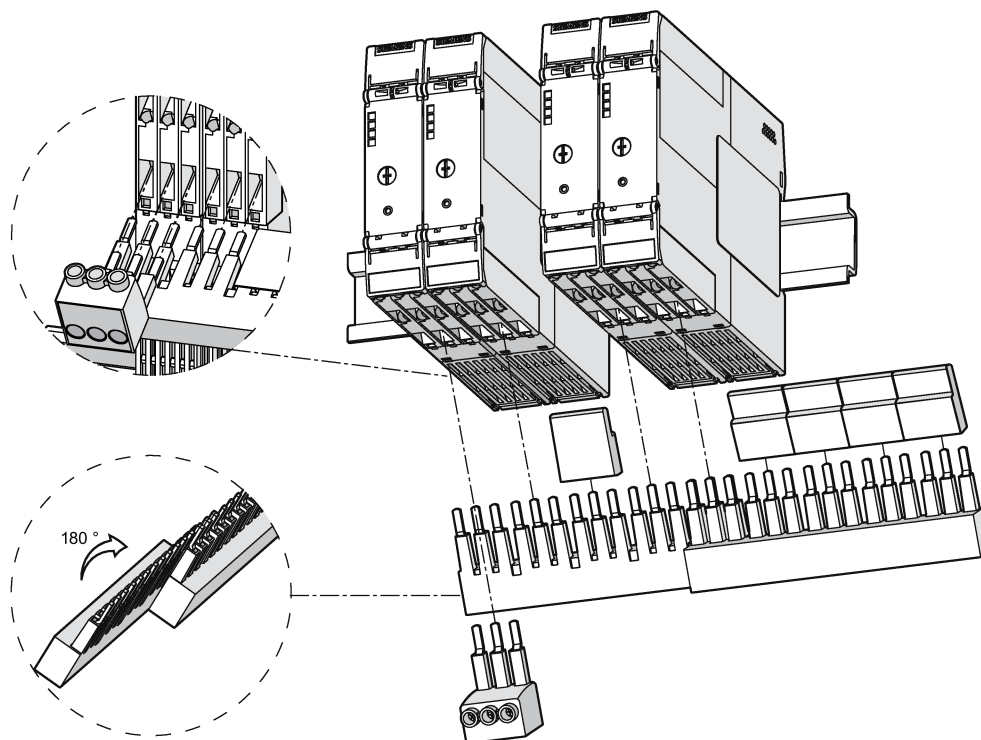


Figure 4-1 Attaching the 3RM19 infeed system to the 3RM1 motor starters



! DANGER

Hazardous voltage

If free and uncovered terminal lugs are touched when the load voltage is switched on, death or serious physical injury will result.

For this reason, always cover the free terminal lugs (= free slots that are unassigned for thermal reasons, for example) with a cover cap. The cover caps are available as accessories.

Energy is supplied via the infeed terminal 3RM1920-1AA. This means you can connect conductors with a cross-section of 1.5 mm² to 4 mm².

4.5 Configuration with device connectors

The device connector may be used only with the 24 V DC version of the motor starter. There must be no mixing of the Standard and Failsafe variants of the 3RM1 motor starter via the device connector. If both variants are used, these must be fitted on separate device connector systems. This has no influence on the infeed system for the 3RM19 main circuit, which can still be used without restriction for a combination of the variants.

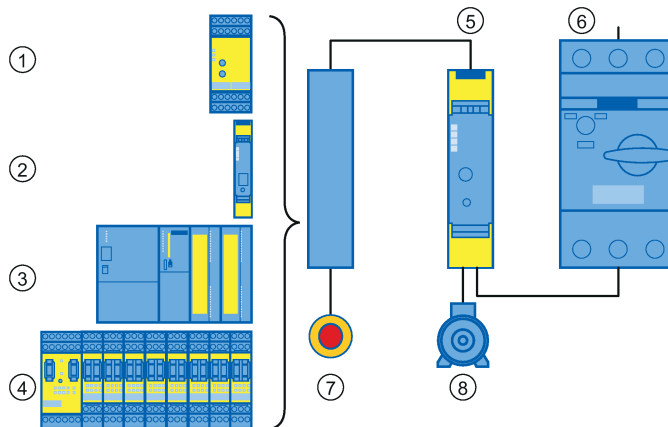
For additional information on using device connectors, refer to "FAQs on the 3RM1 motor starter (<https://support.industry.siemens.com/cs/ww/en/view/81525494>)".

4.6 Examples/applications

4.6.1 Operation with the EMERGENCY-STOP function

The 3RM11 Failsafe and 3RM13 Failsafe safety-related motor starters can be used with any safety relays and safety-related controllers in safety-related applications up to SILCL 3 in accordance with IEC 62061, PL e/Cat. 4 in accordance with EN ISO 13849-1.

Basic configuration



- ① ... ④ Instead of the black box, one of the modules ① ... ④ can be used
- ① SIRIUS 3TK28 safety relay
- ② SIRIUS 3SK safety relay
- ③ SIMATIC S7 fail-safe modules
- ④ SIRIUS 3RK3 modular safety system
- ⑤ SIRIUS 3RM11 Failsafe or SIRIUS 3RM13 Failsafe safety-related motor starter
- ⑥ SIRIUS 3RV2 motor starter protector
- ⑦ EMERGENCY-STOP button
- ⑧ Motor

Figure 4-2 Basic configuration with safety relay or safety-related controller

Configuration with 3SK safety relay and device connectors

If you use safety-related motor starters (3RM11...AA04 / 3RM13...AA04) with a control supply voltage of 24 V DC with the 3SK safety relay, you can establish safe connection of the safety-related wiring using the device connectors. Use of device connectors avoids errors when setting up the control circuit wiring and offers savings in cable laying. In addition, the test overhead following completion of the control cabinets is reduced. The 3RM11/3RM13 safety-related motor starters can be used optimally as output expansions in the 3SK system.

When the EMERGENCY-STOP is actuated, the motor starters' control supply voltage is shut down and the motor is isolated from the power supply. Due to the internal self-monitoring of the 3RM11/13 Failsafe motor starters in compliance with SILCL 3/PL e, there is no need for a feedback signal to the 3SK safety relay.

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Possible failure of the safety function

When the product is operated with a 3SK safety relay and a device connector, the supply voltage for 3RM11 / 3RM13 motor starters is fed in via the device connectors.

In this case, do not connect anything to terminals A1 and A2 of the 3RM11/3RM13 Failsafe motor starters, in order to prevent bypassing of the safety function.

WARNING

Failure of the safety function

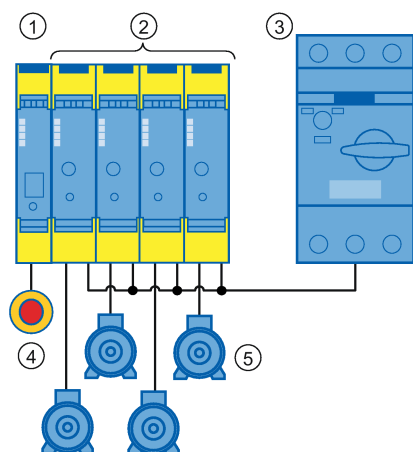
Can result in death, serious physical injury, or damage to property.

Hybrid operation of 3RM10/3RM12 Standard motor starters with 3RM11/3RM13 Failsafe motor starters in safety-related applications is not admissible as this combination may cause the safety functions to fail.

Only ever use safety-related motor starters (3RM11 Failsafe and 3RM13 Failsafe) in safety-related applications.

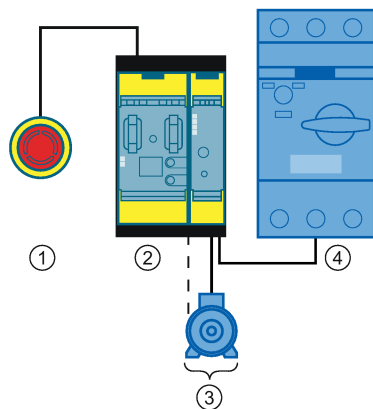
Note

Please observe the information about the control supply voltage and the control inputs in Chapter "Device versions (Page 26)" and in the "Technical Data Sheet (<https://support.industry.siemens.com/cs/ww/en/ps/16311/td>)."



- ① SIRIUS 3SK1 Advanced basic unit
- ② SIRIUS 3RM1 Failsafe (3RM11 Failsafe or 3RM13 Failsafe) safety-related motor starters
- ③ SIRIUS 3RV2 motor starter protector
- ④ SIRIUS 3SU1 EMERGENCY STOP
- ⑤ Motor

Figure 4-3 Configuration with 3SK1 safety relay and device connectors



- ① SIRIUS 3SU1 EMERGENCY STOP
- ② 3SK2 safety relay
- ③ 3RM1 Failsafe motor starter and motor
- ④ SIRIUS RV2 motor starter protector

Figure 4-4 Configuration with 3SK2 safety relay and device connectors

You will find further information on operation in safety-related applications in the 3SK safety relay manuals, see Chapter "Further documentation (Page 11)." Follow the safety instructions and information contained there.

Reference

You will find more typical circuits in the Appendix "Typical circuits (Page 141)."

Mounting

5.1 Warning notices

Warning notices before installation, wiring, and commissioning



DANGER

Hazardous Voltage.

Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

5.2 Mounting the devices on a level surface

Requirements

Please note the following requirements for mounting on a level surface:

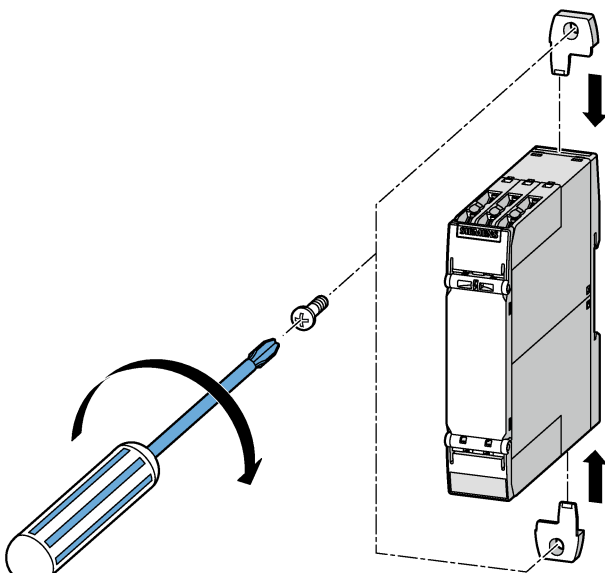
- Please observe the information about the mounting position in the chapter "Mounting position (Page 60)".
- Two properly executed drill holes with thread or plug on the level surface

For details of the distances between the drilled holes, please refer to the relevant dimension drawings in the chapter "Dimension drawings (Page 127)".

- Two screws to fit the holes M4 x 12 in accordance with DIN 784
- Two plastic securing brackets

Refer to the accessories list for the relevant article number in the chapter "Accessories and article number scheme (Page 45)".

Procedure

Step	Operating instruction	Figure
1	Insert the securing brackets into the openings provided on the device until they engage.	
2	Hold the device up to the level surface prepared for screw fastening.	
3	Insert the screws through the oblong holes in the securing brackets.	
4	Screw the device onto the level surface so that it is secure. Tightening torque: 1 Nm	

5.3 Disassembling the devices from a level surface

DANGER

Hazardous voltage

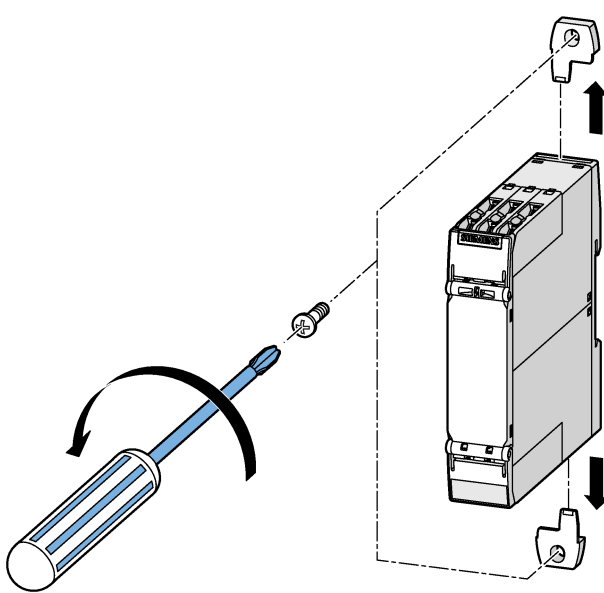
Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

Requirements

- The terminals have been removed or disconnected.

Procedure

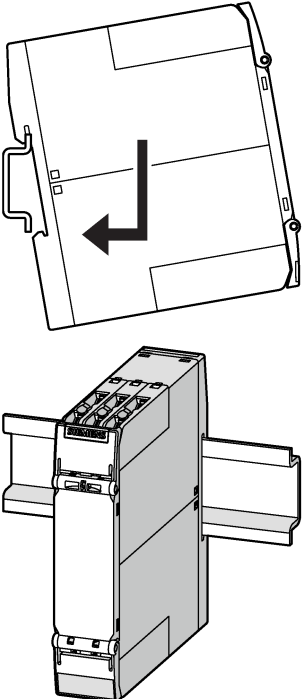
Step	Operating instruction	Figure
1	Hold the device firmly.	
2	Unscrew the cap screws.	
3	Lift the device from the level surface.	
4	Remove the securing brackets from the device.	

5.4 Mounting the devices on a standard mounting rail

Requirements

- At the installation location, a horizontal 35 mm mounting rail in accordance with DIN EN 60715 is properly secured
- Please observe the information about the mounting position in chapter "Mounting position (Page 60)"

Procedure

Step	Operating instruction	Figure
1	Hang the back of the device onto the upper edge of the standard mounting rail	
2	Press the lower half of the device against the rail until the device engages	

5.5 Disassembling devices from a standard mounting rail

DANGER

Hazardous voltage

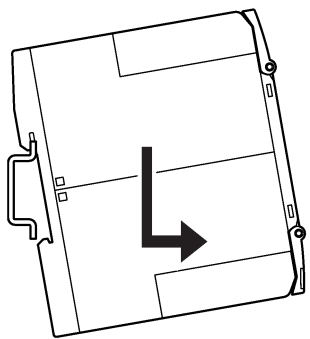
Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

Requirements

- The terminals have been removed or disconnected.

Procedure

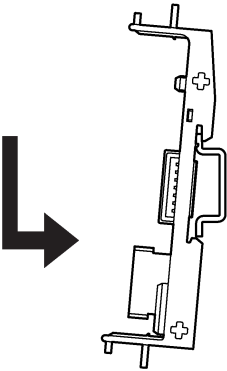
Step	Operating instruction	Figure
1	Press the device downwards.	
2	Pull the lower half of the device away from the standard mounting rail.	
3	Lift the device from the upper edge of the standard mounting rail.	

5.6 Mounting the devices with device connectors on a standard mounting rail

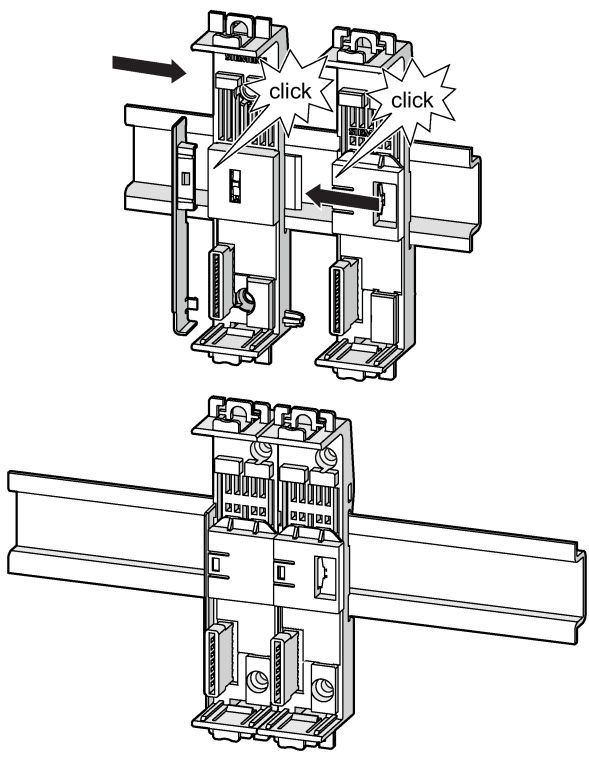
Requirements

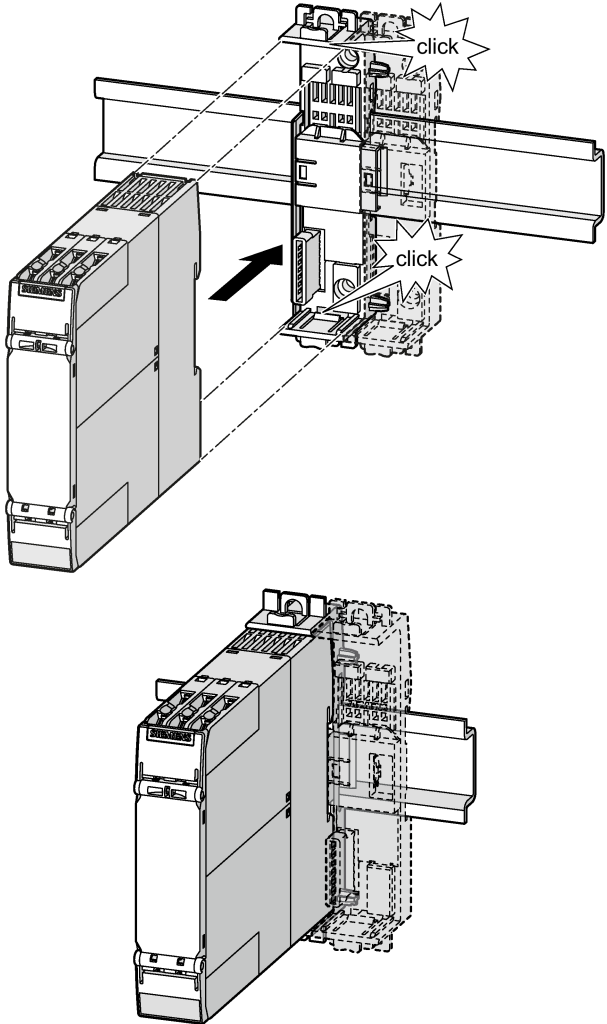
- At the installation location, a horizontal 35 mm mounting rail in accordance with DIN EN 60715 is properly secured
- Please observe the information about the mounting position in chapter "Mounting position (Page 60)"

Procedure

Step	Operating instruction	Figure
1	Place the back of the device connector on the upper edge of the standard mounting rail.	
2	Press the lower half of the device connector against the standard mounting rail until it engages.	
3	Repeat the procedure with all the required device connectors.	

5.6 Mounting the devices with device connectors on a standard mounting rail

Step	Operating instruction	Figure
4	Push the device connectors together until they engage.	
5	Mount the cover on the left on the first device connector in the system configuration. The cover is included in the scope of supply of the device terminator.	

Step	Operating instruction	Figure
6	Mount the device on the device connector.	
7	Mount all the devices required for the system configuration in accordance with the respective installation guidelines on the device connectors.	 The figure consists of two parts. The top part is a perspective view showing a device being inserted into a rail. Two 'click' labels with arrows point to the top and bottom connection points of the device. The bottom part is a perspective view showing the device fully mounted on the rail.

5.7 Disassembling the devices with device connectors from a standard mounting rail

DANGER

Hazardous voltage

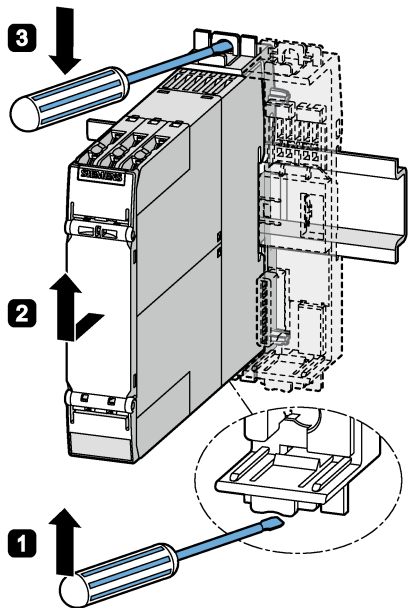
Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

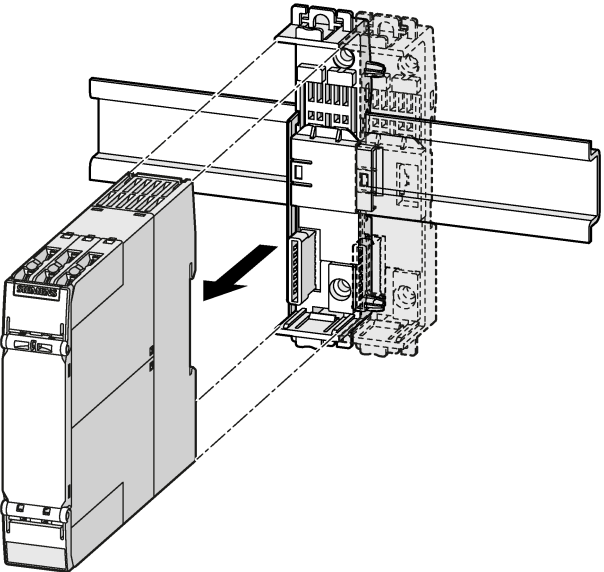
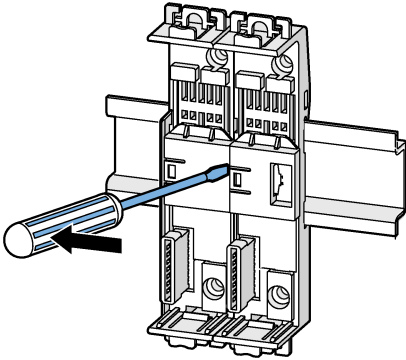
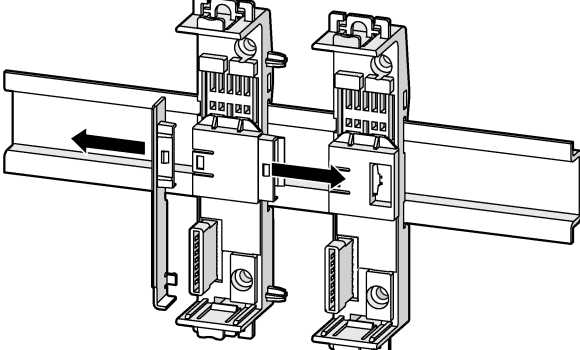
Requirements

- The terminals have been removed or disconnected.

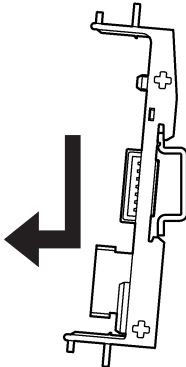
Procedure

Step	Operating instruction	Figure
1	Unlock the device using a screwdriver.	
2	Pull the lower half of the device away from the device connector.	
3	Unlock the device using a screwdriver.	

5.7 Disassembling the devices with device connectors from a standard mounting rail

Step	Operating instruction	Figure
4	Pull the device away from the device connector.	
5	Separate the device connectors using a screwdriver.	
6	Remove the cover to the left of the first device connector.	

5.7 Disassembling the devices with device connectors from a standard mounting rail

Step	Operating instruction	Figure
7	Press the device connector down.	
8	Pull the lower half of the device connector away from the standard mounting rail.	
9	Lift the device connector from the upper edge of the standard mounting rail.	

5.8 Mounting the devices with device connectors on a wall

Requirements

Please note the following requirements for mounting on a level surface:

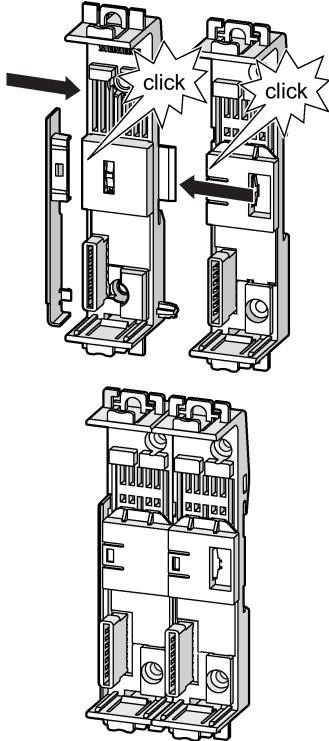
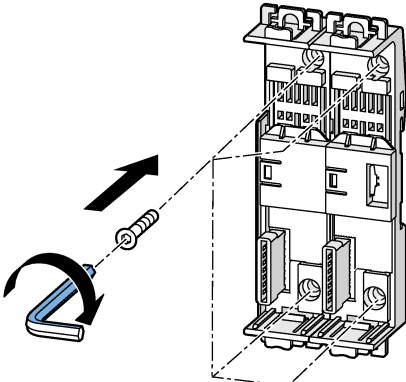
- Please observe the information about the mounting position in the chapter "Mounting position (Page 60)".
- Two properly executed drill holes with thread or plug on the level surface

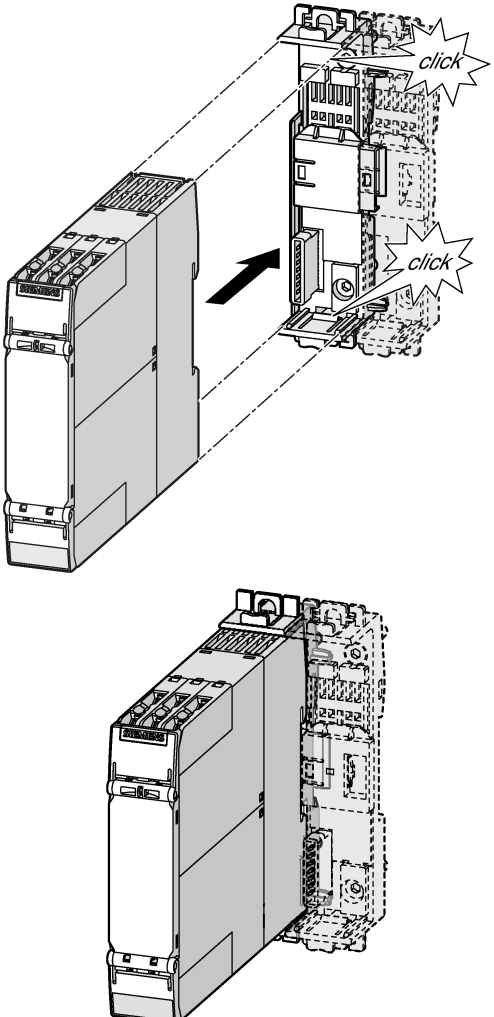
For details of the distances between the drilled holes, please refer to the relevant dimension drawings in the chapter "Dimension drawings (Page 127)".

- Two screws to fit the M4 x 12 holes in accordance with DIN 784.
- Device connector

Refer to the accessories list for the relevant article number in the chapter "Accessories and article number scheme (Page 45)".

Procedure

Step	Operating instruction	Figure
1	Push the device connectors together until they engage.	
2	Mount the cover on the left on the first device connector in the system configuration. The cover is included in the scope of supply of the device terminator.	
3	Hold the device connector against the level surface prepared for screw fastening.	
4	Insert the screws through the holes in the device connectors.	
5	Screw the device connector securely onto the level surface. Tightening torques: • Top: < 0.1 Nm • Bottom: 1 Nm	

Step	Operating instruction	Figure
6	Mount the device on the device connector.	
7	Mount all the devices required for the system configuration in accordance with the respective installation guidelines on the device connectors.	

5.9 Disassembling the devices with device connectors from a wall

DANGER

Hazardous voltage

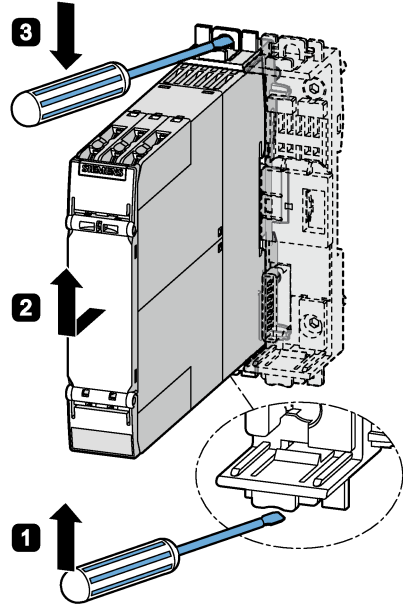
Will result in death, serious physical injury, or damage to property.

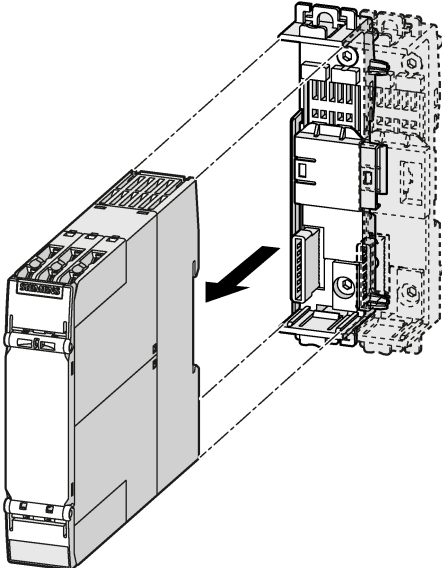
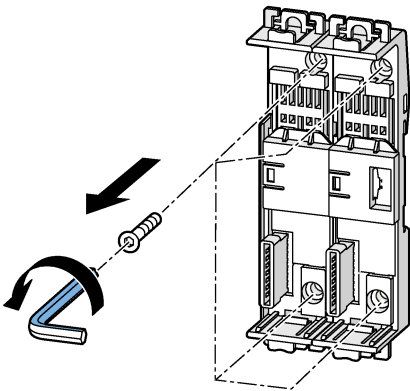
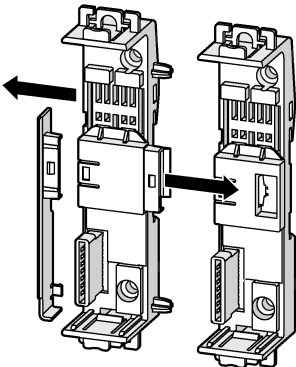
Turn off and lock out all power supplying this device before working on this device.

Requirements

- The terminals have been removed or disconnected.

Procedure

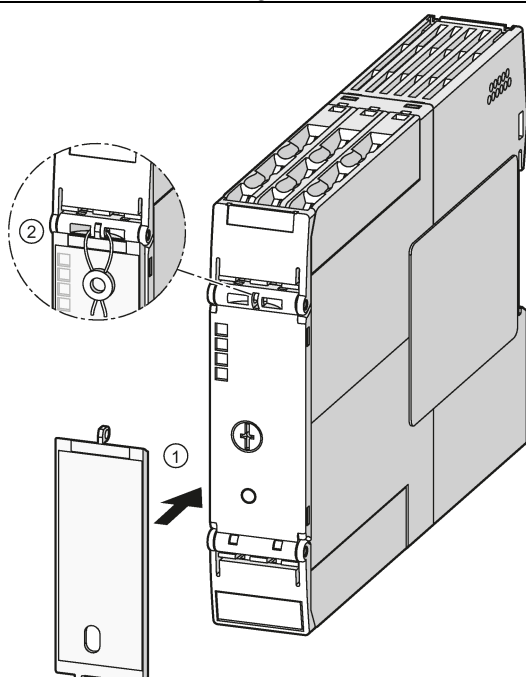
Step	Operating instruction	Figure
1	Unlock the device on the underside using a screwdriver.	
2	Pull the lower half of the device away from the device connector.	
3	Unlock the device on the top using a screwdriver.	

Step	Operating instruction	Figure
4	Pull the device away from the device connector.	 The diagram shows a 3D perspective of the SIRIUS 3RM1 motor starter. A dashed line indicates the device being pulled away from the terminal block. A black arrow points from the device towards the terminal block, indicating the direction of movement.
5	Release the screws.	 The diagram shows a close-up of the terminal block. A black arrow points to a screw being removed. A blue arrow indicates the direction of the screw being turned counter-clockwise. A dashed line shows the screw being removed from the terminal block.
6	Separate the device connectors.	
7	Remove the cover to the left of the first device connector.	 The diagram shows a close-up of the terminal block. A black arrow points to the cover being removed from the first device connector. A dashed line shows the cover being removed from the terminal block.

5.10 Mounting the sealable cover

The sealable cover is used to protect the set rated operational current of the load (current setting I_e) against unauthorized or unintentional manipulation.

Procedure

Step	Operating instruction	Figure
1	Attach the two bottom hooks on the cover to the openings on the device and fold the cover up	
2	Seal the cover to secure it against unauthorized removal.	

Connection

6.1 Connecting the screw-type terminals

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

NOTICE

Absence of overvoltage protection if wiring is incorrect can cause damage to property

Observe the wiring's terminal assignments:

For optimum response in the event of short circuits and for protection against overvoltages, the main circuit's infeed must be at terminals 1/L1 to 5/L3.

Requirements

- Cross-tip screwdriver size PZ 1 x 80
- For suitable connection cross-sections of the cables, see chapter Connection cross sections (Page 122).

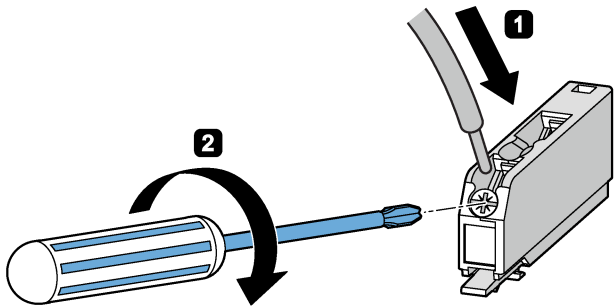
Procedure

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

To ensure protection against an electric shock hazard with the hinged cover open at the signaling contacts 95, 96, 98 at a voltage of ≥ 50 V, screw in all terminal screws that are not needed to clamp conductors.

Step	Operating instruction	Figure
1	Insert the relevant cable into square on the screw terminal until it engages.	
2	Hold the cable in the screw terminal.	
3	Tighten the screw with a torque of 0.6 to 0.8 Nm.	
4	Pull on the cable to ensure it is screwed tight.	

6.2 Disconnecting the screw-type terminals

DANGER

Hazardous voltage

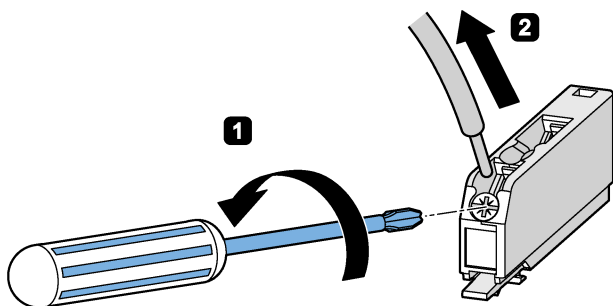
Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

Requirements

- Cross-tip screwdriver size PZ 1 x 80

Procedure

Step	Operating instruction	Figure
1	Unscrew the screw of the screw-type terminal.	
2	Remove the cable from the unscrewed screw-type terminal.	

6.3 Connecting the push-in terminals

6.3.1 Wiring rules for spring-loaded terminals (with push-in technology)

Wiring rules for ...		Control current terminals (top of enclosure):	Main current terminals (bottom of enclosure)
Connectable cross-sections for solid cables		2 x 0.5 ... 2 x 1.5 mm²	0.5 ... 4 mm²
		(AWG ¹⁾ : 20 ... 16)	AWG ¹⁾ : 20 ... 12
Connectable cross-sections for flexible cables	Without end sleeve	2 x 0.5 ... 2 x 1.5 mm²	0.5 ... 4 mm²
		(AWG ¹⁾ : 20 ... 16)	AWG ¹⁾ : 20 ... 12
	With end sleeve (with and without plastic sleeve)	2 x 0.5 ... 2 x 1.0 mm² ²⁾	0.5 ... 2.5 mm²
		(AWG ¹⁾ : 20 ... 18)	(AWG ¹⁾ : 20 ... 14)
	With TWIN end sleeve	---	2 x 0.5 ... 2x1.5 mm²
		---	(AWG ¹⁾ : 20 ... 16)
Cable stripping length		10 ... 11 mm	
End sleeves according to DIN 46228-4 with plastic sleeve		10 mm	

¹⁾ AWG: American Wire Gauge (AWG does not define use of end sleeves)

²⁾ When 2 x 1.0 mm² end sleeves with a plastic sleeve are used, space problems may arise with the sleeves; as an alternative, you are advised to use end sleeves without plastic sleeves

Notes on handling spring-loaded terminals with push-in technology

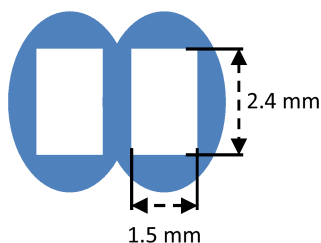
The terminal area of the spring-loaded terminals is rectangular, and the maximum overall dimensions of a conductor to be wired must not exceed 1.5 x 2.4 mm (control current terminals) or 2.4 x 2.8 mm (main current terminals).

Attention must be paid to the orientation of the terminal area, which may call for vertical fitting of rectangularly crimped cables.

To make optimum use of available terminal area, you are advised to choose a form of crimping that creates a corresponding rectangular contour. Trapezoidal crimping is generally very highly suitable in this case.

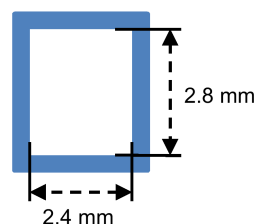
When use is made of a cable that utilizes the full overall height, the terminal's spring is deflected to the maximum. Therefore, removal of this cable may become a problem because it requires further deflection of the spring.

Control current terminals



Terminal area of control current terminals

Main current terminals



Terminal area of main current terminals

6.3.2 Connecting the push-in terminals

! DANGER Hazardous voltage Will result in death, serious physical injury, or damage to property. Turn off and lock out all power supplying this device before working on this device.
NOTICE Absence of overvoltage protection if wiring is incorrect can cause damage to property. Observe the wiring's terminal assignments: For optimum response in the event of short circuits and for protection against overvoltages, the main circuit's infeed must be at terminals 1/L1 to 5/L3.

Push-in connections are a form of spring-loaded terminals allowing wiring without tools for rigid conductors or conductors equipped with end sleeves.

For wiring finely stranded or stranded conductors without end sleeves on push-in connections, a screwdriver is required.

Requirements

- Screwdriver DIN 5264 of the size 0.5 x 3 mm (for finely-stranded conductors only)
- For suitable connection cross-sections of the cables and notes on end sleeve finishing, see Chapter Connection cross sections (Page 122).

6.3 Connecting the push-in terminals

Procedure

Table 6- 1 Rigid conductors or conductors equipped with end sleeves

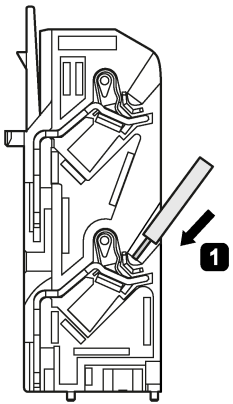
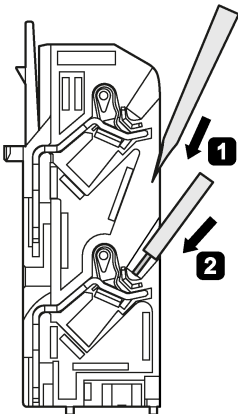
Step	Operating instruction	Figure
1	Insert the cable into the oval opening as far as it will go.	
2	Pull on the cable to ensure it is tight.	

Table 6- 2 Finely-stranded conductors

Step	Operating instruction	Figure
1	Insert the screwdriver in the rectangular opening to open the terminal (oval opening).	
2	Insert the cable as far as it will go into the oval opening and remove the screwdriver.	
3	Pull on the cable to ensure it is tight.	

6.4 Disconnecting the push-in terminals

! DANGER

Hazardous voltage

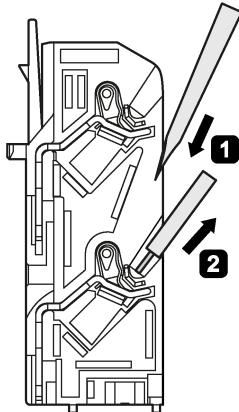
Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

Requirements

- Screwdriver DIN 5264 of the size 0.5 x 3 mm

Procedure

Step	Operating instruction	Figure
1	Insert the flat-head screwdriver into the rectangular opening of the spring-loaded terminal until it engages.	
2	Remove the cable from the oval opening.	
3	Remove the screwdriver.	

6.5 Attaching the terminals

DANGER

Hazardous voltage

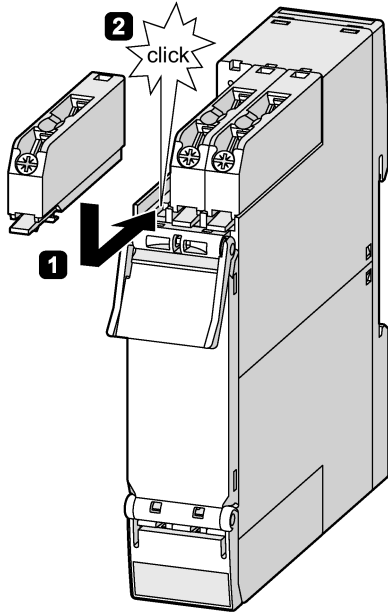
Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

Requirements

- You must have removed the terminals, for the purpose of replacing a device, for example.

Procedure

Step	Operating instruction	Figure
1	Insert the terminal into the guide rail of the device.	
2	Slide the terminal back until it audibly engages.	

6.6 Removing the terminals

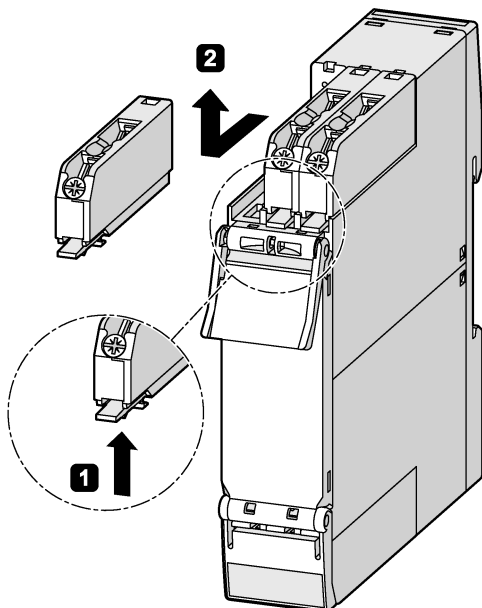
DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

Procedure

Step	Operating instruction	Figure
1	Press the clip of the terminal block upwards.	
2	Pull the terminal out to the front.	
3	Lift the terminal out of the guide rail of the device.	

6.7 Connecting the infeed system (option)



Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

Requirements

Please note the following requirements for mounting the 3RM19 infeed system:

- Make sure the rated operational current of the infeed system (IEC: 25 A; UL: 16 A) is not exceeded.
- The 3RM1 motor starters are mounted next to each other on a mounting rail or a level mounting surface.
- The mounting system is only suitable for main current terminals with screw connections.
- Cross-tip screwdriver size PZ 1 x 80
- For suitable connection cross-sections of the cables for the infeed terminal, see the Chapter "Connection cross sections (Page 122)".

Procedure

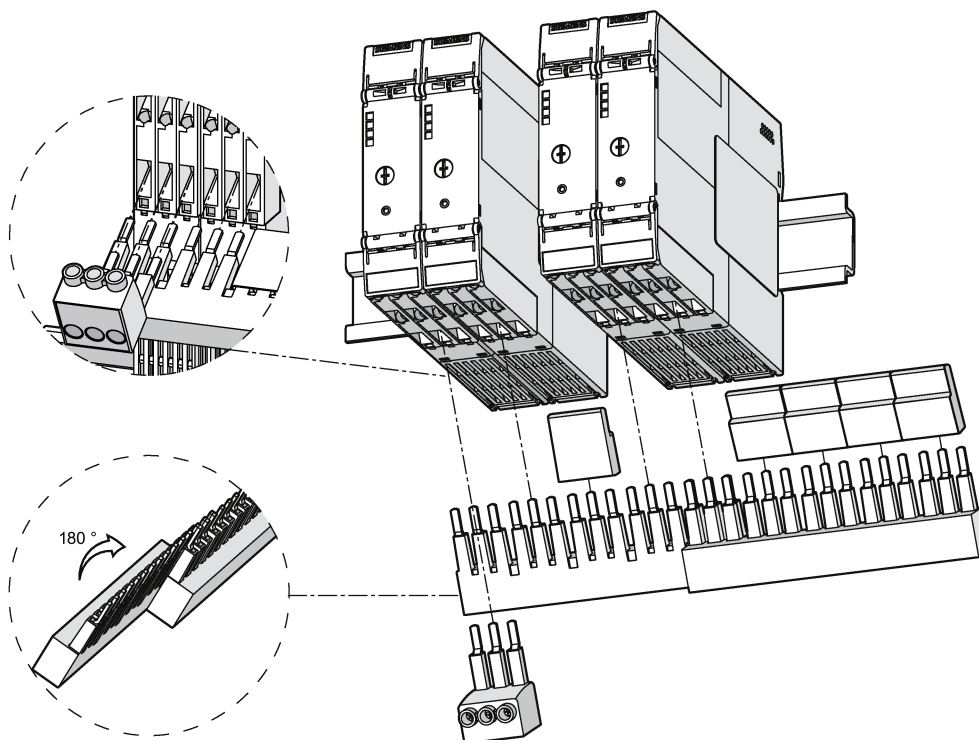


Figure 6-1 Mount the 3RM19 infeed system



! DANGER

Hazardous voltage

If free and uncovered terminal lugs are touched when the load voltage is switched on, death or serious physical injury will result.

For this reason, always cover the free terminal lugs (= free slots that are unassigned for thermal reasons, for example) with a cover cap. The cover caps are available as accessories.

6.7 Connecting the infeed system (option)

Step	Operating instruction
1	Connect the main circuit L1, L2 and L3 to the infeed terminal. Tighten the screw terminals with 0.6 to 0.8 Nm. Pull on the cables to ensure they are screwed tight.
2	Plug the first busbar into the rear screw terminals on the underside of the device. The first three terminal lugs must be assigned here to the first device on the extreme left.
3	Plug the infeed terminal into the screw terminals of the first device together with the busbar from Step 1.
5	Mount the second busbar rotated through 180°. The first three terminal lugs overlap here with the last three terminal lugs of the first busbar, and are mounted in the same device.
4	Tighten the screw terminals with 0.6 to 0.8 Nm. Pull on the busbars to ensure they are screwed tight.

Note

Third busbar

If you use a third busbar, you must rotate it in the same way as the first busbar, and mount it so that it overlaps the second busbar.

Note

Free slots due to derating

If a clearance is left between the devices for thermal reasons, three terminal lugs (= one slot) must always remain free.

Note

Busbar connection points

A motor starter must be plugged into the connection points of the busbars. Observe this in the case of gaps stipulated for derating purposes, and select appropriate busbars.

Operator control and monitoring

7.1 Operator controls

The following operator controls are available on the 3RM1 motor starter for parameter assignment and operation:

- Rotary coding switch with adjusting scale for current setting " I_e ":

You set the rated operational current of the load " I_e " with the rotary coding switch.

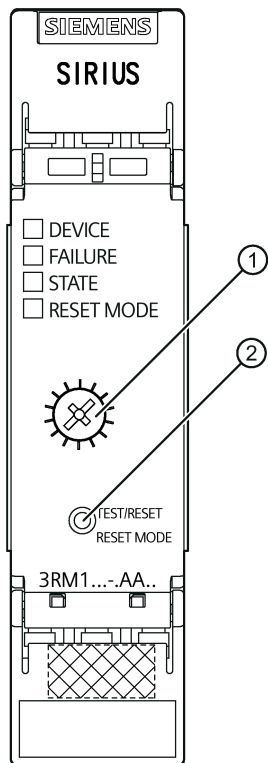
- "TEST/RESET/RESET MODE" button:

You can start different function tests with the "TEST/RESET/RESET MODE" button. You can find details about this in the section "Carrying out the function test (Page 43)".

Following an overload trip, the 3RM1 motor starter can be reset using the button after three minutes have elapsed. You can find details about this in the section "Setting the RESET method (Page 109)".

7.1.1 Parameterizing the 3RM1 motor starter

All the settings are made directly at the 3RM1 motor starter:



- ① Rotary coding switch for rated operational current of the load
- ② TEST/RESET/RESET MODE button

Figure 7-1 Front view of the 3RM1 motor starter

7.1.2 Setting the rated operational current

Use a screwdriver to set the load's rated operational current (current setting I_e) on the rotary coding switch.

A change takes effect immediately. The sealable cover (accessory) prevents unauthorized access to the rotary coding switch.

7.1.3 Setting the RESET method

On the 3RM1 motor starter, you can select either manual RESET or auto RESET for resetting following an overload trip.

Manual RESET

After an overload trip, reset the 3RM1 motor starter manually after a cooling time of three minutes using the "TEST/RESET/RESET MODE" button. As an alternative, in the case of the 3RM10/3RM12 Standard motor starters, you can use the RESET control input IN3 for acknowledgment.

Auto RESET

After an overload trip, the 3RM1 motor starter is reset automatically after a cooling time of three minutes.

WARNING

Hazardous voltage

Can result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart

If the DEVICE LED shows a yellow light and the FAILURE LED a red light, and a manual RESET or auto RESET takes place following an overload trip, the machine starts up immediately if there is a control command at IN1/IN2. Personnel in the danger area can be injured.

Make sure that the danger area of the machine is kept clear of people.

Note

Restriction in the hazardous area

In applications for the protection of motors in hazardous areas, a 3RM11/3RM13 Failsafe motor starter may only be operated with the "Manual RESET" setting. The "RESET Mode" LED must be off.

Procedure

When it is delivered, the 3RM1 motor starter is set to manual RESET. You can change the setting as follows:

1. Press the "TEST/RESET/RESET MODE" button and hold it down. Switch on the supply voltage via A1-A2.
2. The set RESET method is changed to the other RESET method.
 - "RESET MODE" LED lit: Auto RESET selected
 - "RESET MODE" LED not lit: Manual RESET selected

Note

Briefly switching the supply voltage off and on again does not shorten the cooling time and there is no automatic acknowledgment.

7.2 Displays and location of the connections

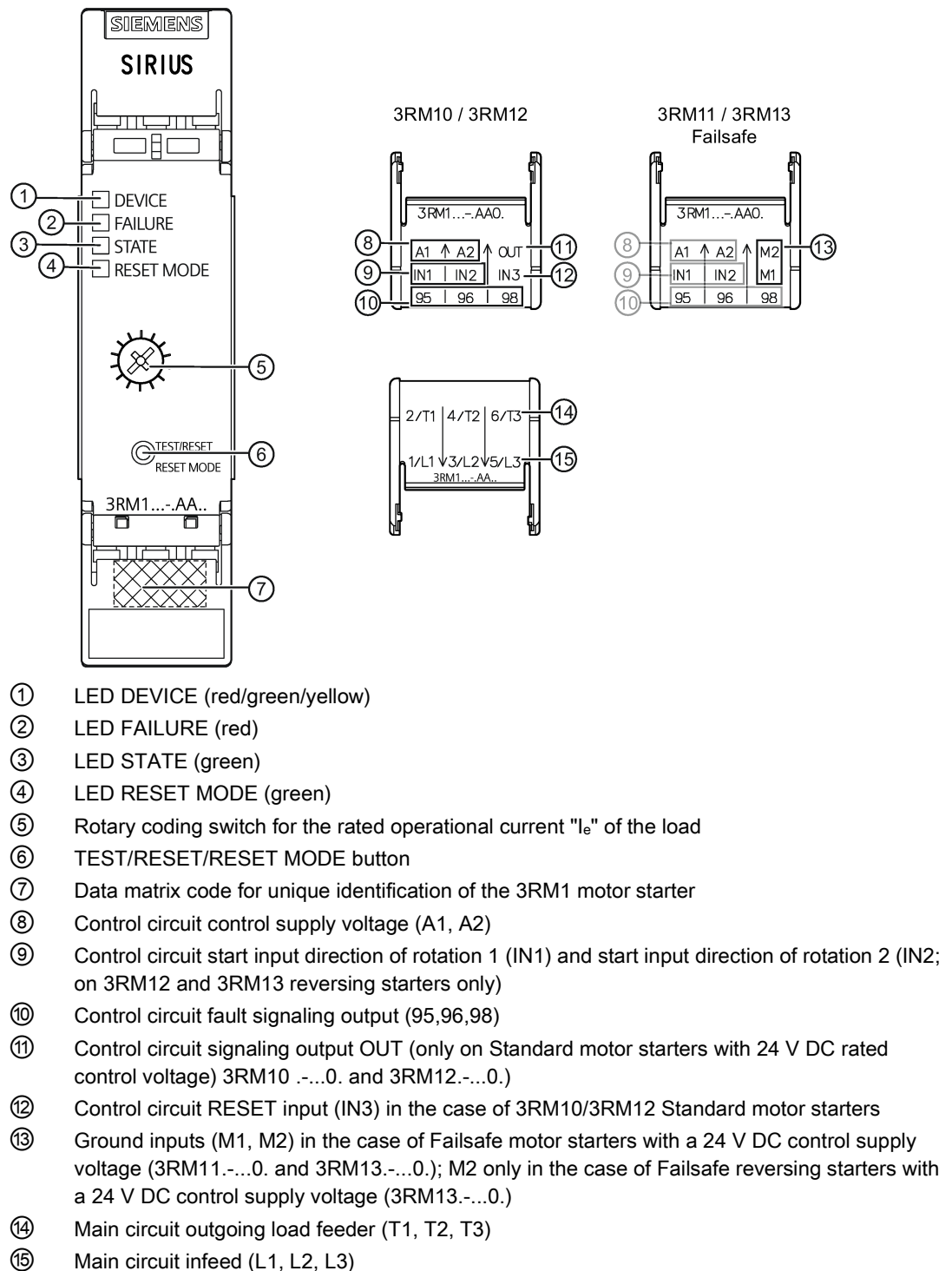


Figure 7-2 Front elements of the 3RM1 motor starter

Location of the connections

The terminal covers for the control and main circuit are labeled on the inside with the designations of the relevant terminals. The position of the label corresponds to the position of the respective terminal.

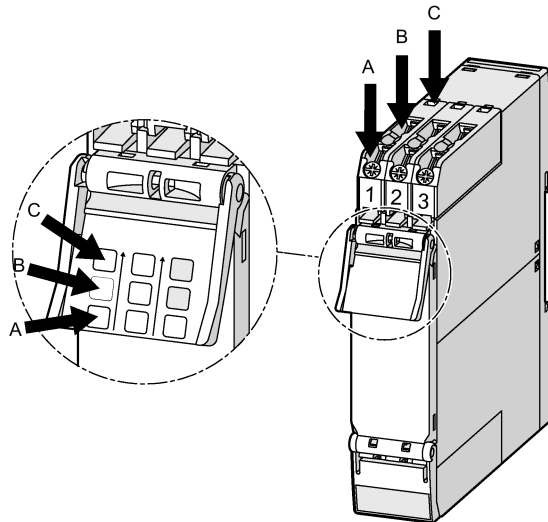


Figure 7-3 Upper terminal cover

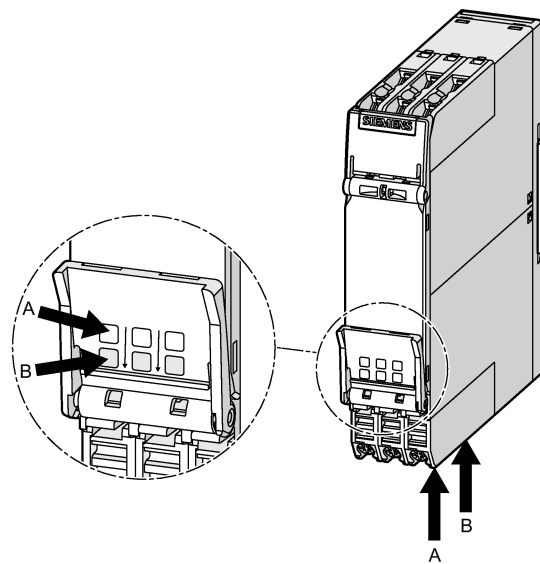
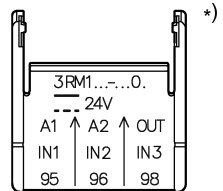
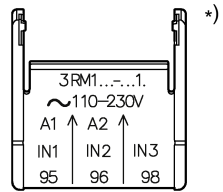
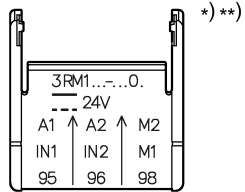
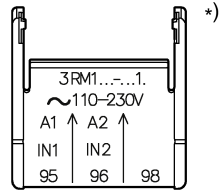
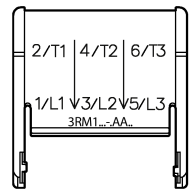


Figure 7-4 Lower terminal cover

Terminal cover

Depending on the version of the 3RM1 motor starter, the device will have the following connections:

Motor starter version		Inscription of the terminal cover
Control circuit (at the top of the device)		
Standard	• 24 V DC • Direct-on-line starter 3RM10.-...0. (IN1) • Reversing starter 3RM12.-...0. (IN1 and IN2)	
	• 110-230 V AC/110 V DC • Direct-on-line starter 3RM10.-...1. (IN1) • Reversing starter 3RM12.-...1. (IN1 and IN2)	
Failsafe	• 24 V DC • Direct-on-line starter (3RM11.-...0.) (IN1, M1) • Reversing starter (3RM13.-...0.) (IN1 and IN2, M1 and M2)	
	• 110-230 V AC/110 V DC • Direct-on-line starter 3RM11.-...1. (IN1) • Reversing starter 3RM13.-...1. (IN1 and IN2)	
Main circuit (at the bottom of the device)		
All motor starters	All motor starters	

*) IN2 only in the case of reversing starters (3RM12 and 3RM13)

**) M2 only in the case of Failsafe reversing starters (3RM13)

7.2.1 Alarms, faults and system events

7.2.1.1 LEDs

The 3RM1 motor starter provides device status information via four LEDs:

LED	Displayable colors
DEVICE	Red/green/yellow
FAILURE	Red
STATE	Green
RESET MODE	Green

7.2.1.2 Status indicators of the 3RM1 motor starter

The 3RM1 motor starter indicates the status messages for operation, fault and test via the LEDs:

Operation

LED				Fault signaling output	Reaction of feedback circuit	Device status / operating mode
DEVICE	FAILURE	STATE	RESET MODE			
Green	Off	Off	Green/off*	-	-	- Control supply voltage (A1, A2) present; - Motor off; - No fault
Green	Off	Green	Green/off*	-	-	- Motor on; - No fault
X	X	X	Green	-	-	RESET MODE: Automatic
X	X	X	Off	-	-	RESET MODE: Manual

X = LED status not relevant; * = depending on the selected RESET MODE

Fault

LED				Fault signaling output	Reaction of feedback circuit	Possible cause/corrective measure
DEVICE	FAILURE	STATE	RESET MODE			
Yellow	Red	Off	Green/off*	Active	-	- Overload trip motor protection - Motor blocks during starting or operation
Flashing yellow	Red	Off	Green/off*	Active	-	- No current flow despite ON command: Check whether the connected load fulfills the minimum current (Page 61). - Phase failure: Check whether all three phases are applied.
Flashing red	Red	Off	Green/off*	Active	Active	Malfunction: For information about correcting faults, please refer to the section "Fault rectification (Page 116)".

* = depending on the selected RESET MODE

Note

Difference between Standard and Failsafe

3RM11 Failsafe and 3RM13 Failsafe motor starters signal a phase failure immediately.

3RM10 and 3RM12 Standard motor starters signal a phase failure after 5 s.

Test

LED				Fault signaling output	Reaction of feedback circuit	Device status / operating mode	
DEVICE	FAILURE	STATE	RESET MODE				
Yellow	Red	Green	Green			Self-test: Indicator test	2 s
Flickering red						Current flow test: Current not flowing	3 s
Flashing red						Current flow test: Current flowing	
Flashing yellow	Red	Off	Off	Active		Self-test: Overload tripping	>5 s
Flashing red	Red	Off	Off	Active	Active	Overload trip failed	

7.2.1.3 Signaling faults on external I/O

Fault signaling output

Faults are signaled via relay outputs as well as via the LEDs.

In the event of a fault, the fault signaling output becomes active, contact (95-98) closes and contact (95-96) opens. The fault signaling output resumes its original state (contact 95-98 open, contact 95-96 closed) when there are no faults and the control supply voltage is not present.

The faults that result in pick-up of the relays are listed in the chapter "Status indicators of the 3RM1 motor starter (Page 114)".

Note

Behavior in the event of a safety-related shutdown

The fault signaling output resumes its original state in the event of a safety-related shutdown via the control supply voltage.

Feedback output

The Standard motor starters with a control voltage of 24 V DC (3RM10...-AA0. and 3RM12...-AA0.) have a feedback output (OUT Run) referred to M (A2) that outputs a high level when the motor is in operation. Since the current flow is not monitored in the first five seconds after switching on the device, the feedback output outputs the high level at least for this time.

The output is non-floating and can be loaded with up to one control input of a PLC. The maximum continuous current of the output is 15 mA.

The safety-related motor starters (3RM11 Failsafe/3RM13 Failsafe) and the Standard motor starters (3RM10/3RM12) with a control voltage of 110 to 230 V/110 V DC (3RM1...-AA1.) do not have a feedback output.

7.2.1.4 Fault rectification

Malfunction

A malfunction (Page 40) is acknowledged by switching the control supply voltage back on and then pressing the "TEST/RESET/RESET MODE" pushbutton.

The following malfunctions can occur:

- Damage to the power semiconductors
- Damage to the relay contacts
- Faults in the control electronics

 DANGER**Checking the safety function**

Will result in death, serious physical injury, or damage to property.

Check the safety function after acknowledging a disruption in the case of 3RM11/3RM13 Failsafe motor starters.

Note

If a malfunction is repeatedly indicated, the 3RM1 motor starter is defective and must be replaced.

Other faults

You acknowledge all other faults using the "TEST/RESET/RESET MODE" button. As an alternative, in the case of the 3RM10/3RM12 Standard motor starters, you can use the RESET control input IN3 for acknowledgment. You will find information about setting the reset method in the chapter Setting the RESET method (Page 109).

Note

If the possible fault causes have been rectified and resetting is still not possible, the 3RM1 motor starter must be replaced.

Service and maintenance

8.1 Maintenance and service

The 3RM1 motor starters are maintenance-free.

The motor starter indicates a detected malfunction with LEDs and the fault output.

Note

If a malfunction is repeatedly indicated, the 3RM1 motor starter is defective and must be replaced.

You will find information about the status indicators of the 3RM1 motor starter in the chapter Alarms, faults and system events (Page 114).

8.2 Device replacement

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

Turn off and lock out all power supplying this device before working on this device.

When replacing a device, you do not need to re-wire it. The terminals can be disconnected from the defective device and then connected to the new device.

Replace device

WARNING

Hazardous voltage

Can result in death, serious physical injury, or damage to property.

You must run a function test of the system after replacing the 3RM11 Failsafe and 3RM13 Failsafe safety-related motor starters.

To ensure the safety of the system, in the case of 3RM11 Failsafe and 3RM13 Failsafe safety-related motor starters a full function test must be conducted and a successful result obtained after a device has been replaced.

Note

A defective device must only be replaced with a device that has exactly the same article number.

1. Disconnect the defective device at the terminals.
2. Remove the defective device.
3. Mount the new device.
4. Connect the new device at the terminals.

You can find more information on disassembling/mounting and disconnecting/connecting in the chapters "Mounting (Page 77)" and "Connection (Page 95)".

Technical data

9.1 Data sheet

You can find all the technical data of the product in the Siemens Industry Online Support (<https://support.industry.siemens.com/cs/ww/en/ps/16311/td>).

1. Enter the full article number of the desired device in the "Product" field, and confirm with the Enter key.
2. Click the "Technical data link.

The screenshot shows the Siemens Industry Online Support search interface. At the top, there is a "Product tree" dropdown set to "All" and a search bar labeled "Enter keyword...". Below this, the search results are displayed. The "Product" field contains "3RV2031-4BA10" with a magnifying glass icon and a close button. The "Entry type" dropdown is set to "Technical data (1)" with a close button. The "Date" field has "From" and "To" input boxes. Below the search filters, there is a section for the product details. On the left, there is a placeholder image for the product. To the right of the image, the product name "3RV2031-4BA10" is displayed in red, followed by its description: "CIRCUIT BREAKER, SCREW TYPE, 20 A CIRCUIT BREAKER SIZE S2 FOR MOTOR PROTECTION, CLASS 10, A-RELEASE 14...20A, N-RELEASE 20A, SCREW TERMINAL, STANDARD BREAKING CAPACITY". At the bottom of this section, there are two links: "> Product details" and "> Technical data", with the latter being highlighted by a red box.

See also

- Device versions (Page 26)
- Rated operational current and derating (Page 55)
- Ambient conditions (Page 60)

9.2 Connection cross sections

Connectable cable cross-sections depend on the motor starter's connection system.

Connection system	Article number
Screw connection	3RM1...-1AA.4
Push-in spring-type terminals	3RM1...-2AA.4
Hybrid connection method Control circuit realized as push-in spring-loaded terminal and main circuit as screw terminal	3RM1...-3AA.4

The table below contains the suitable conductor cross-sections for the control circuit and signaling contacts (= auxiliary contacts) and for the main circuit (= main contacts).

	3RM1...-1....	3RM1...-2....	3RM1...-3....
Type of the connectable conductor cross-section			
• for main contacts			
– solid	1x (0,5 ... 4 mm ²), 2x (0,5 ... 2,5 mm ²)	1x (0,5 ... 4 mm ²)	1x (0,5 ... 4 mm ²), 2x (0,5 ... 2,5 mm ²)
– finely stranded			
– with conductor end processing	1x (0,5 ... 2,5 mm ²), 2x (0,5 ... 1,5 mm ²)	1x (0,5 ... 2,5 mm ²)	1x (0,5 ... 2,5 mm ²), 2x (0,5 ... 1,5 mm ²)
– without conductor final cutting	—	1x (0,5 ... 4 mm ²)	—
• for auxiliary contacts			
– solid	1x (0,5 ... 2,5 mm ²), 2x (1,0 ... 1,5 mm ²)	1x (0,5 ... 1,5 mm ²), 2x (0,5 ... 1,5 mm ²)	1x (0,5 ... 1,5 mm ²), 2x (0,5 ... 1,5 mm ²)
– finely stranded			
– with conductor end processing	1x (0,5 ... 2,5 mm ²), 2x (0,5 ... 1 mm ²)	1x (0,5 ... 1,0 mm ²), 2x (0,5 ... 1,0 mm ²)	1x (0,5 ... 1,0 mm ²), 2x (0,5 ... 1,0 mm ²)
– without conductor final cutting	—	1x (0,5 ... 1,5 mm ²), 2x (0,5 ... 1,5 mm ²)	1x (0,5 ... 1,5 mm ²), 2x (0,5 ... 1,5 mm ²)
• for AWG conductors ¹⁾			
– for main contacts	1x (20 ... 12), 2x (20 ... 14)	1x (20 ... 12)	1x (20 ... 12), 2x (20 ... 14)
– for auxiliary contacts	1x (20 ... 14), 2x (18 ... 16)	1x (20 ... 16), 2x (20 ... 16)	1x (20 ... 16), 2x (20 ... 16)

¹⁾ All AWG data applies to solid conductors.

9.3 Number of starting operations

Permissible number of starting operations per hour 1/h

Parameter: on-load factor = 50 %

The number of starting operations 1/h is calculated for an effective motor current of 100 %, and the motor starter's actual performance can be higher.

Motor starter		3RM1.07...	3RM1.02...	3RM1.01...	Unit
Max. rated operating current I _e		7 A	2 A	0.5 A	
Motor					
Starting current	Starting time				
250 %	100 ms	1000	3300	3600	1/h
400 %	100 ms	1000	1000	1000	1/h
400 %	300 ms	400	400	400	1/h
400 %	500 ms	220	220	220	1/h
500 %	100 ms	750	750	750	1/h
500 %	300 ms	250	250	250	1/h
500 %	500 ms	150	150	150	1/h
600 %	100 ms	500	500	500	1/h
600 %	300 ms	150	150	150	1/h
600 %	500 ms	100	100	100	1/h
700 %	100 ms	360	360	360	1/h
700 %	300 ms	100	100	100	1/h
700 %	500 ms	60	60	60	1/h

9.4 Overload protection/device protection characteristic

Overload protection

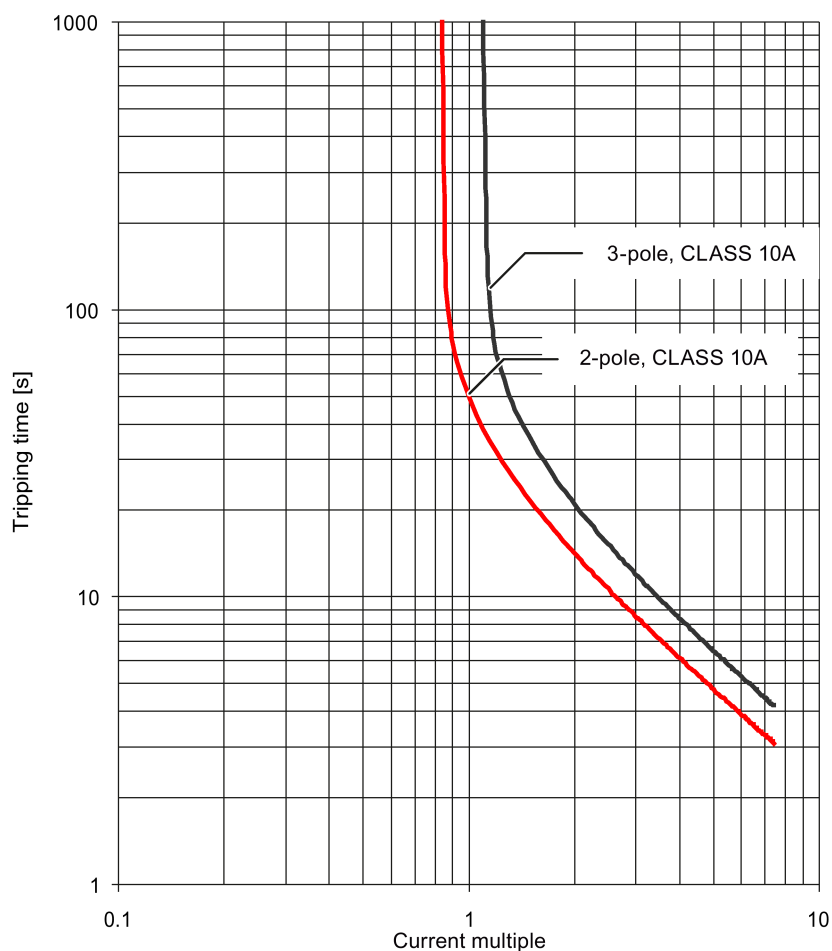
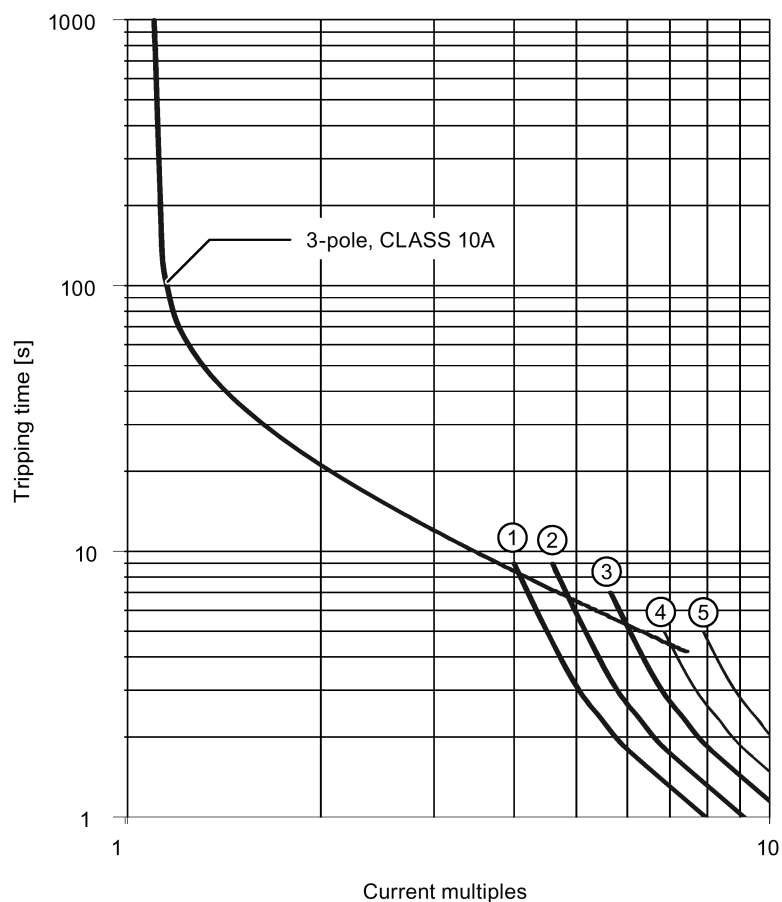


Figure 9-1 Motor overload protection characteristic

Device protection against overload trips at current setting values from 4.6 A. This results in shorter tripping times; see figure below.

Equipment protection

In addition to the motor protection function, the 3RM1 motor starters also protect themselves against overload. This means that in the case of 3RM1 motor starters with a rated operational current of 7 A, an overload trip may occur in the upper current range before the motor protection trips. If, for example, 8-fold current flows in the case of an unloaded motor model and a set current of 7 A, tripping takes place after approximately one second.



- ① Device protection 7 A
- ② Device protection 6.1 A
- ③ Device protection 5.2 A
- ④ Device protection 4.6 A
- ⑤ Device protection 4.0 A

Figure 9-2 Device protection characteristic

Resistive loads

The tripping characteristic shown does not apply for resistive loads at a setting value of 10 A.

In this case, tripping takes place as follows:

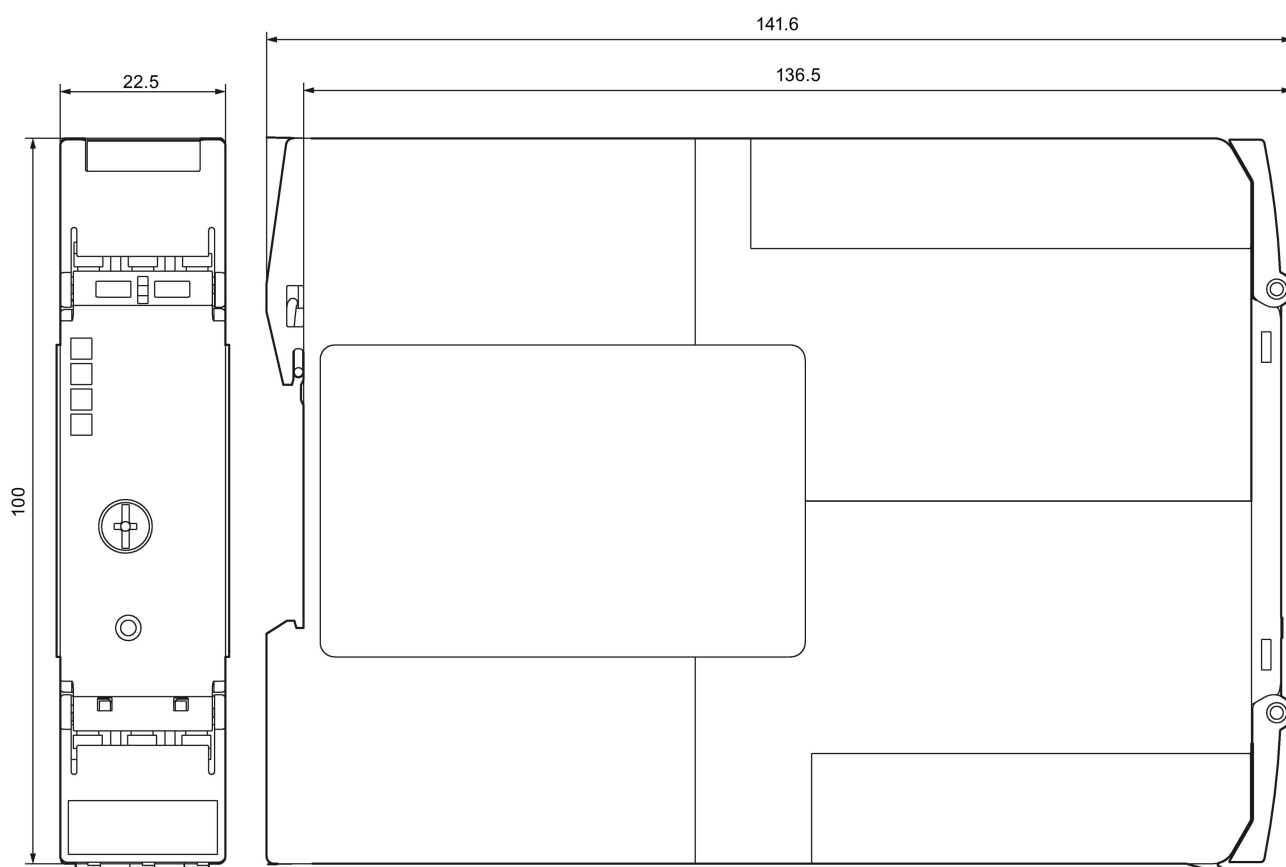
Table 9- 1 Tripping for resistive loads at a setting value of 10 A.

	Tripping after
10 A < load current < 15 A	20 s
Load current > 15 A	1 s

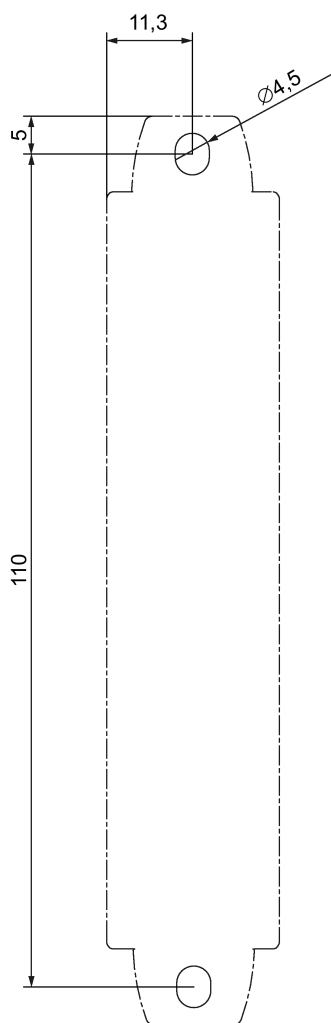
Dimension drawings

10.1 3RM1 dimension drawings

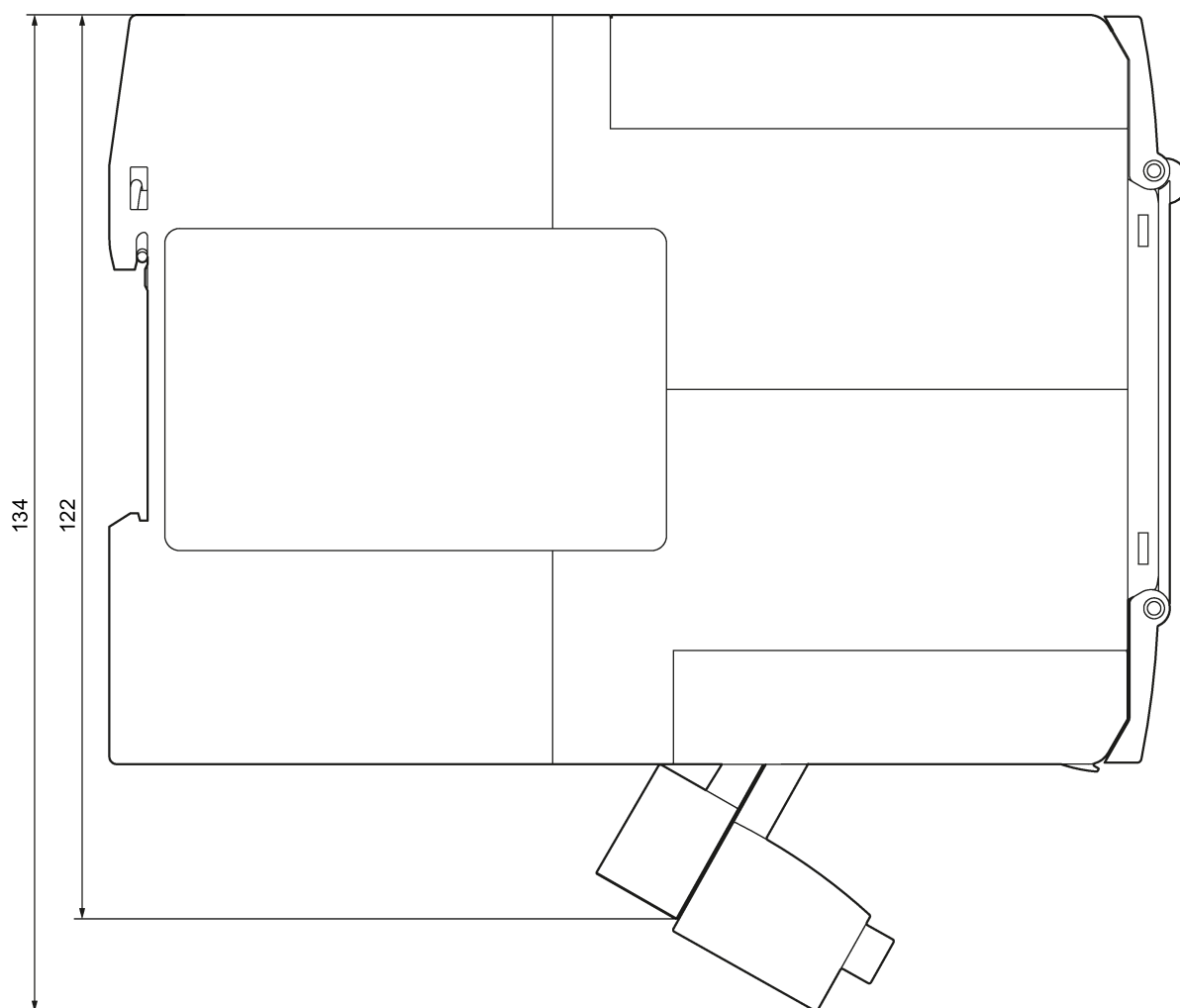
Dimension drawing for 3RM1 motor starter



Drilling diagram for push-in lugs for wall mounting

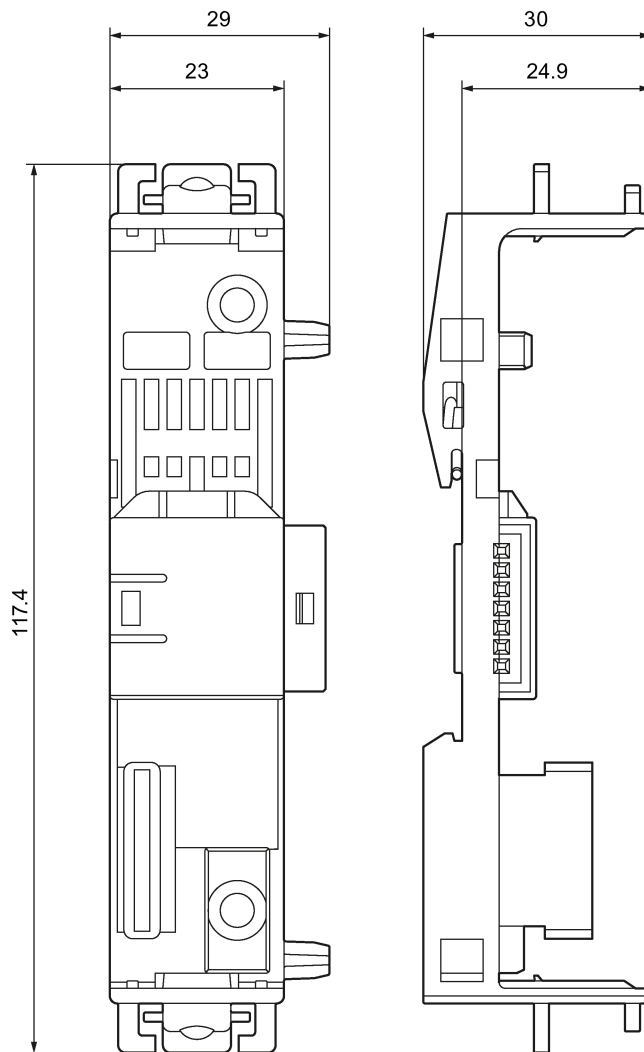


Dimension drawing for 3RM1 motor starter with infeed terminal and system

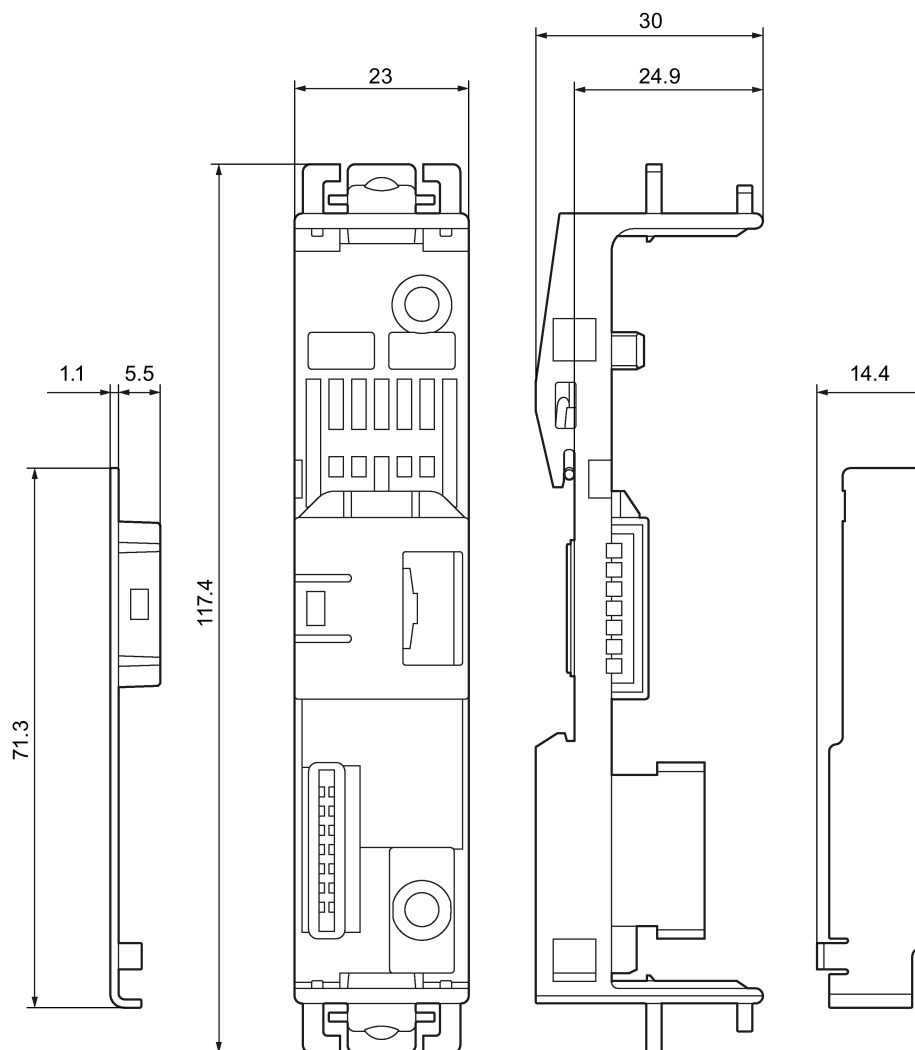


10.2 Dimension drawings for 3RM1 device connectors

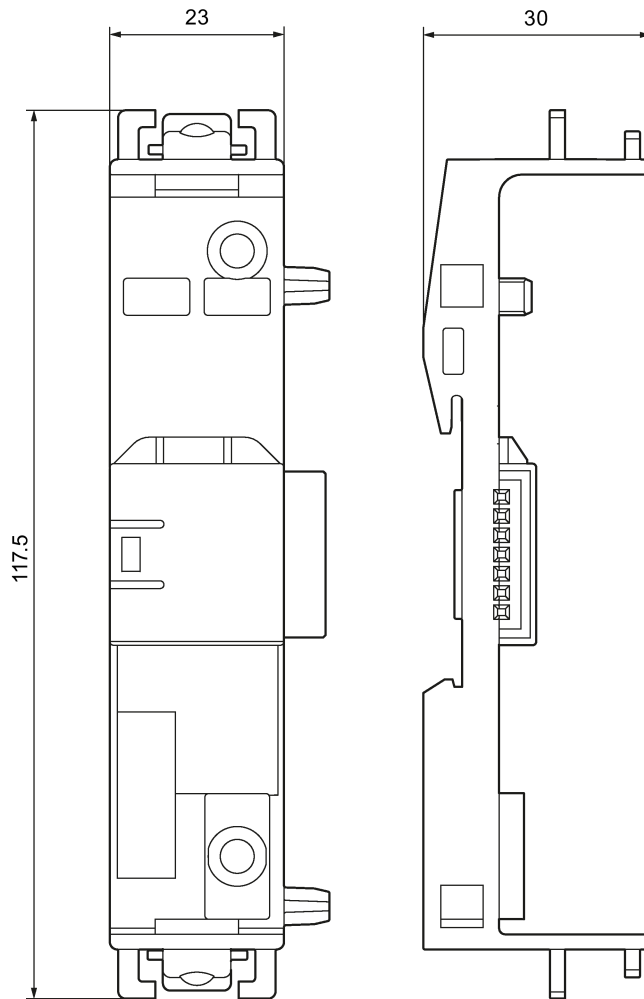
Dimension drawing of device connectors for 3RM1 motor starter



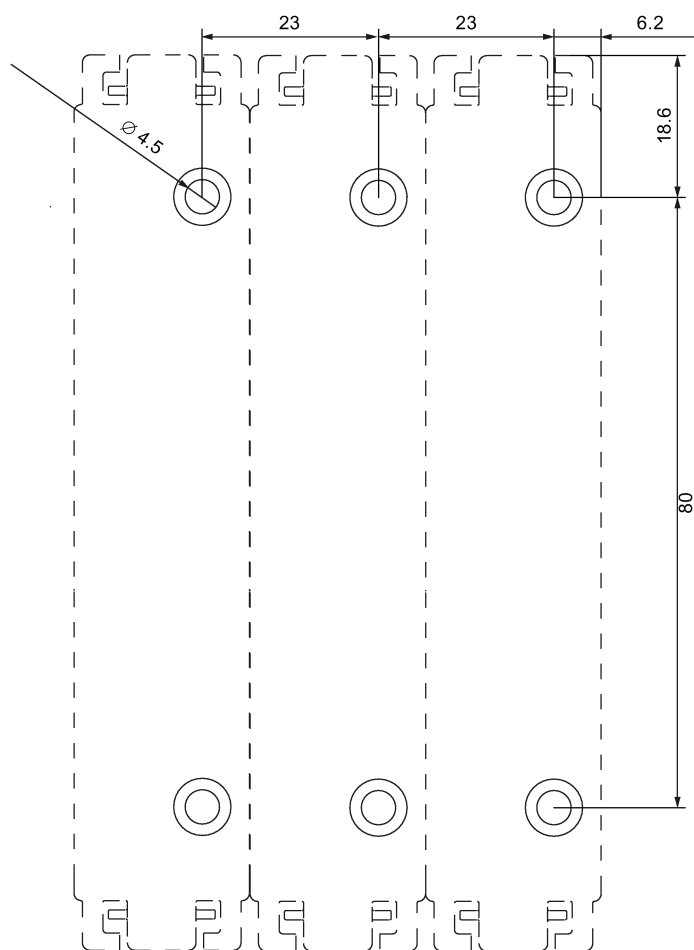
Dimension drawing of device termination connectors for 3RM1 motor starter



Dimension drawing of device connectors for signal loop-through



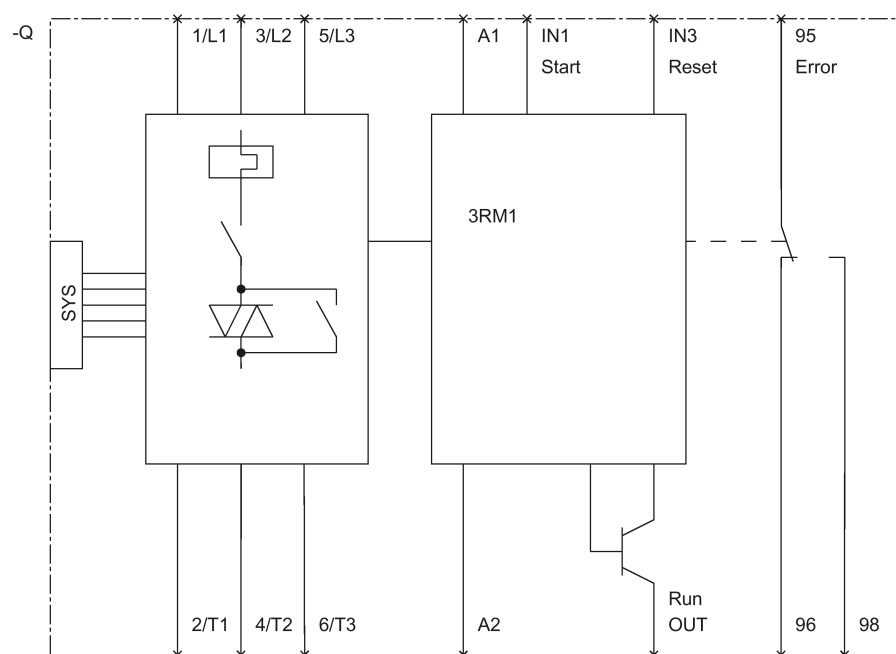
Drilling diagram for device connectors with 22 mm width



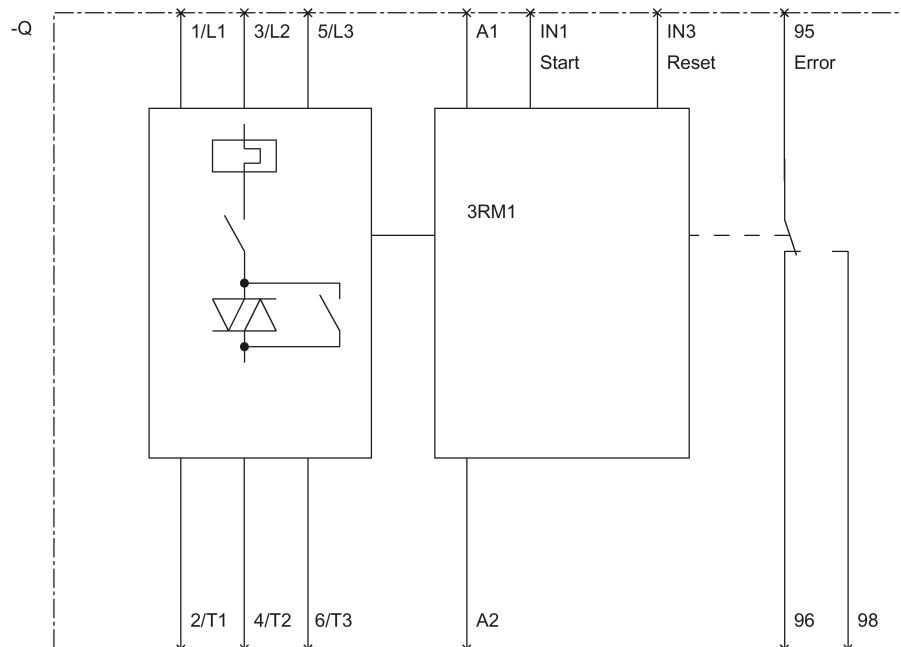
Circuit diagrams

11.1 3RM10 circuit diagrams (direct-on-line starter, Standard)

Versions with 24 V DC control supply voltage



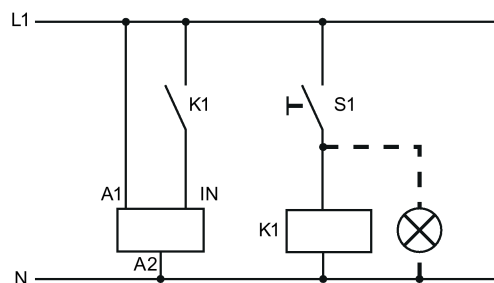
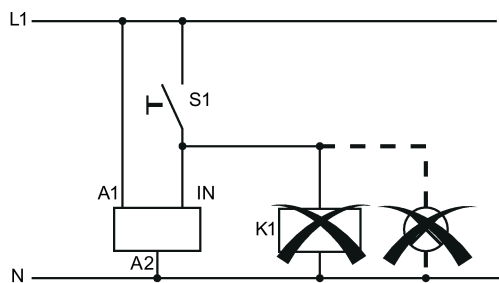
Versions with 110 ... 230 V AC; 110 V DC control supply voltage



Note

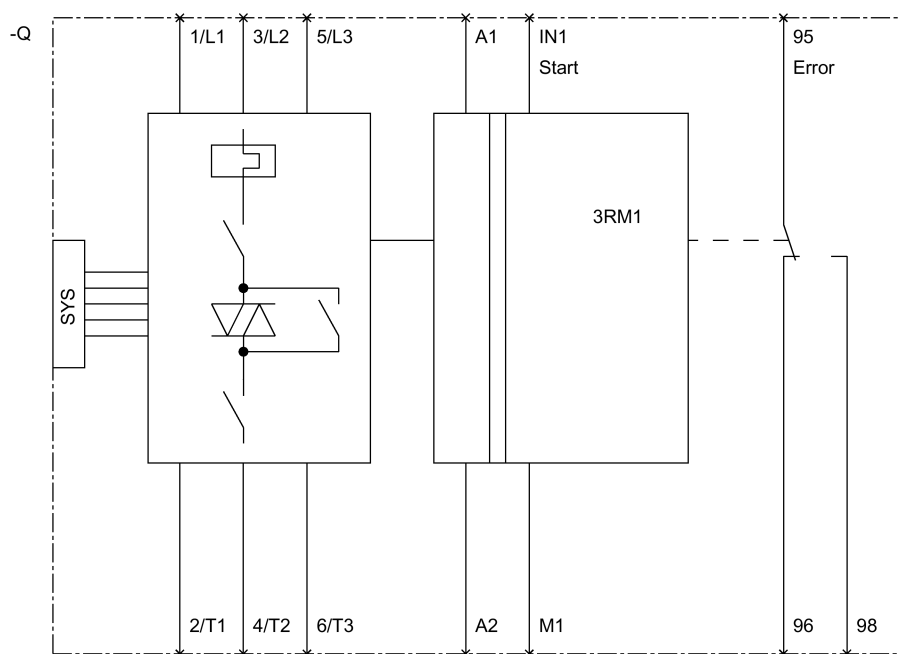
Parallel loads in the case of motor starters with product version E01

In the case of motor starters with product version E01, no parallel loads can be connected to the IN terminal; see diagram below. This is admissible as of product version E02.

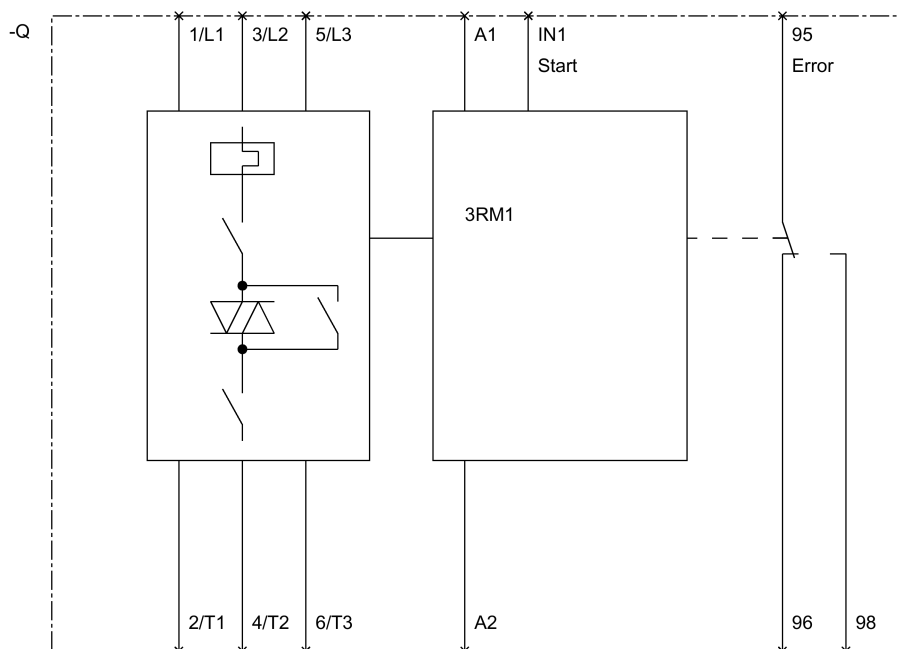


11.2 3RM11 circuit diagrams (direct-on-line starter, Failsafe)

Versions with 24 V DC control supply voltage

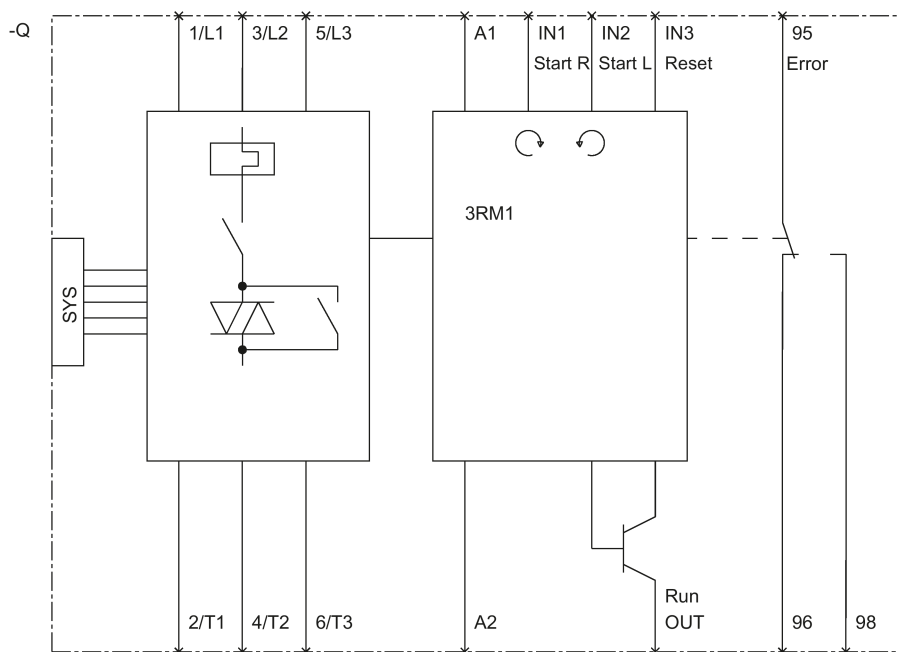


Versions with 110 ... 230 V AC; 110 V DC control supply voltage

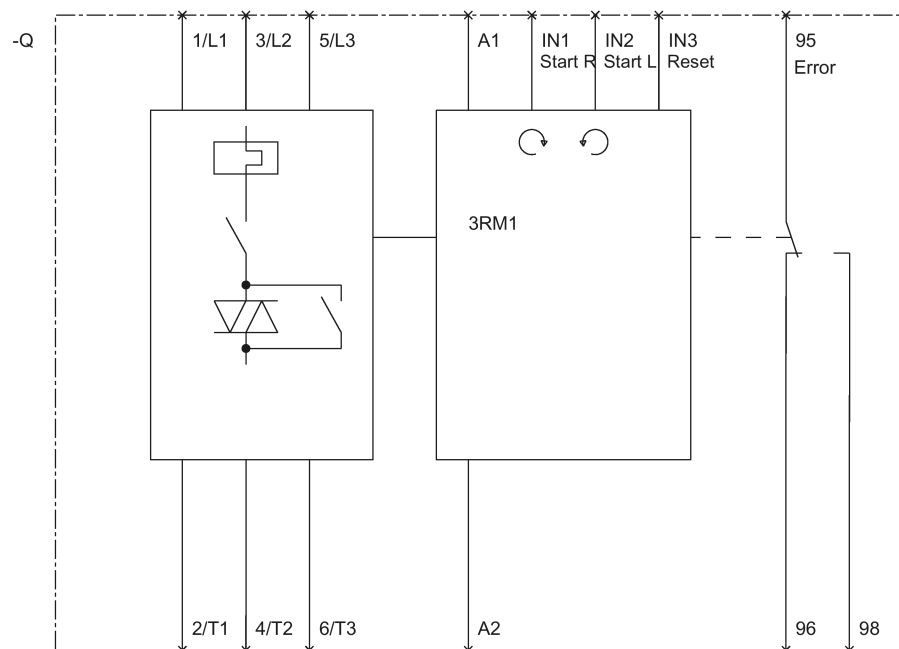


11.3 3RM12 circuit diagrams (reversing starter, Standard)

Versions with 24 V DC control supply voltage

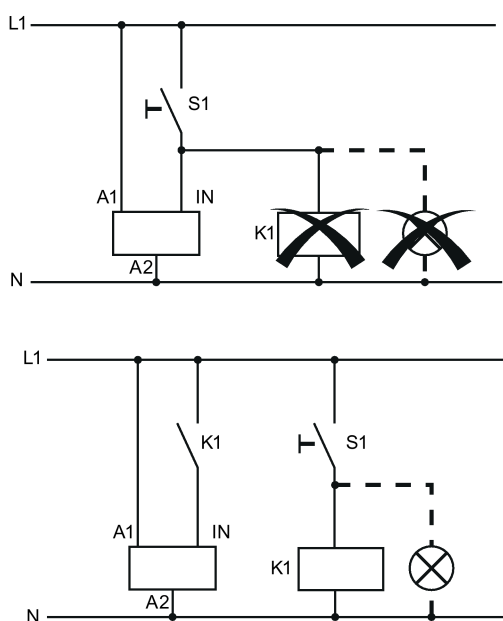


Versions with 110 ... 230 V AC; 110 V DC control supply voltage



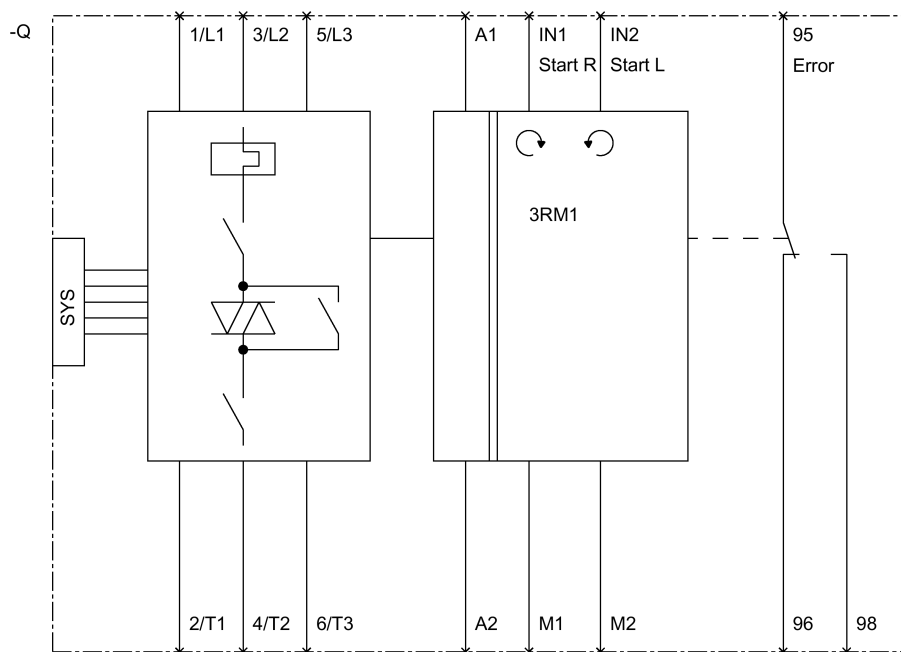
Note**Parallel loads in the case of motor starters with product version E01**

In the case of motor starters with product version E01, no parallel loads can be connected to the IN terminal; see diagram below. This is admissible as of product version E02.

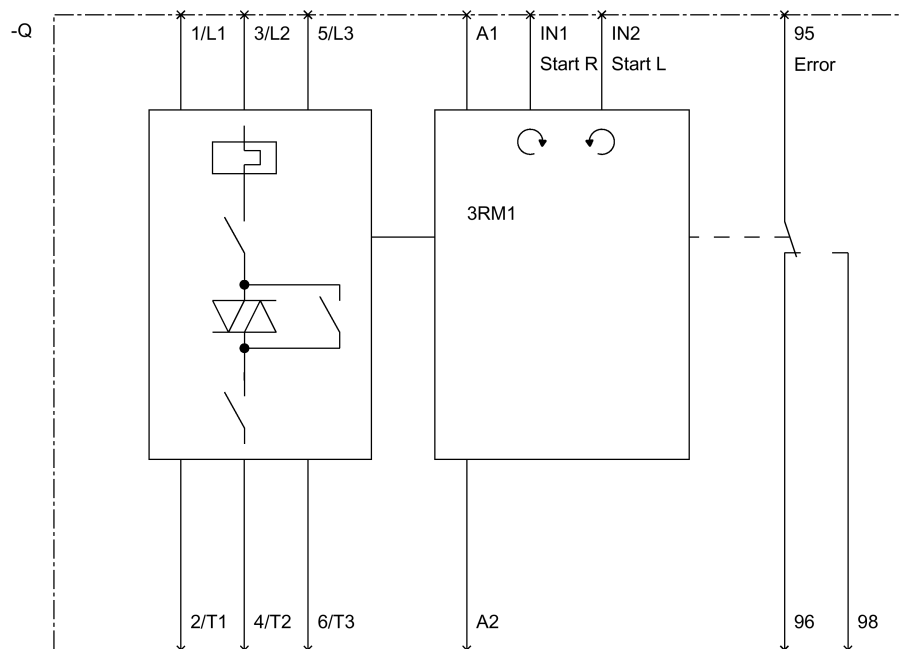


11.4 3RM13 circuit diagrams (reversing starter, Failsafe)

Versions with 24 V DC control supply voltage



Versions with 110 ... 230 V AC; 110 V DC control supply voltage



Typical circuits

A.1 Typical circuits for 3RM1

A.1.1 Direct-on-line starter 24 V DC with switch operation

The 3RM1 motor starter is controlled with a switch. The 3RM1 motor starter is operated as a direct-on-line starter.

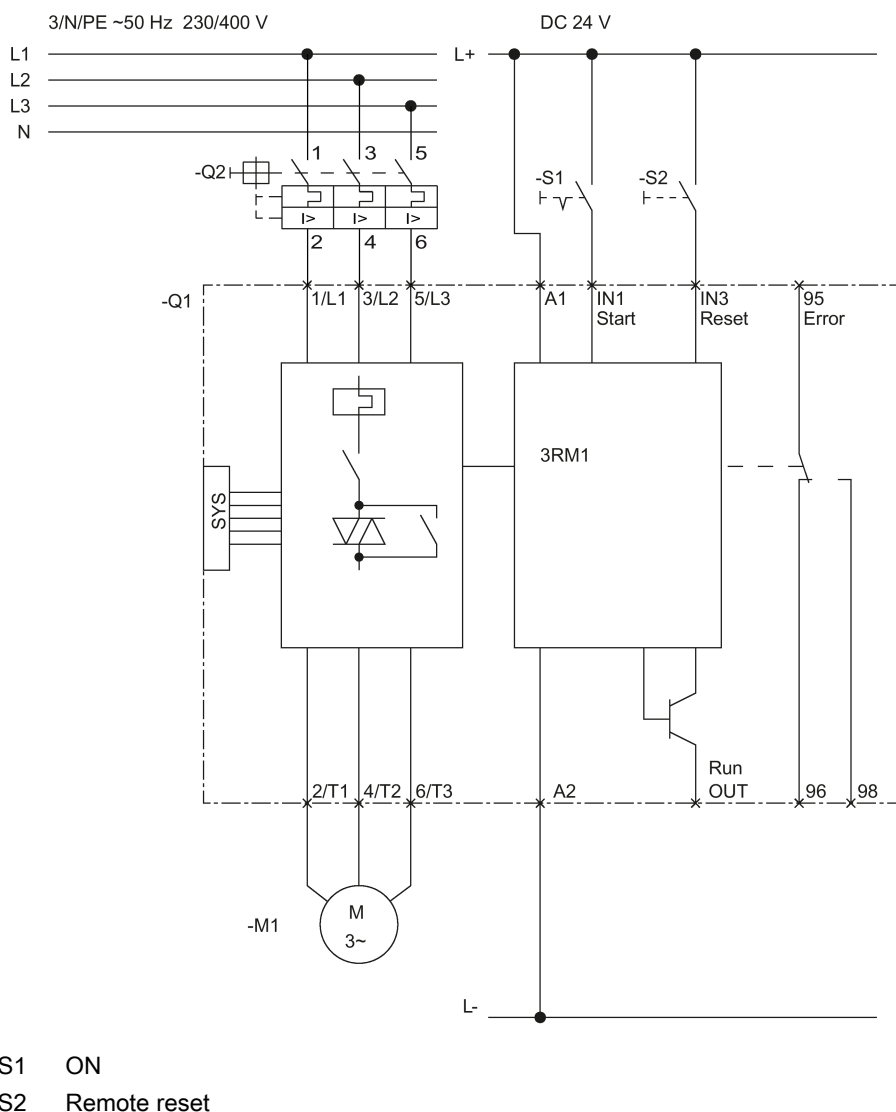


Figure A-1 Standard 24 V DC direct-on-line starter with switch operation

A.1.2 Direct-on-line starter 24 V DC with switch operation and 230 V brake

The current for the brake facility is drawn via only one phase. Connection is via T2.

Note

This configuration is admissible for standard variants 3RM10 and 3RM12 only.

This operating mode is not suitable for 3RM11 Failsafe and 3RM13 Failsafe motor starters since there is a connection to the neutral conductor.

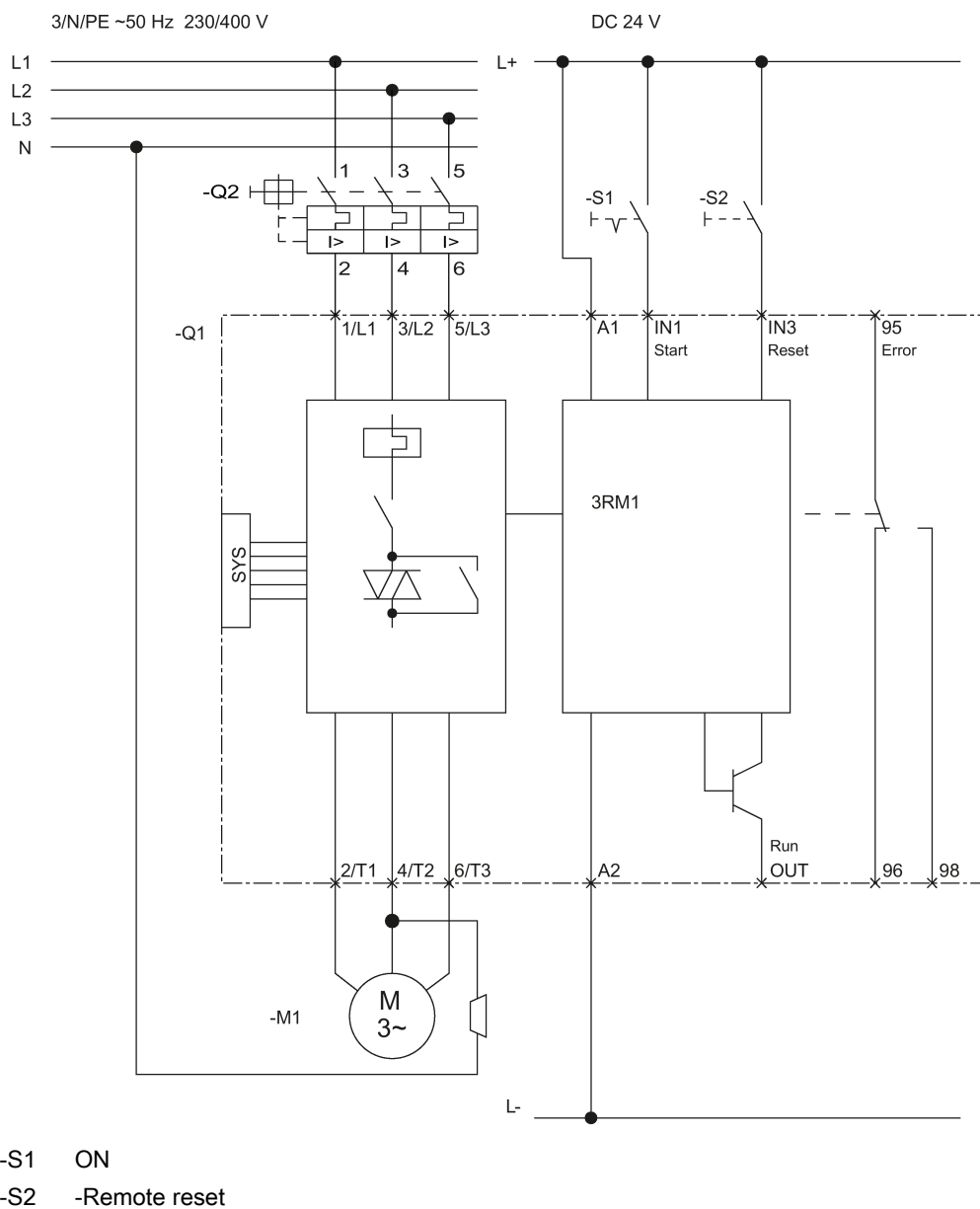


Figure A-2 Standard 24 V DC direct-on-line starter with switch operation and 230 V brake

A.1.3 Direct-on-line starter 24 V DC with switch operation and 400 V brake

The current for the brake facility is drawn via two phases. Connection is via T1 and T3.

Make allowance for the additional current in the current setting value.

The configuration is shown based on the example of a standard motor starter. The configuration with a 400 V brake is also admissible for the Failsafe variants 3RM11 and 3RM13. The notes on safety-related shutdown (Page 149) must be observed.

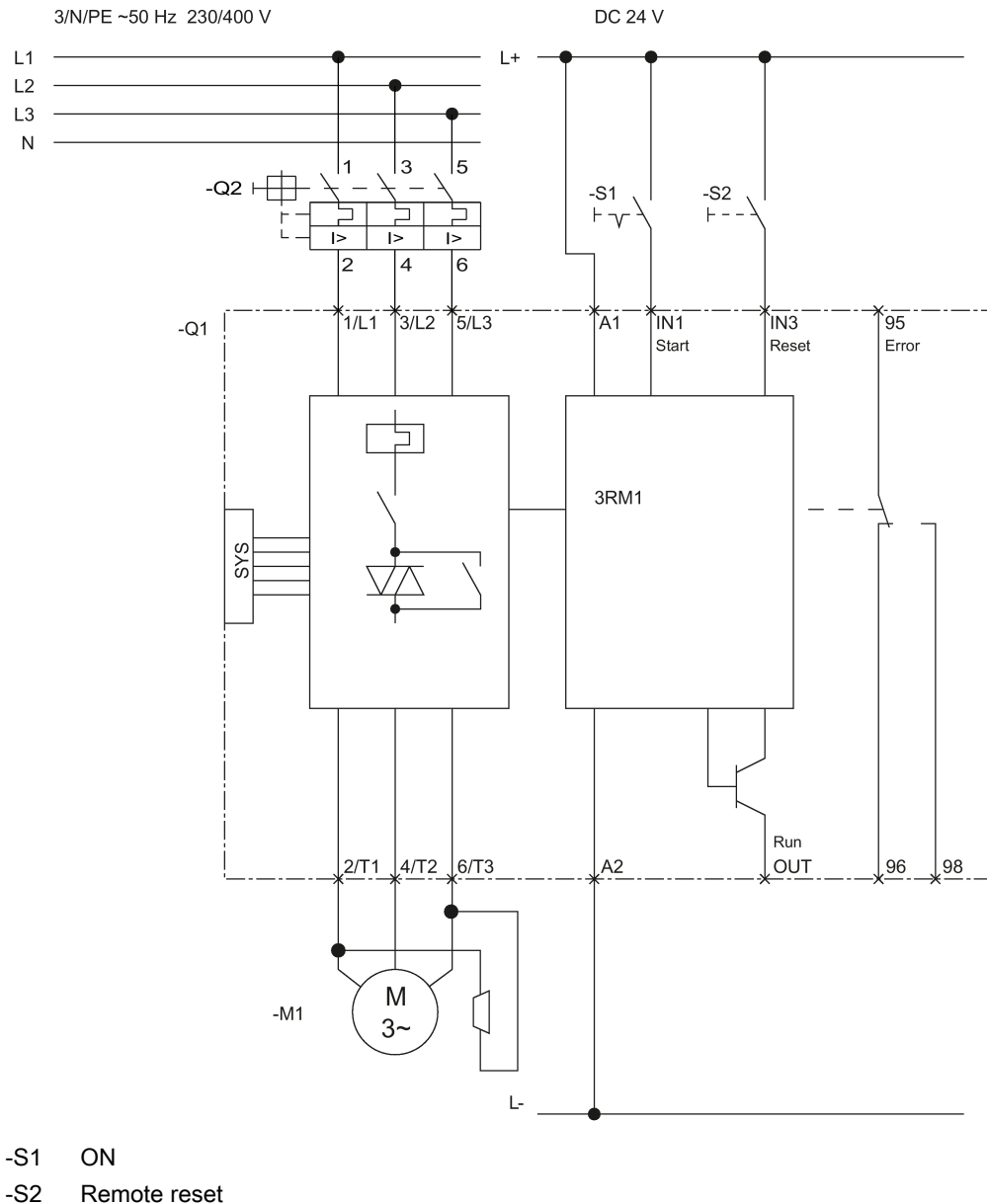


Figure A-3 Standard 24 V DC direct-on-line starter with switch operation and 400 V brake

A.1.4 Direct-on-line starter 24 V DC with switch operation and single-phase motor

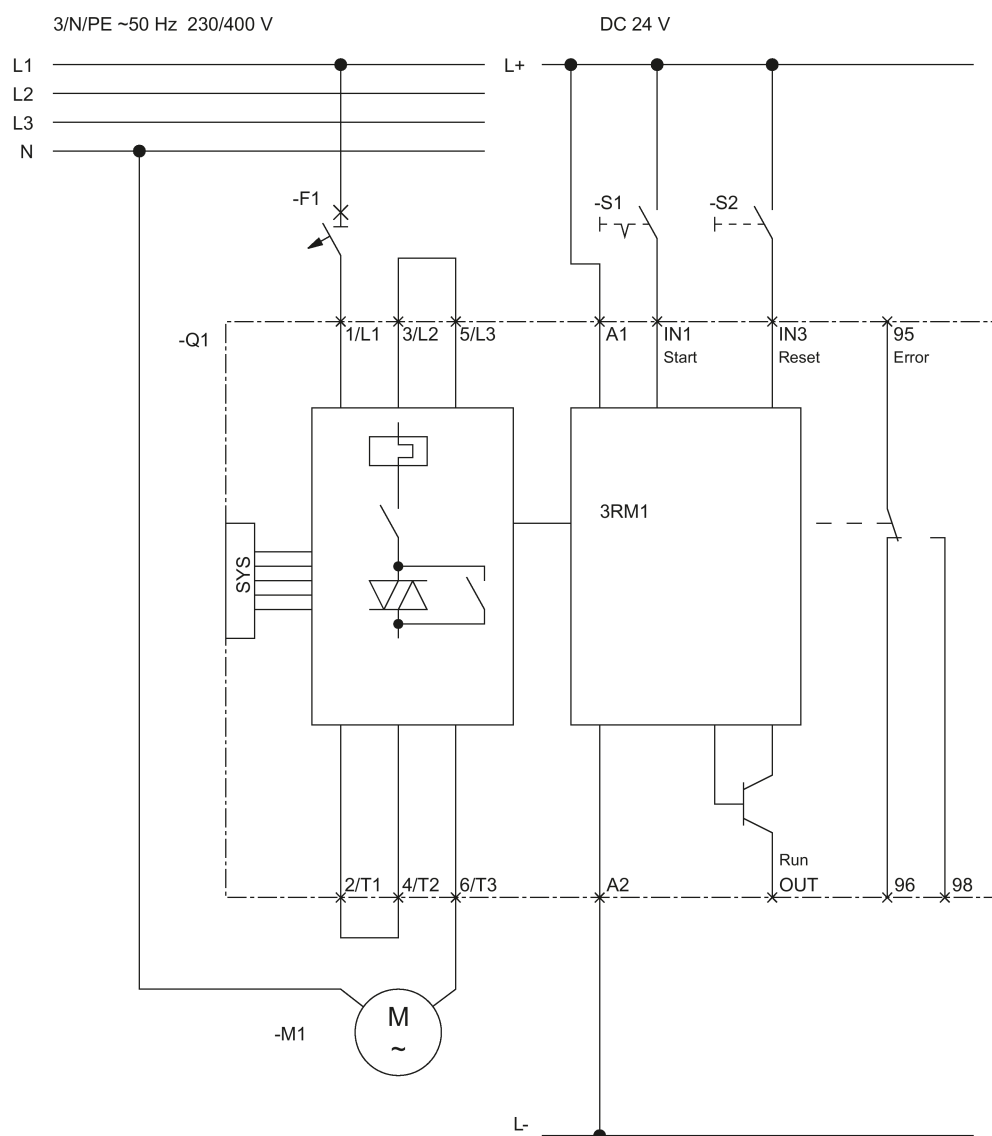
The current is drawn via only one phase. The motor is connected at T3. The second connection is directly at the neutral conductor. On the 3RM1 motor starter, T1 is wired direct with T2, and L2 direct with L3.

The configuration is shown based on the example of a standard motor starter. The configuration for single-phase loads is also admissible for the Failsafe variants 3RM11 and 3RM13. The notes on safety-related shutdown (Page 149) must be observed.

NOTICE
Damage to property can occur
Connection IN2 on reversing starters must not be used in the case of single-phase motors.

NOTICE
Damage to property can occur from operating capacitive loads
When using capacitive loads, the switching components in the 3RM1 motor starter can be damaged by excessively high making currents.
Do not operate capacitive loads such as frequency converters on the 3RM1 motor starter. Operation of single-phase capacitor motors is admissible.

In this operating mode, switching on and off is carried out only with connection IN1.



-S1 ON
-S2 Remote reset

Figure A-4 Standard direct-on-line starter 24 V DC with switch operation and single-phase motor

A.1.5 Direct-on-line starter with group protection, 24 V DC and PLC operation

Several 3RM1 motor starters can be connected by means of 3-phase busbars on the main current side. Control takes place via digital outputs.

The configuration is shown based on the example of standard motor starters. In the main circuit, this configuration is also admissible for the Failsafe variants 3RM11 and 3RM13. The notes on safety-related shutdown (Page 149) must be observed.

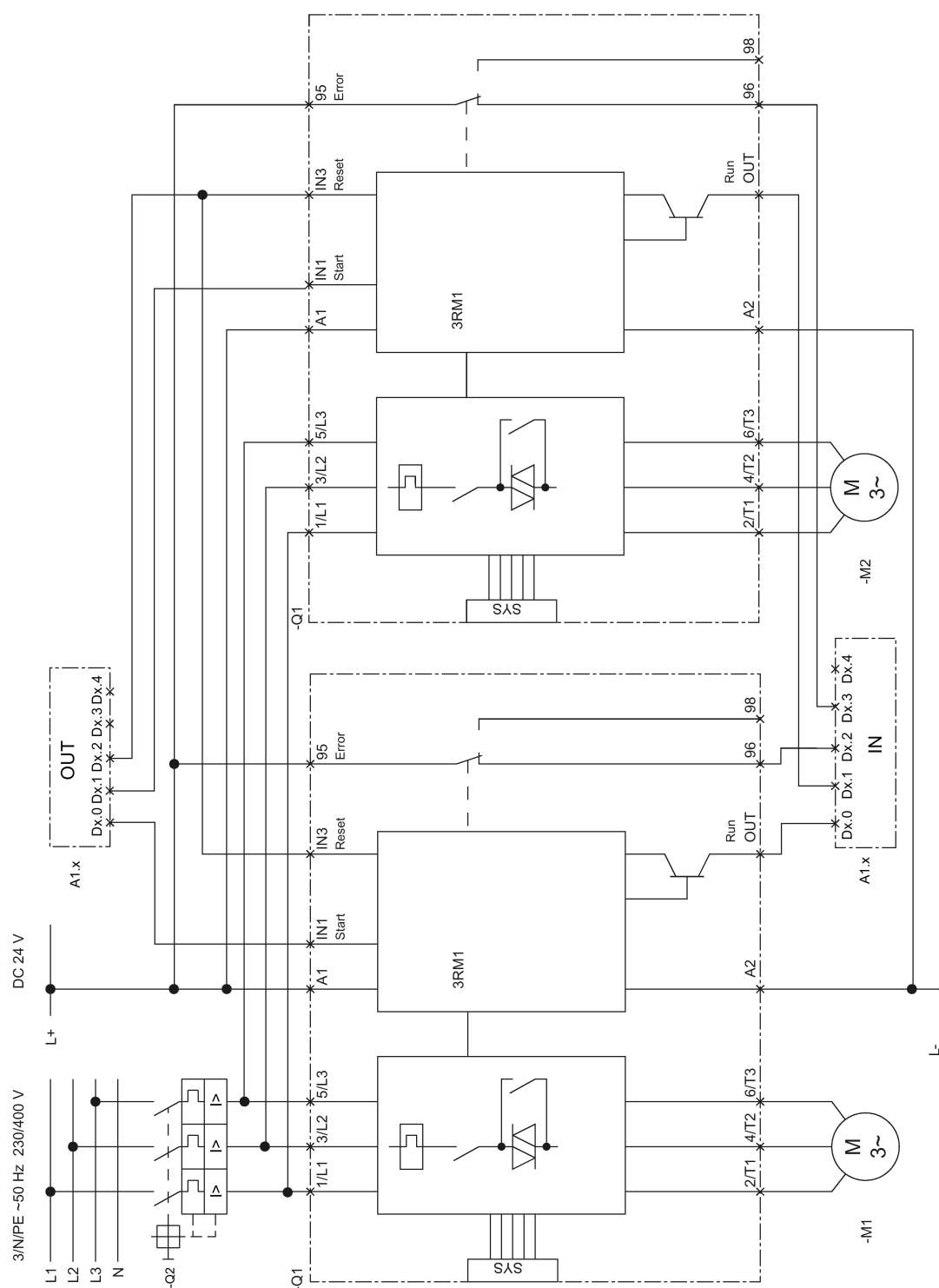


Figure A-5 Standard direct-on-line starter with group protection, 24 V DC and PLC operation

A.1.6 Reversing starter 24 V DC with PLC operation

The motor is controlled via a PLC. The 3RM1 motor starter is operated as a reversing starter.

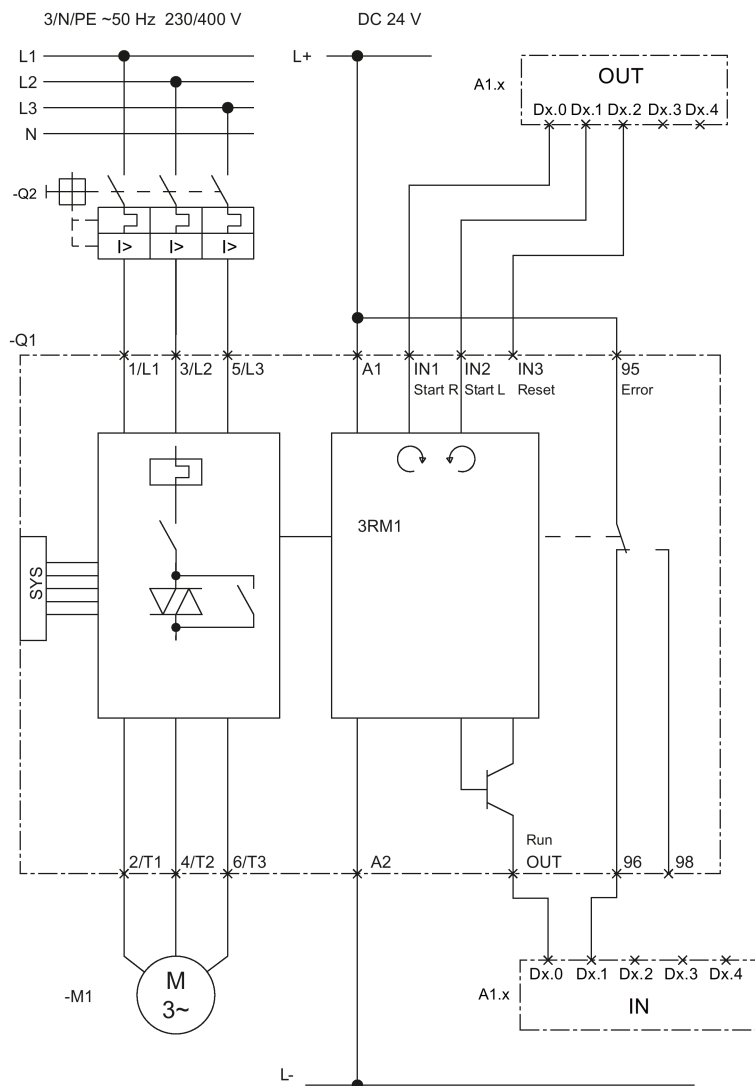


Figure A-6 Reversing starter 24 V DC with PLC operation

A.2 Typical circuits for safety-related shutdown

A.2.1 General safety notes

Note

SILCL 3 in accordance with EN 62061/PL e in accordance with EN ISO 13849-1

The 3RM11/3RM13 motor starters are designed in such a way as to allow implementation of applications up to SILCL 3 in accordance with EN 62061 and PL e/Cat. 4 in accordance with EN ISO 13849-1.

To achieve SILCL 3/PL e/Cat. 4 for the safety-related application, all components of the safety functions (detecting/evaluating/reacting) must be designed accordingly.

Note

Safe state

The basis of the safety function is the definition of the safe state. In the case of the 3RM11 and 3RM13 Failsafe motor starters, this is the "OFF state".

DANGER

Hazardous voltage on the motor

Will result in death, serious physical injury, or damage to property.

Health hazard from automatic hot restart.

The safety function must be checked after every tripping of the motor starter protector (also in the case of overload).

After every short circuit in the case of short-circuit protection according to **type of coordination 1**, a 3RM1 motor starter is defective.

Replace a 3RM1 motor starter including the terminals after a short circuit in the case of short-circuit protection according to type of coordination 1.

With 3RM11/3RM13 Failsafe motor starters, you must check the safety function in addition to replacing the unit.

With 3RM11/3RM13 Failsafe motor starters, also check the safety function after a short circuit in the case of short-circuit protection according to **type of coordination 2**.

DANGER

Hazardous voltage

Will result in death, serious physical injury, or damage to property.

To ensure protection against an electric shock hazard with the hinged cover open at the signaling contacts 95, 96, 98 at a voltage of ≥ 50 V, screw in all terminal screws that are not needed to clamp conductors.

 WARNING

Function test interval for 3RM11/3RM13 Failsafe motor starters

Can result in death, serious physical injury, or damage to property.

In continuous operation, the key safety values apply in the case of a function test interval (state change of the outputs) ≤ 1 year.

Annual function test

- Actuate the connected sensors.
- Check their effect on the safe evaluation unit and the downstream actuators.
- Activate the safe evaluation unit via the connected sensors.
- Check their effect on the safe evaluation unit and the downstream actuators.
- Defective devices must be replaced.

 WARNING

Bypassing the safety function in the event of a fault in the case of 3RM11/3RM13 Failsafe motor starters with 110 to 230 V AC/110 V DC control supply voltage

Can result in death, serious physical injury, or damage to property.

The control signal for the control inputs in the case of 3RM11/3RM13 Failsafe motor starters with 110 to 230 V AC control supply voltage must come from A1. Otherwise, the safety function is bypassed in the event of a fault. Thus, only relay outputs are admissible when using a PLC.

Do not use a separate control voltage. Only use the relay outputs when using a PLC.

 WARNING

Bypassing the safety function when using device connectors

Can result in death, serious physical injury, or damage to property.

When the product is operated with a 3SK safety relay and a device connector, the supply voltage for 3RM1 motor starters is fed in via the device connectors.

In this case, do not connect anything to terminals A1 and A2 of the 3RM1 motor starters, in order to prevent bypassing of the safety function.

 WARNING

Failure of the safety function during hybrid operation of Failsafe and Standard motor starters

Can result in death, serious physical injury, or damage to property.

Hybrid operation of 3RM10/3RM12 Standard motor starters with 3RM11/3RM13 Failsafe motor starters in safety-related applications is not admissible.

Only ever use safety-related motor starters (3RM11 Failsafe and 3RM13 Failsafe) in safety-related applications.

! WARNING**Protection against electrically conductive pollution**

Can result in death, serious physical injury, or damage to property.

The devices must be protected against conductive contamination while taking account of the ambient conditions. One way you can do this is to install the devices in a control cabinet with the appropriate degree of protection. You will find further information in IEC 60529, for example.

! WARNING**Safe functional extra low voltage**

Can result in death, serious physical injury, or damage to property.

Can result in death, serious physical injury, or damage to property.

The 3RM11/3RM13 Failsafe motor starters with a supply voltage of 24 V DC must be operated with safe functional extra-low voltage (SELV, PELV). This means these modules must only be subjected to a voltage of U_m even in the event of a fault. The following applies to these 3RM1 motor starters: **$U_m < 60.0 \text{ V}$** .

You will find more detailed information about safe functional extra-low voltage in the data sheets of the power supplies to be used.

NOTICE**Protection against electrostatic charge**

When handling and installing the 3RM11/3RM13 Failsafe motor starters, ensure that the components are protected against electrostatic charge. Changes to the system configuration and wiring are only permissible in the voltage-free state.

Connection of 3RM11/3RM13 Failsafe motor starters is only permissible when the power supply units (PELV and SELV) are switched off.

NOTICE**Noise immunity / grounding**

The following must be grounded in accordance with regulations to ensure noise immunity of the 3RM11/3RM13 Failsafe motor starters:

- PELV / SELV power supply units (please also note the documentation for the respective power supply unit in this regard).

Note**Operational faults and malfunctions in communication**

If the EMC Directive 2004/108/EC is not complied with when plants and devices are installed, communication breaks may occur.

Note

Cover the 3ZY12 device connector on the left-hand side using the cover supplied with the device termination connector.

A.2.2 3SK1 safety relay with 3RM13 motor starter via device connector

The 3RM13 Failsafe motor starter is connected to a 3SK1 safety relay via a 3ZY1 device connector.

The two directions of rotation are switched on and off operationally via IN1 and IN2, e.g. with a PLC.

The control supply voltage (L+ and L-) to the motor starters is deactivated by the 3SK1 safety relay and the system is in a safe state.

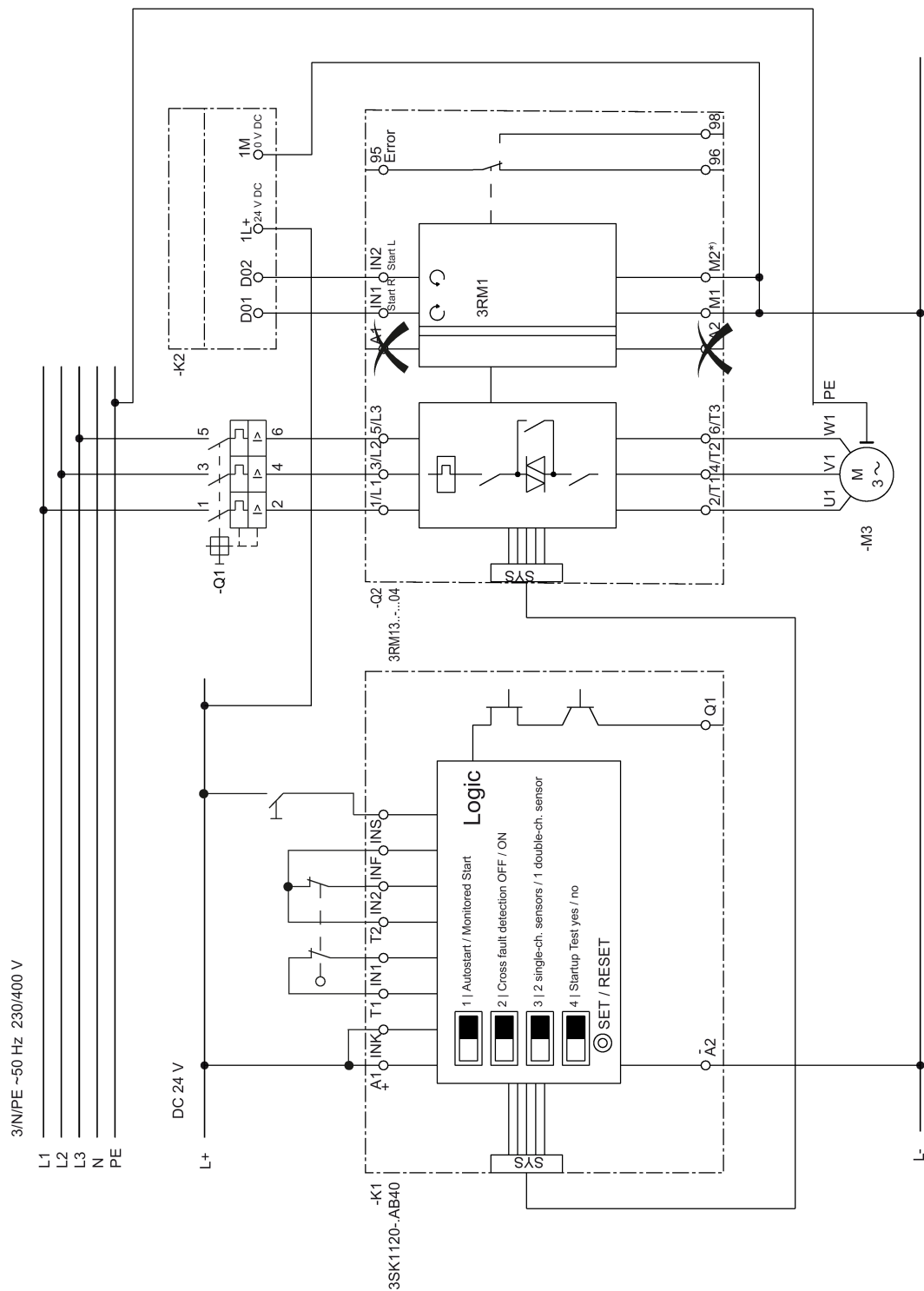
Note

SILCL 3 in accordance with EN 62061/PL e in accordance with EN ISO 13849-1

A safety-related application up to SILCL 3 to EN 62061:2005, PL e/Cat. 4 to EN ISO 13849-1 can be realized in this way.

To achieve SILCL 3/PL e/Cat. 4 for the safety-related application, all components of the safety functions (detecting/evaluating/reacting) must be designed accordingly.

WARNING
Bypassing the safety function when using device connectors
Can result in death, serious physical injury, or damage to property.
When operating with a 3SK1 safety relay and a device connector, the supply voltage for 3RM1 motor starters is established via the device connectors.
In this case, do not connect anything to terminals A1 and A2 of the 3RM1 motor starters, in order to prevent bypassing of the safety function.



*) If there is a common reference potential for the two inputs, it suffices to establish a ground connection.

If there are separate potentials or controls, both connections must be assigned.

Figure A-7 3SK1 safety relay with 3RM13 reversing starter via device connector

A.2.3 3SK1 safety relay wired with 3RM13 motor starter

The 3RM13 Failsafe motor starter (230 V supply voltage) is connected to a 3SK1 safety relay.

The two directions of rotation are switched on and off operationally via IN1 and IN2, e.g. with separate switches.



WARNING

Bypassing the safety function in the event of a fault in the case of 3RM11/3RM13 Failsafe motor starters with 110 to 230 V AC/110 V DC control supply voltage

Can result in death, serious physical injury, or damage to property.

The control signal for the control inputs in the case of 3RM11/3RM13 Failsafe motor starters with 110 to 230 V AC control supply voltage must come from A1. Otherwise, the safety function is bypassed in the event of a fault. Thus, only relay outputs are admissible when using a PLC.

Do not use a separate control voltage. Only use the relay outputs when using a PLC.

The control supply voltage (L1 and N) to the motor starters is deactivated by the 3SK1 safety relay and the system is in a safe state.

Note

SILCL 3 in accordance with EN 62061, PL e/Cat. 4 in accordance with EN ISO 13849-1

A safety-related application up to SILCL 3 to EN 62061, PL e/Cat. 4 to EN ISO 13849-1 can be realized in this way.

To achieve SILCL 3/PL e/Cat. 4 for the safety-related application, all components of the safety functions (detecting/evaluating/reacting) must be designed accordingly.

Note

A fuse must be fitted as short-circuit protection (4 A gL/gG) in the supply line to the supply voltage of 3RM1 (L1).



A.2.4 3SK2 safety relay with 3RM11 motor starter via device connector

Description

- Shutdown via 3RM1 Failsafe motor starter
- "Reacting" subsystem up to SILCL 3 as per EN 62061 and PL e / Cat. 4 as per EN ISO 13849-1
- Safety-related control of up to five 3RM1 Failsafe motor starters via 3ZY12 device connectors
- Operational, non-safety-related switching of the 3RM1 Failsafe motor starter through its local input IN1 by means of the 3SK2 safety relay

WARNING

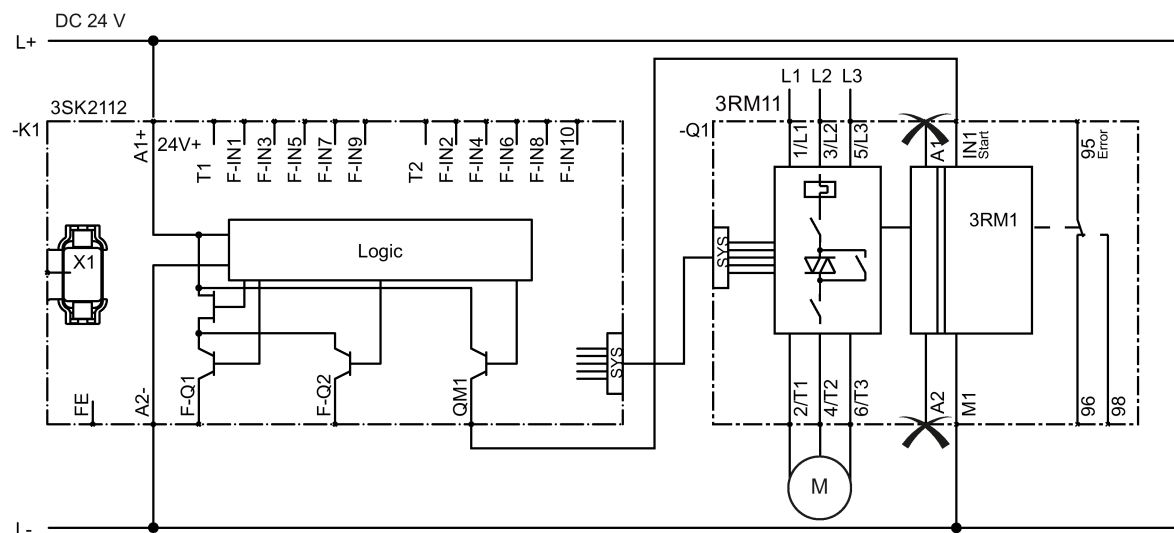
Bypassing the safety function when using device connectors

Can result in death, serious physical injury, or damage to property.

When operating with a 3SK2 safety relay and 3ZY12 device connectors, the supply voltage for 3RM1 Failsafe motor starters is established via the 3ZY12 device connectors.

To prevent bypassing of the safety function, do not connect anything to terminals A1 and A2 of the 3RM1 Failsafe motor starters in this case.

Circuit diagram



-K1 3SK2 safety relay 22.5 mm

-Q1 3RM11 Failsafe motor starters (direct-on-line starters)

You will find detailed information, for example, about "logic" or "parameters" in the manual "SIRIUS 3SK2 Safety Relays

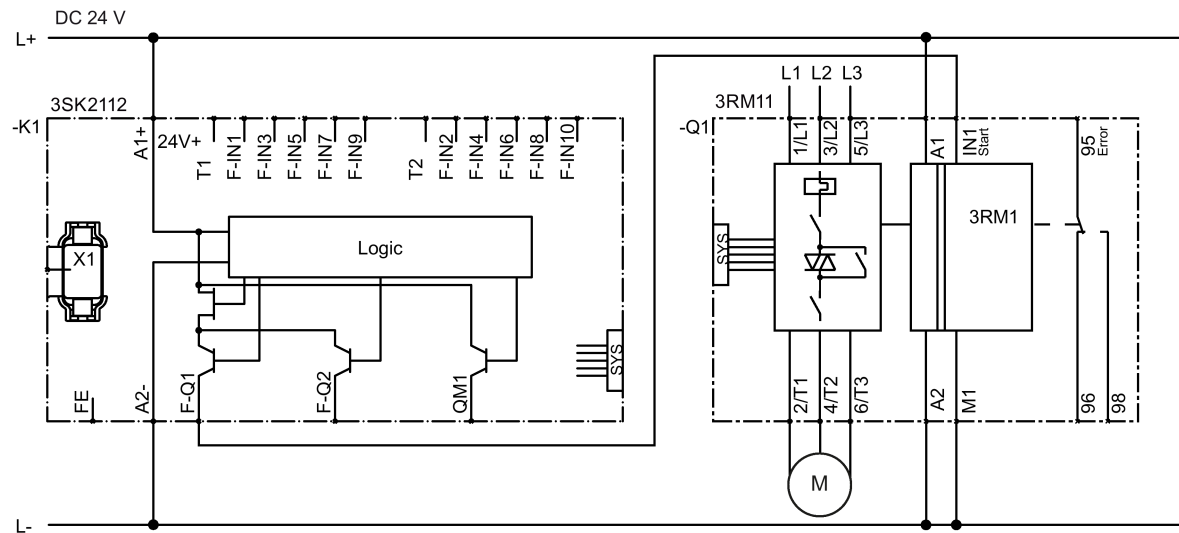
(<https://support.industry.siemens.com/cs/ww/de/view/109444336>)."

A.2.5 3SK2 safety relay wired to 3RM11 motor starter

Description

- Shutdown via 3RM1 Failsafe motor starter
- "Reacting" subsystem up to SILCL 3 as per EN 62061 and PL e / Cat. 4 as per EN ISO 13849-1
- Component certified in accordance with EN 62061/EN ISO 13849-1 in the actuator circuit (3RM1 Failsafe motor starter)
- Device supply of the 3RM1 Failsafe motor starter via the terminals A1/A2
- Safety-related switching of the local input of the 3RM1 Failsafe motor starter via fail-safe output of the 3SK2 safety relay
- Suitable for frequently requested safety functions (e.g. on indexing tables)
- No use of device connectors
- Protected laying of signal cables between the 3SK2 safety relay and 3RM1 Failsafe motor starter (in the same control cabinet or in armored conduit)

Circuit diagram



-K1 3SK2 safety relay 22.5 mm

-Q1 3RM11 Failsafe motor starters (direct-on-line starters)

You will find detailed information, for example, about "logic" or "parameters" in the manual "SIRIUS 3RM1 Motor Starter": SIRIUS 3SK2 Safety Relays

(<https://support.industry.siemens.com/cs/ww/de/view/109444336>)

A.2.6 3RM13 reversing starter with PLC control, F-DO pp-switching

As an alternative to the 3SK safety relay, 3RM11/3RM13 Failsafe motor starters can be controlled by an F-DO module. In this typical circuit, shutdown is realized by a pp-switching F-DO module in 1 channel.

Up to SILCL 3 in accordance with EN 62061, PL e/Cat. 4 in accordance with EN ISO 13849-1 can be achieved. To this end, the controlling outputs must also fulfill SILCL3 in accordance with EN 62061/PL e/Cat. 4 in accordance with EN ISO 13849-1.

Note

SILCL 3 in accordance with EN 62061, PL e/Cat. 4 in accordance with EN ISO 13849-1

SILCL 3 in accordance with EN 62061, PL e/Cat. 4 in accordance with EN ISO 13849-1 can be achieved by laying the control lines from the safe output to IN1 and IN2 of the 3RM1 motor starter in such a way that they are protected against a cross-circuit or short-circuit to a power cable, e.g. as separately sheathed cable or by laying in separate cable ducts.

This application is available for S7-300, ET 200M, ET 200MP and MSS 3RK3. This application is not available for ET 200S.

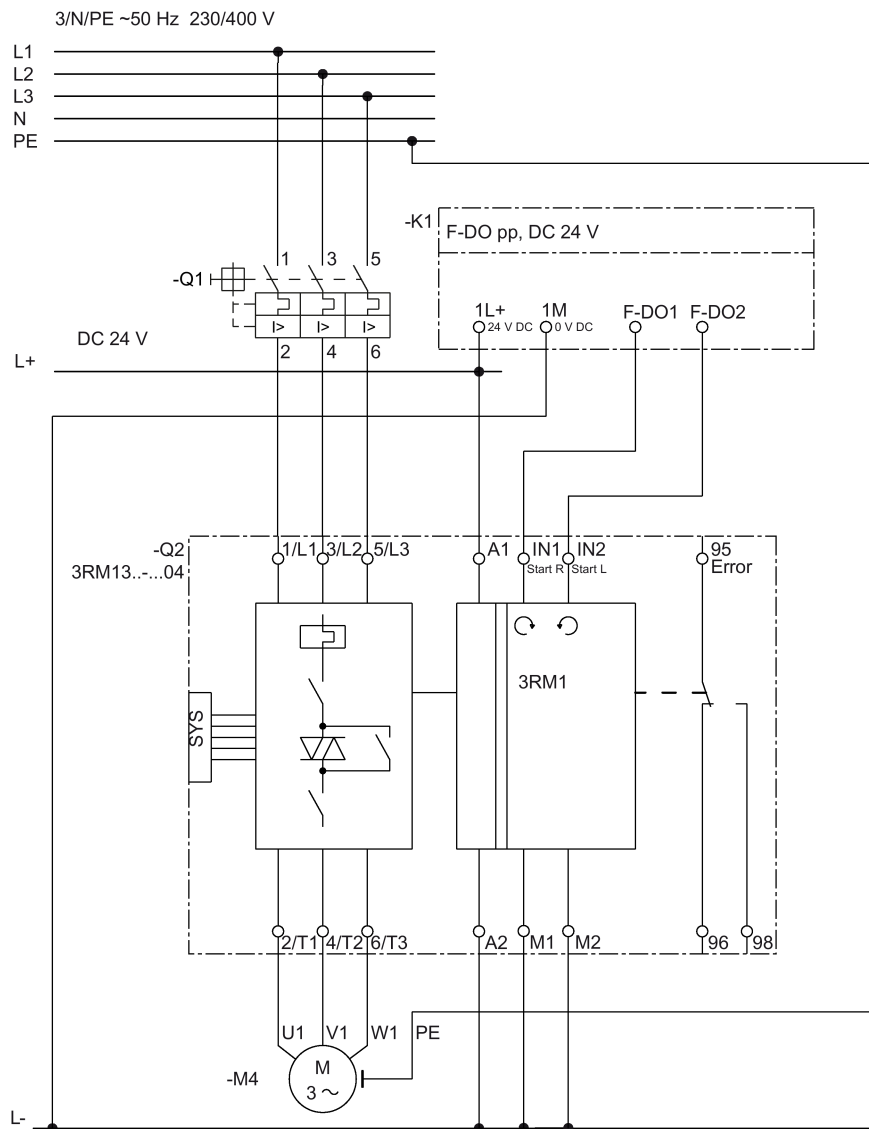


Figure A-9 3RM13 reversing starter with PLC control, F-DO pp-switching

A.2.7 3RM13 reversing starter with PLC control, F-DO pm-switching

As an alternative to the 3SK safety relay, 3RM11/3RM13 Failsafe motor starters can be controlled by an F-DO module. In this typical circuit, shutdown is realized by a pm-switching F-DO module in 2 channels.

Up to SILCL 3 in accordance with EN 62061, PL e/Cat. 4 in accordance with EN ISO 13849-1 can be achieved. To this end, the controlling outputs must also fulfill SILCL3 in accordance with EN 62061/PL e/Cat. 4 in accordance with EN ISO 13849-1.

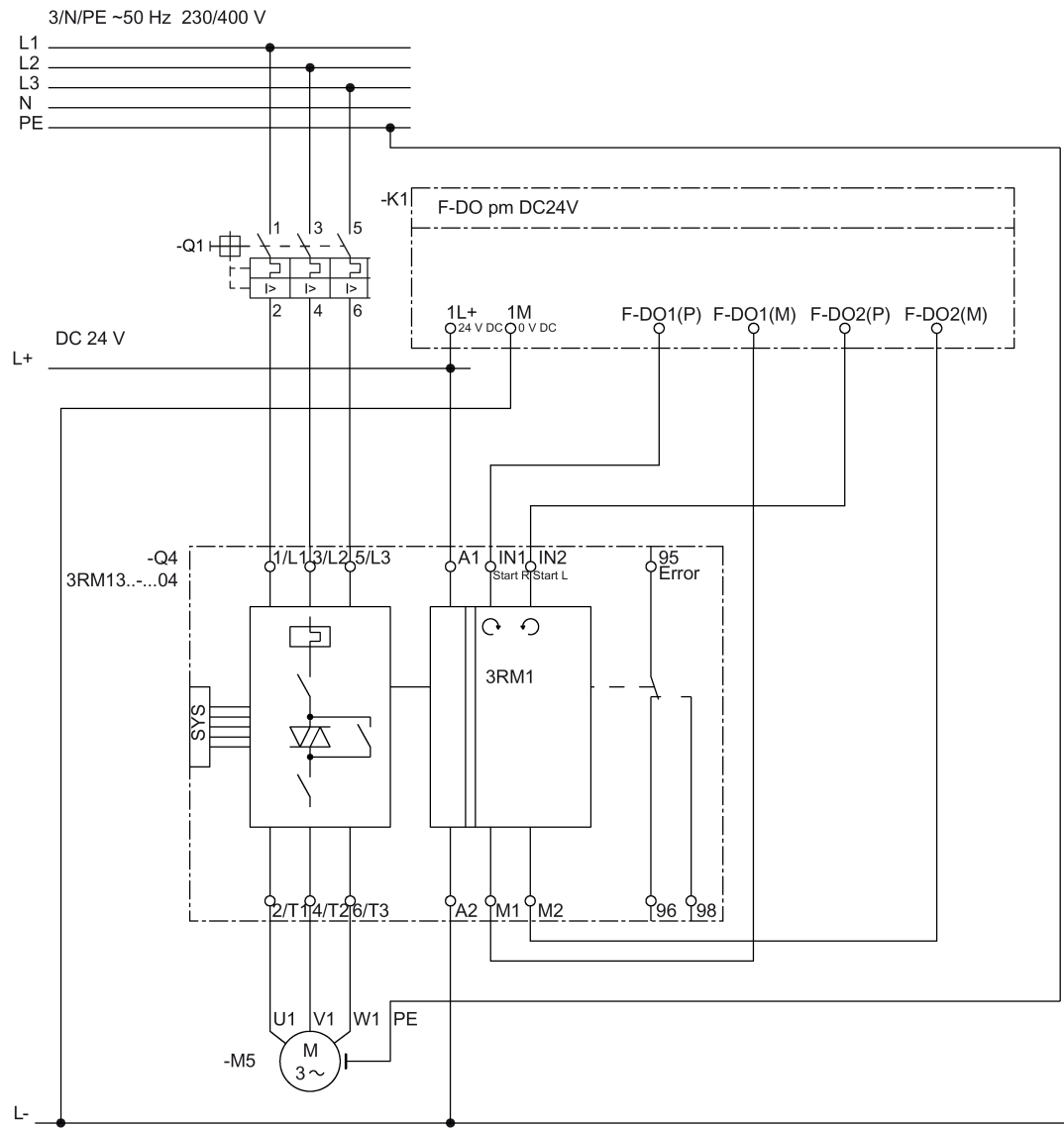


Figure A-10 3RM13 reversing starter with PLC control, F-DO pm-switching

Directives

B.1 ESD Guidelines

Definition of ESD

All electronic modules are equipped with large-scale integrated ICs or components. Due to their design, these electronic elements are highly sensitive to overvoltage, and thus to any electrostatic discharge.

The electrostatic sensitive components/modules are commonly referred to as ESD devices. This is also the international abbreviation for such devices.

ESD modules are identified by the following symbol:



NOTICE
ESD devices can be destroyed by voltages well below the threshold of human perception. These static voltages develop when you touch a component or electrical connection of a device without having drained the static charges present on your body. The electrostatic discharge current may lead to latent failure of a module, that is, this damage may not be significant immediately, but in operation may cause malfunction.

Electrostatic charging

Anyone who is not connected to the electrical potential of their surroundings can be electrostatically charged.

The figure below shows the maximum electrostatic voltage which may build up on a person coming into contact with the materials indicated. These values correspond to IEC 801-2 specifications.

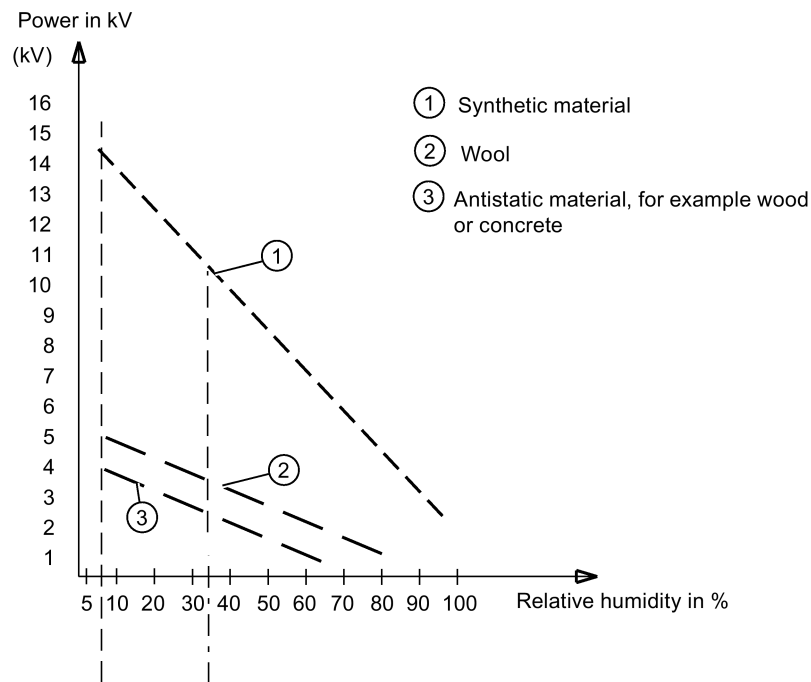


Figure B-1 Electrostatic voltages on an operator

Basic protective measures against electrostatic discharge

- **Ensure good equipotential bonding:**
When handling electrostatic sensitive devices, ensure that your body, the workplace and packaging are grounded. This prevents electrostatic charge.
- **Avoid direct contact:**
As a general rule, only touch electrostatic sensitive devices when this is unavoidable (e.g. during maintenance work). Handle the modules without touching any chip pins or PCB traces. In this way, the discharged energy can not affect the sensitive devices.

Discharge your body before you start taking any measurements on a module. Do so by touching grounded metallic parts. Always use grounded measuring instruments.

Glossary

3-phase busbar

The 3-phase busbar enables several 3RM1 motor starters to be supplied using a single infeed terminal.

Approval

Approval of switching devices and switchgear in accordance with national standards, some of which must be met on a mandatory basis, which exist in addition to "IEC", "CENELEC", and "CEE" guidelines, for example. The North American market (the USA and Canada), for example, requires UL or CSA approvals. There is also an obligation to label the equipment accordingly, i.e. the approval mark must be inscribed on the device itself.

AWG (American Wire Gauge)

A standard measure for conductors used in the USA, which is assigned to a specific cross-sectional area of a conductor or wire. Each AWG number represents a jump of 26% in the cross-sectional area. The thicker the wire, the smaller the AWG number.

CLASS (time)

See "Trip class".

Connection system

The 3RM1 motor starter is connected using screw terminals, push-in connections and hybrid connection technology.

Explosion protection

Prerequisite for using electrical equipment in hazardous areas conforming to DIN EN 50014 (VDE 0170/0171). For explosion protection, you must ensure that equipment which may generate explosive arcs (plasma) during operation is enclosed in a flameproof casing. This means that, although the potentially explosive mixture could penetrate the enclosure, if an explosion were to occur inside it then no explosive flame could escape.

Infeed system for 3RM1

The infeed system for 3RM1 enables simple and fast establishment of the main circuit connections when using several 3RM1 motor starters arranged side-by-side.

Main switch

Every industrial machine which falls under the scope of DIN EN 60204 Part 1 (VDE 0113, Part 1) must be equipped with a main switch which disconnects all electrical equipment from the network while cleaning, maintenance, and repair work is being carried out, as well as during long periods of downtime. Usually a switch which can be operated by hand is stipulated in order to prevent electrical or mechanical hazards. The main switch can also function as an EMERGENCY-OFF device.

It must meet the following requirements:

1. Externally accessible handle.
2. Only one "Off" position and one "On" position with allocated stops.
3. Two positions labeled "0" and "I".
4. Lockable "Off" position.
5. Cover for the power supply terminals to protect against accidental contact.
6. The switching capacity must correspond to AC-23 for motor switches and AC-22 for load-break switches (utilization category).
7. Switch position displayed automatically.

Motor protection

Motor protection protects electric motors against thermal overload due to mechanical overload or in the event of failure of a single or two phase(s).

Motor starter protector/circuit breaker

Generally a latch-operated switch that can make, carry, and break currents in the circuit under normal operating conditions; and that can make current under specified conditions that are not normal up to the point of a short circuit, conduct the current for a defined period, and break it.

Operating range

Range within which the control circuit voltage of a switching device may deviate from the rated control circuit voltage without impairing the operational reliability of the switching device (e.g. shutdown of the 3RM1 motor starter).

Overload relay

Inverse-time delayed relay which responds to an overload in accordance with a time/current characteristic, thus protecting the switching device and any load against overloads.

Overload release

Overcurrent release that provides protection against overload.

Push-in lugs for wall mounting

The adapters for screw mounting enable the 3RM1 motor starter to be mounted on a level surface.

Rated conditional short-circuit current I_q

The assured short-circuit breaking capacity of switchgear assemblies and load feeders, also called "rated conditional short-circuit current".

Rating data for the main circuit

The most important rating data for the main circuit in terms of selecting a switching device are the rated operational current I_e (current specified by the conditions of use) or the rated power (motor power), as well as the corresponding rated voltage U_e .

Recovery time

When a protection function in a switching device has been tripped (e.g. 3RM1 motor starter) the motor cannot be restarted until a recovery time has elapsed. The recovery time is dependent on the cause of the fault. For further information, refer to the relevant product documentation.

Response delay

The response delay is the time from the start of entering a command until the time that the first contact connection is made, at the contactor, for example. The switch-on delay is the time between application of the control voltage and the first current flow. The switch-off delay is the time between removal of the control voltage and interruption of the current flow.

RoHS

EC Directive 2002/95/EC concerning the restriction of the use of certain hazardous substances in electrical and electronic equipment regulates the use of hazardous substances in devices and components. The English abbreviation RoHS is used to refer to this directive: (Restriction of the use of certain hazardous substances), as well as all related measures for implementing it into national legislation.

Service life

Period of time for which the switching device will work properly under normal operating conditions. This is specified as the number of operating cycles, the electrical durability (e.g. contact erosion), and the mechanical durability (e.g. operating cycles without load).

SIL (Safety Integrity Level)

Discrete level (one of a possible three) for the determination of the safety integrity requirements of safety-related control functions. Safety integrity level 3 is the highest possible level, level 1 the lowest.

Trip class (CLASS)

The trip class of an inverse time-delay overload relay specifies the maximum tripping time from cold at a particular load. The trip class number (e.g. CLASS 10A) represents the maximum permissible tripping time in seconds when the 3RM1 motor starter is subjected to a symmetrical 3-pole load from cold with 7.2 times the current setting (IEC 60947-4-1; DIN VDE 0660 Part 107).

Tripping characteristic

The tripping characteristic (curve) is the graphical representation of the relationship between the tripping time and the influencing variable. The time/current diagram can be used to find out, for example, how long it will take for the release or the tripping relay to operate at a particular current.

Tripping current (of an overload release)

The current value at which a release trips within a certain time.

Type of coordination 1

The load feeder may be non-operational after a short circuit has been cleared. Damage to the 3RM1 motor starter is permissible.

Type of coordination 2

The 3RM1 motor starter remains functional after a short-circuit.

Types of coordination

The IEC 60947-4-1 (VDE 0660 Part 102) standard distinguishes between two types of coordination (type of coordination) referred to as coordination type "1" and coordination type "2". The short circuit that needs to be dealt with is cleared reliably and safely with both types of coordination; the only differences are in the extent of the damage sustained by the device following a short circuit.

Utilization category

According to IEC 60947-4-1, the application area of and the load applied to 3RM1 motor starters can be identified by looking at the specified utilization category in conjunction with the specified rated operational current or the motor power and the rated voltage. An example is utilization category AC 53 a for starting and switching off squirrel-cage motors.

Index

3

- 3D model, 13
- 3-phase busbar, 46
- 3-phase infeed terminal, 47
- 3SK safety relay, 42, 50, 75
- 3SK1 safety relay, 11
- 3SK2 safety relays, 11

A

- AC motors, 23
- Acknowledging, 35, 40, 109, 117
- Ambient conditions, 38
 - Application environment, 60
- Ambient temperature, 56, 60
- Applications, 24
- Approvals, 10
- Article No. scheme, 53
- Assembly type, 56
- ATEX, 36
- Auto RESET, 109

B

- Braking device, 33

C

- Cable protection, 38
- Capacitive load, 29, 144
- CAX data, 13
- Certification
 - ATEX, 36
- Characteristic, 10
- Characteristics, 124
- CLASS 10A, 55, 126
- Configuration, 13
- Configurator, 13
- Conformity, 10
- Connecting, 33
- Connection, 112
- Connection system, 26
- Connections, 113
- Control circuit, 111

- Control current terminal, 98
- Control current terminals, 52
- Control input, 30
- Control power supply, 49
- Control supply voltage, 27, 42
- Control voltage, 42
- Cooling down period, 34, 107, 109
- Cover cap, 46, 73
- Current range, 27

D

- Data sheet, 39, 121
- Declaration of conformity, 21
- Degree of protection, 11
- Derating, 49, 55
- Design version, 56
- Device connector, 42, 74, 75
 - Device connector for looping through signals, 49
 - Terminator, 49
- Device connectors for looping through signals, 49
- Device identification label, 52
- Device replacement, 120
- Device status, 114
- Device versions, 26
- Diagnostics, 114
- Dimension drawings, 13
- Direct-on-line starter, 30
- Display, 114

E

- EC declaration of conformity, 21
- End sleeve, 98
- EPLAN macros, 13
- Equipment protection, 32, 37
- ESD guidelines, 161

F

- Fail-safe, 42
- Fault, 115
- Fault rectification, 117
- Fault signaling output, 111, 116
- Feedback output, 116
- Frequency converters, 29

Function test, 43
Fused design
 Load feeder, 65
Fuseless design
 Load feeder, 63

G

Grounding measures, 60
Group installation, 70
Guidelines
 ESD guidelines, 161

H

Hazardous areas, 36, 37
High capacity short circuit current ratings, 67
Hybrid technology, 25

I

Infeed, 72
Infeed system, 46, 72, 104
 Mounting, 73
Installation altitude, 59

L

LED colors, 114
LED display, 114
LED test, 43
Load feeder, 26, 62
 IEC, 62
 UL, 67
Loads, 23

M

Main circuit, 72, 111
Main current terminal, 98
Main current terminals, 52
Maintenance, 39, 119
Malfunction, 40, 117, 119
 Display concept, 41
Manual RESET, 109
Messages, 114
Minimum load current, 61
Monitoring, 40
Motor memory module, 32, 34, 37
Motor power, 27

Motor protection, 32, 37, 125
Motor protection test disconnection, 43
Motorized load, 61
Mounting position, 60

N

Newsletters, 20

O

Online configurator, 13
Operating mode, 114
Operating voltage, 61
Operator controls, 107
Ordering, 13
Overload protection, 32
Overload relay
 solid-state, 23
Overload tripping, 32, 34, 37, 107, 109

P

Parameter assignment, 108
Phase failure, 55, 115
Phase failure protection, 33
PM (PPM)-switching, 43
Power semiconductors, 23
PP-switching, 43
Product data sheet, 121
Protected against a cross-circuit, 43
Protected against a line-to-line fault, 43
Protective cap, 46
Protective function, 38
Pushbutton, 107
Push-in, 98
Push-in connection, 26
Push-in lugs, 48

R

Rated operational current, 27, 37, 55, 72, 107, 109
References, 11
Relay contacts, 23
Repairs, 39
Reset, 35, 109
RESET, 35, 109
Resistive load, 27, 29, 35, 55, 61
Reversing starter, 30
Rotary coding switch, 48, 107, 108, 109, 111

S

- Safety data
 - ATEX, 39
- Safety function, 19, 50, 75, 150
- Safety-related application
 - Example, 74
- Safety-related shutdown, 26, 40, 41, 43
- SCCR, 67
- Screw connection, 26
- Sealable cover, 48, 109
- Sealing, 93
- Service&Support, 11
- Setting, 108
- Short circuit current rating, 67
- Short-circuit protection, 24, 38, 62
- Short-circuit strength, 62
- Shutdown, 40
 - Safety-related, 41
- Solid-state overload relays, 23
- Spring-loaded terminals, 98
- Standards, 10, 19
 - ATEX, 36
- Status messages, 114
- Switching frequency, 35
- System documentation, 13

T

- Technical data, 121
- Terminal cover, 113
- Terminals, 52
- Terminator, 49
- Test, 38, 115
 - Current measurement, 43
- Test reports, 10
- Three-phase infeed system, 72
- Trip class, 37
- Tripping
 - Safety-related, 43
- Tripping characteristic, 37, 126

U

- UL 508, 69, 71
- UL 508a, 67, 70

V

- Voltage source, 27

W

- Wall mounting, 48
- Warranty, 39

